Huawei CloudEngine S6730-H-V2 Series 10GE Switches Datasheet

Huawei CloudEngine S6730-H-V2 series 10GE switches are next-generation enterprise-class core and aggregation switches that provide 24/48-10GE downlink optical ports and 100GE uplink optical ports, and provide one extended slot.

Introduction

Huawei CloudEngine S6730-H-V2 series switches are next-generation enterprise-class core and aggregation switches that offer high performance, high reliability, cloud management, and intelligent operations and maintenance (O&M). They are purpose-built with security, IoT, and cloud in mind. With these traits, CloudEngine S6730-H-V2 can be widely used in enterprise campuses, colleges/universities, data centers, and other scenarios.

CloudEngine S6730-H-V2 switches offer 10GE, 40GE, and 100GE port types, flexibly adapting to diversified network bandwidth requirements. They also support cloud management and implement cloud-managed network services throughout the full lifecycle from planning, deployment, monitoring, experience visibility, and fault rectification, all the way to network optimization, greatly simplifying network management.

CloudEngine S6730-H-V2 support free mobility, enables consistent user experience no matter the user location or IP address, fully meeting enterprises' demands for mobile offices.

CloudEngine S6730-H-V2 switches support VXLAN to implement network virtualization, achieving multi-purpose networks and multi-network convergence for greatly improved network capacity and utilization. As such, CloudEngine S6730-H-V2 switches are an ideal choice for building next-generation IoT converged networks in terms of cost, flexibility, and scalability.

Product Overview

1&6 - S6730-H24X6C-V2(24*10GE SFP+ PORTS, 6*40GE QSFP28 PORTS, OPTIONAL LICENSE FOR UPGRADE TO 6*100GE QSFP28, WITHOUT POWER MODULE)

Models and Appearances

The following models are available in the CloudEngine S6730-H-V2 series.

Appearance	Description
 CloudEngine S6730-H48X6C-V2 CloudEngine S6730-H48X6C-TV2**	• 48 x 1/10 Gig SFP+, 6 x 40/100 Gig QSFP28 • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 490 Mpps • Switching capacity: 2.16Tbps/2.4Tbps* *Note: All ports support 40GE by default. You can purchase right-to-use (RTU) licenses to upgrade the port rate from 40GE to 100GE*
 CloudEngine S6730-H24X6C-V2 CloudEngine S6730-H24X6C-TV2**	• 24 x 1/10 Gig SFP+, 6 x 40/100 Gig QSFP28 • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 490 Mpps • Switching capacity: 1.68Tbps/2.4Tbps*

Appearance	Description
	Note: All ports support 40GE by default. You can purchase right-to-use (RTU) licenses to upgrade the port rate from 40GE to 100GE
 CloudEngine S6730-H48X6CZ-V2 CloudEngine S6730-H48X6CZ-TV2**	• 48 x 10 Gig SFP+, 6 x 40/100 Gig QSFP28 • One extended slot • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 490 Mpps • Switching capacity: 2.16Tbps/2.4Tbps* *Note: All ports support 40GE by default. You can purchase right-to-use (RTU) licenses to upgrade the port rate from 40GE to 100GE*
 CloudEngine S6730-H28X6CZ-V2 CloudEngine S6730-H28X6CZ-TV2**	• 28 x 10 Gig SFP+, 6 x 40/100 Gig QSFP28 • One extended slot • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 490 Mpps • Switching capacity: 1.76Tbps/2.4Tbps* *Note: All ports support 40GE by default. You can purchase right-to-use (RTU) licenses to upgrade the port rate from 40GE to 100GE*

*Note：The value before the slash (/) refers to the device's switching capability, while the value after the slash (/) means the system's switching capability.

**Note：‘-T ’means Hardware Trust Module(HTM), support hardware root of trust and measurement startup.

Fan Module

The following table lists the fan module on the CloudEngine S6730-H-V2 series.

Fan Module	Technical Specifications	Applied Switch Model
 FAN-031A-B	• Dimensions (W x D x H): 40 mm x 100.3 mm x 40 mm • Number of fans: 1 • Weight: 0.1 kg • Maximum power consumption: 21.6 W • Maximum fan speed: 24500±10% revolutions per minute (RPM) • Maximum wind rate: 31 cubic feet per minute (CFM) • Hot swap: Supported	• CloudEngine S6730-H48X6C-V2 • CloudEngine S6730-H48X6C-TV2 • CloudEngine S6730-H24X6C-V2 • CloudEngine S6730-H24X6C-TV2 • CloudEngine S6730-H48X6CZ-V2 • CloudEngine S6730-H48X6CZ-TV2 • CloudEngine S6730-H28X6CZ-V2 • CloudEngine S6730-H28X6CZ-TV2

Power Supply

The following table lists the power supplies on the CloudEngine S6730-H-V2 series.

Power Module	Technical Specifications	Applied Switch Model

Power Module	Technical Specifications	Applied Switch Model
 PAC600S12-CB	• Dimensions (H x W x D): 40 mm x 90 mm x 215 mm • Weight: 0.95 kg (2.09 lb) • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 65 Hz – 190 V DC to 290 V DC • Maximum input current: – 100 V AC to 240 V AC: 8 A – 240 V DC: 4 A • Maximum output current: 50 A • Rated output voltage: 12 V • Maximum output power: 600 W • Hot swap: Supported	• CloudEngine S6730-H48X6C-V2 • CloudEngine S6730-H24X6C-V2 • CloudEngine S6730-H48X6C-TV2 • CloudEngine S6730-H24X6C-TV2
 PAC600S12-EB 2&7- 600W AC Power Module(Back to Front, Power panel side exhaust)	• Dimensions (H x W x D): 40 mm x 90 mm x 215 mm • Weight: 0.985 kg • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 65 Hz – 190 V DC to 290 V DC • Maximum input current: – 100 V AC to 240 V AC: 8 A – 240 V DC: 4 A • Maximum output current: 50 A • Rated output voltage: 12 V • Maximum output power: 600 W • Hot swap: Supported	• CloudEngine S6730-H48X6C-V2 • CloudEngine S6730-H24X6C-V2 • CloudEngine S6730-H48X6C-TV2 • CloudEngine S6730-H24X6C-TV2
 PAC600S12-DB	• Dimensions (H x W x D): 40 mm x 90 mm x 215 mm • Weight: 0.95 kg • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 65 Hz – 190 V DC to 290 V DC • Maximum input current: – 100 V AC to 240 V AC: 8 A – 240 V DC: 4 A • Maximum output current: 50 A • Rated output voltage: 12 V • Maximum output power: 600 W	• CloudEngine S6730-H48X6C-V2 • CloudEngine S6730-H24X6C-V2 • CloudEngine S6730-H48X6C-TV2 • CloudEngine S6730-H24X6C-TV2

Power Module	Technical Specifications	Applied Switch Model
	• Hot swap: Supported	
 PDC1000S12-DB	• Dimensions (H x W x D): 40 mm x 90 mm x 215 mm • Weight: 1.02 kg (2.25 lb) • Rated input voltage range: -48 V DC to -60 V DC • Maximum input voltage range: -38.4 V DC to -72 V DC • Maximum input current: 30 A • Maximum output current: 83.3 A • Maximum output power: 1000 W • Hot swap: Supported	• CloudEngine S6730-H48X6C-V2 • CloudEngine S6730-H24X6C-V2 • CloudEngine S6730-H48X6C-TV2 • CloudEngine S6730-H24X6C-TV2
 PAC600S12-PB	• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm (1.6 in. x 2.6 in. x 8.5 in.) • Weight: 1 kg (2.2 lb) • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 65 Hz – 190 V DC to 290 V DC • Maximum input current: – 100 V AC to 240 V AC: 8 A – 240 V DC: 4 A • Maximum output current: 50 A • Rated output voltage: 12 V • Maximum output power: 600 W • Hot swap: Supported	• CloudEngine S6730-H48X6CZ-V2 • CloudEngine S6730-H48X6CZ-TV2 • CloudEngine S6730-H28X6CZ-V2 • CloudEngine S6730-H28X6CZ-TV2
 PAC1K2S12-PB	• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm • Weight: 0.84 kg • Rated input voltage range: – 200V AC～240V AC；50/60Hz – 100V AC～130V AC；50/60Hz – 240V DC • Maximum input voltage range: – AC：90V AC～290V AC；45Hz～65Hz – HVDC：190V DC～290V DC • Maximum input current: – 100V AC～130V AC：10A – 200V AC～240V AC：8A – 240V DC：8A • Maximum output current: 100 A • Maximum output power: 1200 W • Hot swap: Supported	• CloudEngine S6730-H48X6CZ-V2 • CloudEngine S6730-H48X6CZ-TV2 • CloudEngine S6730-H28X6CZ-V2 • CloudEngine S6730-H28X6CZ-TV2

Power Module	Technical Specifications	Applied Switch Model
 PDC1K2S12-CE	• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm (1.6 in. x 2.6 in. x 8.5 in.) • Weight: 1.5 kg (3.31 lb) • Rated input voltage range: -48 V DC to -60 V DC • Maximum input voltage range: -38.4 V DC to -72 V DC • Maximum input current: 38 A • Maximum output current: 83.3 A • Maximum output power: 1200 W • Hot swap: Supported	• CloudEngine S6730- H48X6CZ-V2 • CloudEngine S6730- H48X6CZ-TV2 • CloudEngine S6730- H28X6CZ-V2 • CloudEngine S6730- H28X6CZ-TV2

The S6730-H-V2 uses pluggable power modules. It can be configured with a single power module or double power modules for 1+1 power redundancy.

Product Features and Highlights

Enabling Networks to be More Agile for Services

- Built-in high-speed and flexible processor chips, with their flexible packet processing and traffic control capabilities, CloudEngine S6730-H-V2 series switches are close to services, meeting current and future challenges, and helping customers build scalable networks.
- CloudEngine S6730-H-V2 series switches support fully customizing the forwarding mode, forwarding behavior, and search algorithm of traffic. New services are implemented through microcode programming. Customers do not need to replace new hardware and new services can be rolled out within six months.
- CloudEngine S6730-H-V2 series switches provide open interfaces and user-defined forwarding processes to meet customized service requirements of enterprises. Enterprises can use multi-layer open interfaces to develop new protocols and functions independently. They can also hand over their requirements to vendors and jointly develop them to build an enterprise-dedicated campus network.

Delivering Abundant Services More Agilely

- With the unified user management function, the CloudEngine S6730-H-V2 series switches authenticates both wired and wireless users, ensuring a consistent user experience no matter whether they are connected to the network through wired or wireless access devices. The unified user management function supports various authentication methods, including 802.1x, MAC address, and is capable of managing users based on user groups, domains, and time ranges. These functions visualize user and service management and boost the transformation from device-centric management to user experience-centric management.
- The CloudEngine S6730-H-V2 series switches provide excellent quality of service(QoS) capabilities and supports queue scheduling and congestion control algorithms. Additionally, it adopts innovative priority queuing and multi-level scheduling mechanisms to implement fine-grained scheduling of data flows, meeting service quality requirements of different user terminals and services.

Fine-Grained Network Management and Visualized Fault Diagnosis

- In-situ Flow Information Telemetry (IFIT) is an in-band Operations, Administration, and Maintenance (OAM) measurement technology that uses service packets to measure real performance indicators of an IP network, such as the packet loss rate and delay. IFIT can significantly improve the timeliness and effectiveness of network O&M, thereby promoting the development of intelligent O&M.
- Three IFIT modes are available: application-level quality measurement, tunnel-level quality measurement, and native-IP IFIT measurement. Currently, CloudEngine S6730-H-V2 series switches support native-IP IFIT measurement only. By providing in-band measurement capabilities, CloudEngine S6730-H-V2 series switches can monitor indicators such as the delay and packet loss rate of service flows in real time. CloudEngine S6730-H-V2 series switches also offer visualized O&M capabilities to centrally manage and control networks and graphically display performance data. Designed with IFIT capabilities featuring high measurement precision and easy deployment, CloudEngine S6730-H-V2 series switches are ideal for constructing an intelligent O&M system and stand out with future-proof scalability.

Flexible Ethernet Networking

- In addition to traditional Spanning Tree Protocol (STP), Rapid Spanning Tree Protocol (RSTP), and Multiple Spanning Tree Protocol (MSTP), the CloudEngine S6730-H-V2 series switches support Huawei-developed Smart Ethernet Protection (SEP) technology and the latest Ethernet Ring Protection Switching (ERPS) standard. SEP is a ring protection protocol specific to the Ethernet link layer, and applies to various ring network topologies, such as open ring topology, closed ring topology, and cascading ring topology. This protocol is reliable, easy to maintain, and implements fast service switching within 50 ms. ERPS is defined in ITU-T G.8032. It implements millisecond-level protection switching based on traditional Ethernet MAC and bridging functions.
- CloudEngine S6730-H-V2 series switches support Smart Link and Virtual Router Redundancy Protocol (VRRP), which implement backup of uplinks. One CloudEngine S6730-H-V2 series switches can connect to multiple core switches through multiple links, significantly improving reliability of aggregation devices.

Mature IPv6 Features

- The CloudEngine S6730-H-V2 series swithes are developed based on the mature, stable VRP and supports IPv4/IPv6 dual stacks, IPv6 routing protocols (RIPng, OSPFv3, BGP4+, and IS-IS for IPv6). With these IPv6 features, the CloudEngine S6730-H-V2 can be deployed on a pure IPv4 network, a pure IPv6 network, or a shared IPv4/IPv6 network, helping achieve IPv4-to-IPv6 transition.

Intelligent Stack (iStack)

- CloudEngine S6730-H-V2 series switches support the iStack function that combines multiple switches into a logical switch. Member switches in a stack implement redundancy backup to improve device reliability and use inter-device link aggregation to improve link reliability. iStack provides high network scalability. You can increase a stack's ports, bandwidth, and processing capability by simply adding member switches. iStack also simplifies device configuration and management. After a stack is set up, multiple physical switches can be virtualized into one logical device. You can log in to any member switch in the stack to manage all the member switches in it.

VXLAN Features

- VXLAN is used to construct a Unified Virtual Fabric (UVF). As such, multiple service networks or tenant networks can be deployed on the same physical network, and service and tenant networks are isolated from each other. This capability truly achieves 'one network for multiple purposes'. The resulting benefits include enabling data transmission of different services or customers, reducing the network construction costs, and improving network resource utilization.
- This series switches are VXLAN-capable and allow centralized and distributed VXLAN gateway deployment modes. These switches also support the BGP EVPN protocol for dynamically establishing VXLAN tunnels and can be configured using NETCONF/YANG.

Link Layer Security

- CloudEngine S6730-H28X6CZ-TV2/S6730-H28X6CZ-V2/S6730-H48X6CZ-TV2/S6730-H48X6CZ-V2 models support MACsec. MACsec protects transmitted Ethernet data frames through identity authentication, data encryption, integrity check, and anti-replay protection, reducing the risks of information leakage and malicious network attacks. With MACsec, these switch models are able to address strict information security requirements of customers in industries such as government and finance.

Intelligent O&M

- This series switches provides telemetry technology to collect device data in real time and send the data to Huawei campus network analyzer(iMaster NCE-CampusInsight). The CampusInsight analyzes network data based on the intelligent fault identification algorithm, accurately displays the real-time network status, effectively demarcates and locates faults in a timely manner, and identifies network problems that affect user experience, accurately guaranteeing user experience.

Intelligent Upgrade

- Switches support the intelligent upgrade feature. Specifically, switches obtain the version upgrade path and download the newest version for upgrade from the Huawei Online Upgrade Platform (HOUP). The entire upgrade process is highly automated and achieves one-click upgrade. In addition, preloading the version is supported, which greatly shortens the upgrade time and service interruption time.
- The intelligent upgrade feature greatly simplifies device upgrade operations and makes it possible for the customer to upgrade the version independently. This greatly reduces the customer's maintenance costs. In addition, the upgrade policies on the HOUP platform standardize the upgrade operations, which greatly reduces the risk of upgrade failures.

Cloud-based Management

- The Huawei cloud management platform allows users to configure, monitor, and inspect switches on the cloud, reducing on-site deployment and O&M manpower costs and decreasing network OPEX.

Open Programmability System(OPS)

- Open Programmability System (OPS) is an open programmable system based on the Python language. IT administrators can program the O&M functions of a switch through Python scripts to quickly innovate functions and implement intelligent O&M.

Product Specifications

The following table describes the functions and features available on the CloudEngine S6730-H-V2 series.

Functions and Features

Category	Service Features
User management	Unified user management
	802.1X authentication
	MAC authentication
	Traffic- and duration-based accounting
	User authorization based on user groups, domains, and time ranges
MAC	Automatic MAC address learning and aging
	384K MAC entries (MAX)
	Static, dynamic, and blackhole MAC address entries
	Source MAC address filtering
	MAC address learning limiting based on ports and VLANs
VLAN	4K VLANs simultaneously
	4K VLANif interface simultaneously
	Access mode, Trunk mode and Hybrid mode
	Default VLAN
	Private VLAN
	QinQ and enhanced selective QinQ
	VLAN Stacking
	Dynamic VLAN assignment based on MAC addresses
ARP	ARP Snooping
DHCP	DHCPv4 Client, DHCPv4 Relay, DHCPv4 Server, DHCPv4 Snooping
	DHCPv6 Client, DHCPv6 Relay, DHCPv6 Server, DHCPv6 Snooping
IP routing	Maximum number of IPv4 routing entries: 256,000(MAX)
	Maximum number of IPv6 routing entries: 80,000(MAX)
	IPv4 dynamic routing protocols such as RIP v1/v2, OSPF v1/v2, IS-IS, and BGP
	IPv6 dynamic routing protocols such as RIPng, OSPFv3, ISISv6, and BGP4+

Category	Service Features
	Routing Policy, Policy-Based Routing
	VRF
Segment Routing	SRv6 BE (L3 EVPN)
	BGP EVPN
	SRv6 configuration through NETCONF
Multicast	IGMPv1/v2/v3 and IGMP v1/v2/v3 Snooping
	PIM-DM, PIM-SM, and PIM-SSM
	Fast-leave mechanism
	Multicast traffic control
	Multicast querier
	Multicast protocol packet suppression
MPLS	MPLS-LDP
	MPLS-L3VPN
	MPLS Qos
VXLAN	Centralized gateway
	Distributed gateway
	BGP-EVPN
	Configures VXLANs through NETCONF
QoS	Number of ACL rules: 6,656(MAX)
	Traffic classification based on Layer 2 headers, Layer 3 protocols(IP), Layer 4 protocols(TCP/UDP), and 802.1p priority
	Actions such as ACL, Committed Access Rate (CAR), re-marking, and scheduling
	Queuing algorithms, such as PQ, DRR, WDRR, and PQ+DRR, PQ+WDRR
	Congestion avoidance mechanisms such as WRED and tail drop
	Traffic shaping
	8 queues on each interface
	Network Slicing
Native-IP IFIT	Marks the real service packets to obtain real-time count of dropped packets and packet loss ratio
	The statistical period can be modified
	Two-way frame delay measurement
Ethernet loop protection	STP (IEEE 802.1d), RSTP (IEEE 802.1w), and MSTP (IEEE 802.1s).
	VLAN-based Spanning Tree (VBST)
	BPDU protection, root protection, and loop protection
	G.8032 Ethernet Ring Protection Switching (ERPS)

Category	Service Features
Reliability	M-LAG
	Service interface-based stacking
	Maximum number of stacked devices: 9
	Stack bandwidth (Bidirectional)
	Link Aggregation Control Protocol (LACP) and E-Trunk
	Virtual Router Redundancy Protocol (VRRP) and Bidirectional Forwarding Detection (BFD) for VRRP
	BFD for BGP/IS-IS/OSPF/static routes
	Eth-OAM 802.1ag(CFM)
	Smartlink
	LLDP, LLDP-MED
System management	Console terminal service
	Telnet/IPv6 Telnet terminal service
	SSH v1.5
	SSH v2.0
	SNMP v1/v2c/v3
	FTP、TFTP、SFTP
	BootROM upgrade and remote in-service upgrade
	Hot patch
	User operation logs
	Open Programmability System (OPS)
	Streaming Telemetry
	JSON/XML/GPB encoding
	CRUD(create, read, update, and delete) Operations
Security and management	NAC
	Port security
	RADIUS and HWTACACS authentication for login users
	MACsec-256 (IEEE 802.1ae)
	Management by Command Line Interface(CLI)
	Command line authority control based on user levels, preventing unauthorized users from using command configurations
	Defense against DoS attacks, Transmission Control Protocol (TCP) SYN Flood attacks, User Datagram Protocol (UDP) Flood attacks, broadcast storms, and heavy traffic attacks
	IPv6 RA Guard
	CPU hardware queues to implement hierarchical scheduling and protection for protocol packets on the control plane

Category	Service Features
	Remote Network Monitoring (RMON)
	Secure boot
	Netstream
	Port mirroring
	Dynamic ARP Inspection
	IP Source Guard
	Auto MDI and MDI-X

NOTE

This content is applicable only to regions outside mainland China. Huawei reserves the right to interpret this content.

Hardware Specifications

The following table lists hardware specifications of the CloudEngine S6730-H-V2 series.

Item		CloudEngine S6730-H48X6CZ-V2 CloudEngine S6730-H48X6CZ-TV2	CloudEngine S6730-H28X6CZ-V2 CloudEngine S6730-H28X6CZ-TV2
Physical specifications	Dimensions (H x W x D)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.4 in. x 16.5 in.)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.4 in. x 16.5 in.)
	Chassis height	1 U	1 U
	Chassis weight (full configuration weight, including weight of packaging materials)	7.48 Kg(16.49 lb)	7.06 Kg(15.56 lb)
Fixed port	GE port	48	28
	10GE port	48	28
	25GE port	-	-
	40GE port	6	6
	100GE port	6 (*RTU upgrade*)	6 (*RTU upgrade*)
Extended slot		One extended slot, support 2 x 100GE, 2 x 40GE and 8 x 25/10GE cards in the future	
Management port	ETH management port	Supported	Supported
	Console port (RJ45)	Supported	Supported
	USB port	USB 2.0	USB 2.0
CPU	Frequency	1.4 GHz	1.4 GHz
	Cores	4	4
Memory	Memory (RAM)	4GB	4GB
	Flash	Hardware: 2 GB	Hardware: 2 GB
Power supply	Power supply type	• 600 W AC (pluggable)	• 600 W AC (pluggable)

Item		CloudEngine S6730-H48X6CZ-V2 CloudEngine S6730-H48X6CZ-TV2	CloudEngine S6730-H28X6CZ-V2 CloudEngine S6730-H28X6CZ-TV2
system		• 1200 W AC (pluggable) • 1200 W DC (pluggable)	• 1200 W AC (pluggable) • 1200 W DC (pluggable)
	Rated voltage range	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC
	Maximum voltage range	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC
	Maximum input current	• AC 600W：8A • DC 1000W：38A	• AC 600W：8A • DC 1000W：38A
	Typical power consumption (30% of traffic load, tested according to ATIS standard)	234 W	203 W
	Maximum power consumption (100% throughput, full speed of fans)	327 W	263 W
Heat dissipation system	Heat dissipation mode	Air-cooled heat dissipation and intelligent fan speed adjustment	Air-cooled heat dissipation and intelligent fan speed adjustment
	Number of fan modules	3, Fan modules are pluggable	3, Fan modules are pluggable
	Airflow	Air flows in from the front side and exhausts from the rear panel	Air flows in from the front side and exhausts from the rear panel
Environment parameters	Long-term operating temperature	• 0-1800 m: -5°C to 45°C • 1800-5000 m: The operating temperature decreases 1°C for every 220 m increase in altitude.	• 0-1800 m: -5°C to 45°C • 1800-5000 m: The operating temperature decreases 1°C for every 220 m increase in altitude.
	Storage temperature	-40°C to +70°C	-40°C to +70°C
	Relative humidity	5% to 95%, noncondensing	5% to 95%, noncondensing
	Operating altitude	0-5000 m	0-5000 m
	Acoustics (LpA) maximum	39.72 dB(A)	39.72 dB(A)
	Acoustics (LwA) maximum	5.34B	5.34B
	Surge protection specification (power port)	• Using AC power modules: ±6 kV in differential mode, ±6 kV in common mode • Using DC power modules: ±2 kV in differential mode, ±4 kV in	• Using AC power modules: ±6 kV in differential mode, ±6 kV in common mode • Using DC power modules: ±2 kV in differential mode, ±4 kV in

Item		CloudEngine S6730-H48X6CZ-V2 CloudEngine S6730-H48X6CZ-TV2	CloudEngine S6730-H28X6CZ-V2 CloudEngine S6730-H28X6CZ-TV2
		common mode	common mode
Reliability	MTBF (year)[2]	48.99	52.24
	MTTR (hour)	2	2
	Availability	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification NOTE For details about certifications, see the section Safety and Regulatory Compliance.	• EMC certification • Safety certification • Manufacturing certification NOTE For details about certifications, see the section Safety and Regulatory Compliance.

Item		CloudEngine S6730-H24X6C-V2 CloudEngine S6730-H24X6C-TV2	CloudEngine S6730-H48X6C-V2 CloudEngine S6730-H48X6C-TV2
Physical specifications	Dimensions (H x W x D)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.4 in. x 16.5 in.)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.4 in. x 16.5 in.)
	Chassis height	1 U	1 U
	Chassis weight (full configuration weight, including weight of packaging materials)	8.9 kg (19.62 lb)	9.2 kg (20.28 lb)
Fixed port	GE port	24(GE and 10GE auto-sensing)	48(GE and 10GE auto-sensing)
	10GE port	24	48
	25GE port	-	-
	40GE port	6	6
	100GE port	6 (*RTU upgrade*)	6 (*RTU upgrade*)
Extended slot		NA	
Management port	ETH management port	Supported	Supported
	Console port (RJ45)	Supported	Supported
	USB port	USB 2.0	USB 2.0
CPU	Frequency	1.4 GHz	1.4 GHz
	Cores	4	4
Memory	Memory (RAM)	4GB	4GB
	Flash	Hardware: 2 GB	Hardware: 2 GB
Power supply	Power supply type	• 600 W AC (pluggable)	• 600 W AC (pluggable)

Item		CloudEngine S6730-H24X6C-V2 CloudEngine S6730-H24X6C-TV2	CloudEngine S6730-H48X6C-V2 CloudEngine S6730-H48X6C-TV2
system		• 1000 W DC (pluggable)	• 1000 W DC (pluggable)
	Rated voltage range	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC
	Maximum voltage range	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC
	Maximum input current	• AC 600W：8A • DC 1000W：30A	• AC 600W：8A • DC 1000W：30A
	Typical power consumption (30% of traffic load, tested according to ATIS standard)	149 W	165 W
	Maximum power consumption (100% throughput, full speed of fans)	254 W	291 W
Heat dissipation system	Heat dissipation mode	Air-cooled heat dissipation and intelligent fan speed adjustment	Air-cooled heat dissipation and intelligent fan speed adjustment
	Number of fan modules	4, Fan modules are pluggable	4, Fan modules are pluggable
	Airflow	Air flows in from the front side and exhausts from the rear panel	Air flows in from the front side and exhausts from the rear panel
Environment parameters	Long-term operating temperature	• 0-1800 m: -5°C to 45°C • 1800-5000 m: The operating temperature decreases 1°C for every 220 m increase in altitude.	• 0-1800 m: -5°C to 45°C • 1800-5000 m: The operating temperature decreases 1°C for every 220 m increase in altitude.
	Storage temperature	-40°C to +70°C	-40°C to +70°C
	Relative humidity	5% to 95%, noncondensing	5% to 95%, noncondensing
	Operating altitude	0-5000 m	0-5000 m
	Acoustics (LpA) maximum	52 dB(A)	52 dB(A)
	Acoustics (LwA) maximum	6.5B	6.5B
	Surge protection specification (power port)	• Using AC power modules: ±6 kV in differential mode, ±6 kV in common mode • Using DC power modules: ±2 kV in differential mode, ±4 kV in common mode	• Using AC power modules: ±6 kV in differential mode, ±6 kV in common mode • Using DC power modules: ±2 kV in differential mode, ±4 kV in common mode

Item		CloudEngine S6730-H24X6C-V2 CloudEngine S6730-H24X6C-TV2	CloudEngine S6730-H48X6C-V2 CloudEngine S6730-H48X6C-TV2
Reliability	MTBF (year)[2]	62.27	56.87
	MTTR (hour)	2	2
	Availability	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification NOTE For details about certifications, see the section Safety and Regulatory Compliance.	• EMC certification • Safety certification • Manufacturing certification NOTE For details about certifications, see the section Safety and Regulatory Compliance.

NOTE

1: The power consumption under different load conditions is calculated according to the ATIS standard. Additionally.

2: The reliability parameter values are calculated based on the typical configuration of the device. The parameter values vary according to the modules configured by the customer.

Licensing

Licensing

This series switches supports both the traditional feature-based licensing mode and the latest Huawei IDN One Software (N1 mode for short) licensing mode. The N1 mode is ideal for deploying Huawei CloudCampus Solution in the on-premises scenario, as it greatly enhances the customer experiences in purchasing and upgrading software services with simplicity.

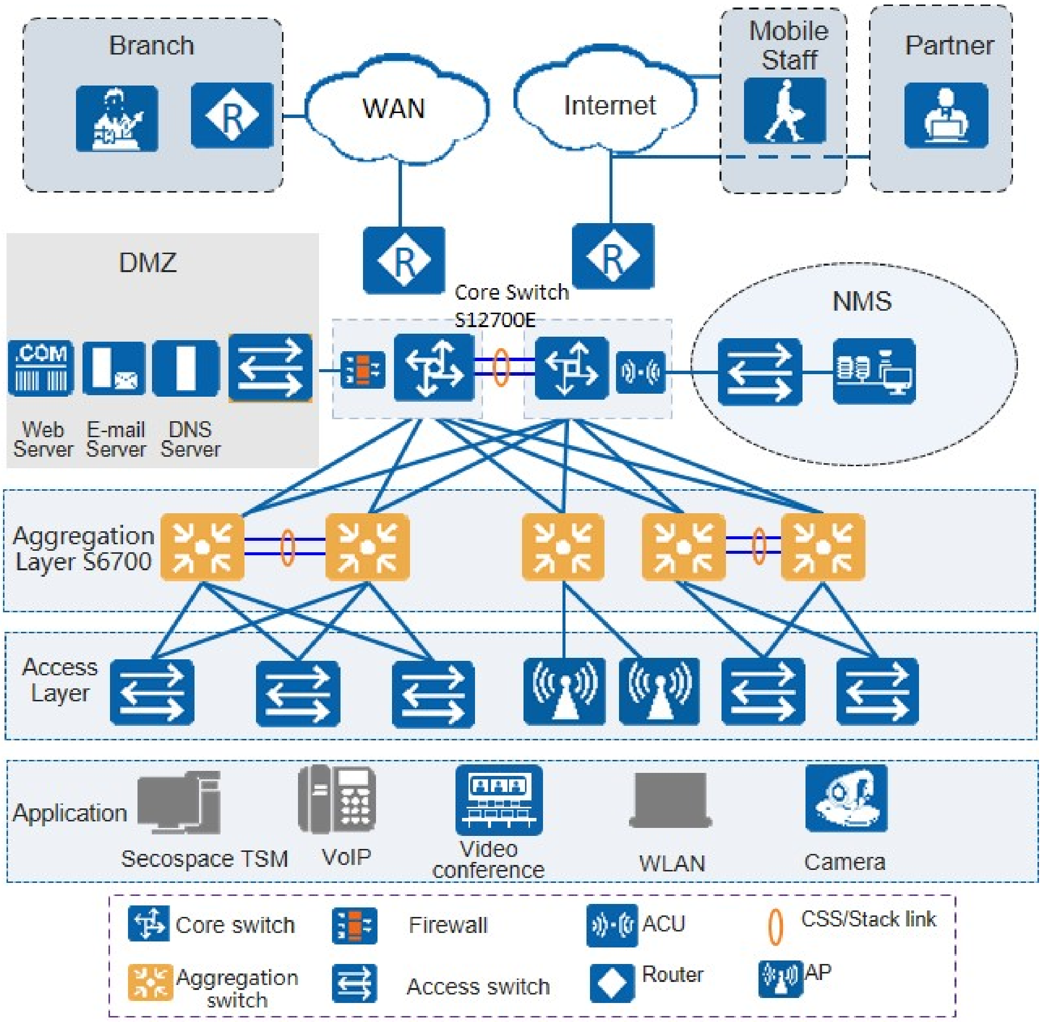
Software Package Features in N1 Mode

Switch Functions	N1 Basic Software	N1 Foundation Software Package	N1 Advanced Software Package
Basic network functions: Layer 2 functions, IPv4, IPv6, SVF, and others Note: For details, see the Service Features	√	√	√
Basic network automation based on the iMaster NCE-Campus: • NE management: Device management, topology management and discovery • User access authentication	×	√	√
Advanced network automation and intelligent O&M: VXLAN, Free Mobility, IPCA, CampusInsight basic functions	×	×	√

Networking and Applications

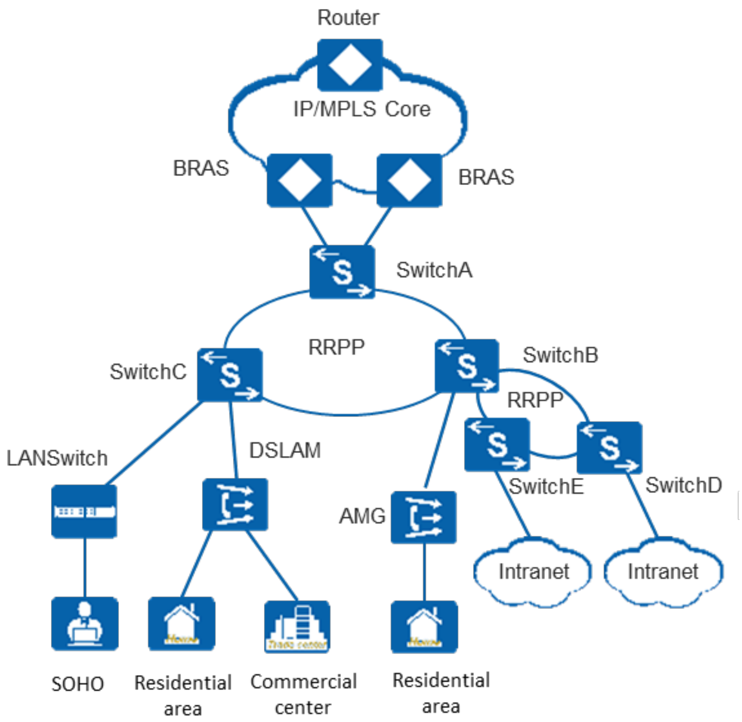
Large-scale Enterprise Campus Network

CloudEngine S6730-H-V2 series switches can be deployed at the aggregation layer of a large-scale enterprise campus network, creating a highly reliable, scalable, and manageable enterprise campus network.



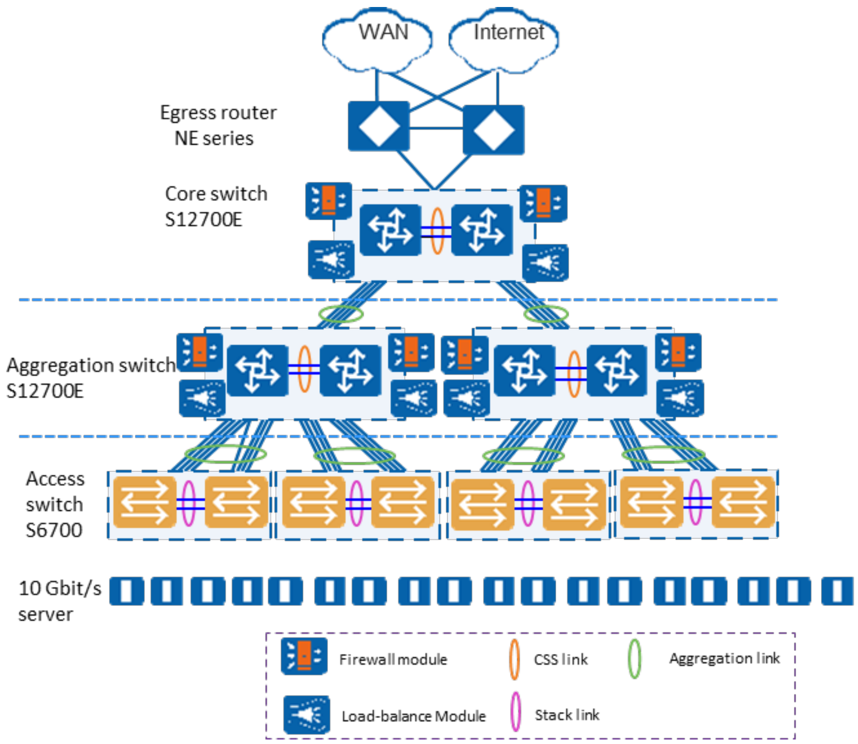
Application on a MAN

CloudEngine S6730-H-V2 series switches can be deployed at the access layer of a MAN(Metropolitan Area Network) to build a high-performance, multi-service, and highly reliable ISP MAN network.



Data Center

CloudEngine S6730-H-V2 switches can be deployed at the access layer build a virtualized, highly reliable, non-blocking, and energy conservative data center network.



Product Accessories

Optical Modules and Fibers

10GE SFP+ ports support optical modules and cables

- GE optical module

- GE-CWDM optical module
- GE-DWDM optical module
- GE copper module
- 10GE SFP+ optical module (OSXD22N00 not supported)
- 10GE-CWDM optical module
- 10GE-DWDM optical module
- 1 m, 3 m, 5 m, and 10 m SFP+ high-speed copper cables
- 3 m and 10 m SFP+ AOC cables
- 0.5 m and 1.5 m SFP+ dedicated stack cables (supported by the last 16 SFP+ ports and used only for zero-configuration stacking)

25GE SFP28 ports support optical modules and cables

- GE eSFP optical module
- GE SFP optical module
- GE-CWDM optical module
- GE-DWDM optical module
- 10GE SFP+ optical module (OSXD22N00 not supported)
- 10GE-CWDM optical module
- 10GE-DWDM optical module
- 25GE SFP28 optical module
- 1 m, 3 m, 5 m, and 10 m SFP+ high-speed cables
- 3 m and 10 m SFP+ AOC cables
- 1 m, 3 m, 5 m SFP28 high-speed cables
- 3 m, 5 m, 7 m, and 10 m SFP28 AOC cables

40GE/100GE QSFP28 ports support optical modules and cables

- QSFP+ optical module
- QSFP28 optical module
- 1 m, 3 m, and 5 m QSFP+ to QSFP+ high-speed copper cables
- 10 m QSFP+ to QSFP+ AOC cable
- 1 m QSFP28 to QSFP28 high-speed copper cable
- 10 m QSFP28 to QSFP28 AOC cable

Stack Cables

The CloudEngine S6730-H-V2 Series switches support service port stacking. The applicable stack cables are as follows:

Port Supporting Stacking	Stack Cable	Rate of a Single Port
10GE ports on the front panel	• 1 m, 3 m, and 5 m SFP+ passive high-speed cables • 10 m SFP+ active high-speed copper cables • 3 m and 10 m AOC cables • 10GE SFP+ optical module and optical fiber • 0.5 m and 1.5 m SFP+ dedicated stack cable	10 Gbit/s
40GE/100GE ports on the front panel	• 1 m QSFP28 high-speed copper cables • 10 m QSFP28 AOC cables • QSFP28 optical module and optical fiber	100Gbit/s

Safety and Regulatory Compliance

The following table lists the safety and regulatory compliance of the CloudEngine S6730-H-V2.

Certification Category	Description
Safety	• IEC 60950-1 and all country deviations • EN 60950-1 • UL 60950-1 • CAN/CSA 22.2 No.60950-1 • GB 4943
Electromagnetic Compatibility (EMC)	• EMI • FCC CFR47 Part 15 Class A • EN55022 Class A • CISPR 22 Class A • EN61000-3-2/IEC-1000-3-2, Power line harmonics • EN61000-4-3/IEC-1000-4-3, Radiated immunity • EN61000-4-2/IEC-1000-4-2, ESD • EN61000-4-4/IEC-1000-4-4, EFT • EN61000-4-5/IEC-1000-4-5, Surge Signal Port • EN61000-4-6/IEC-1000-4-6, Low frequency conducted immunity • EN61000-4-11/IEC-1000-4-11, Voltage dips and sags • EN61000-4-29/IEC61000-4-29, Voltage dips and sags • EMC Directive 89/336/EEC • EMC Directive 2004/108/EC • VCCI V-3 Class A • ICES-003 Class A • AS/NZS CISPR 22 Class A • CNS 13438 Class A • GB9254 Class A

NOTE

- EMC: electromagnetic compatibility
- CISPR: International Special Committee on Radio Interference
- EN: European Standard
- ETSI: European Telecommunications Standards Institute
- CFR: Code of Federal Regulations
- FCC: Federal Communication Commission
- IEC: International Electrotechnical Commission
- AS/NZS: Australian/New Zealand Standard
- VCCI: Voluntary Control Council for Interference
- UL: Underwriters Laboratories
- CSA: Canadian Standards Association
- IEEE: Institute of Electrical and Electronics Engineers

MIB and Standards Compliance

Supported MIBs

Category	MIB
Public MIB	• BRIDGE-MIB • DISMAN-NSLOOKUP-MIB • DISMAN-PING-MIB • DISMAN-TRACEROUTE-MIB • ENTITY-MIB • EtherLike-MIB • IF-MIB • IP-FORWARD-MIB • IPv6-MIB • LAG-MIB • LLDP-EXT-DOT1-MIB • LLDP-EXT-DOT3-MIB • LLDP-MIB • NOTIFICATION-LOG-MIB • NQA-MIB • OSPF-TRAP-MIB • P-BRIDGE-MIB • Q-BRIDGE-MIB • RFC1213-MIB • RIPv2-MIB • RMON2-MIB • RMON-MIB • SAVI-MIB • SNMP-FRAMEWORK-MIB • SNMP-MPD-MIB • SNMP-NOTIFICATION-MIB • SNMP-TARGET-MIB • SNMP-USER-BASED-SM-MIB • SNMPv2-MIB • SNMPv3-MIB • TCP-MIB • UDP-MIB
Huawei-proprietary MIB	• HUAWEI-AAA-MIB • HUAWEI-ACL-MIB • HUAWEI-ALARM-MIB • HUAWEI-ALARM-RELIABILITY-MIB • HUAWEI-BASE-TRAP-MIB • HUAWEI-BRAS-RADIUS-MIB • HUAWEI-BRAS-SRVCFG-EAP-MIB • HUAWEI-BRAS-SRVCFG-STATICUSER-MIB

Category	MIB
	• HUAWEI-CBQOS-MIB • HUAWEI-CDP-COMPLIANCE-MIB • HUAWEI-CONFIG-MAN-MIB • HUAWEI-CPU-MIB • HUAWEI-DAD-TRAP-MIB • HUAWEI-DC-MIB • HUAWEI-DATASYNC-MIB • HUAWEI-DEVICE-MIB • HUAWEI-DHCPR-MIB • HUAWEI-DHCPS-MIB • HUAWEI-DHCP-SNOOPING-MIB • HUAWEI-DIE-MIB • HUAWEI-DNS-MIB • HUAWEI-DLDP-MIB • HUAWEI-ELMI-MIB • HUAWEI-ERPS-MIB • HUAWEI-ERRORDOWN-MIB • HUAWEI-ENERGYMNGT-MIB • HUAWEI-EASY-OPERATION-MIB • HUAWEI-ENTITY-EXTENT-MIB • HUAWEI-ENTITY-TRAP-MIB • HUAWEI-ETHARP-MIB • HUAWEI-ETHOAM-MIB • HUAWEI-FLASH-MAN-MIB • HUAWEI-FWD-RES-TRAP-MIB • HUAWEI-GARP-APP-MIB • HUAWEI-GTSM-MIB • HUAWEI-HGMP-MIB • HUAWEI-HWTACACS-MIB • HUAWEI-IF-EXT-MIB • HUAWEI-INFOCENTER-MIB • HUAWEI-IPPOOL-MIB • HUAWEI-IPV6-MIB • HUAWEI-ISOLATE-MIB • HUAWEI-L2IF-MIB • HUAWEI-L2MAM-MIB • HUAWEI-L2VLAN-MIB • HUAWEI_LDT-MIB • HUAWEI-LLDP-MIB • HUAWEI-MAC-AUTHEN-MIB • HUAWEI-MEMORY-MIB • HUAWEI-MFF-MIB • HUAWEI-MFLP-MIB • HUAWEI-MSTP-MIB • HUAWEI-MULTICAST-MIB

Category	MIB
	• HUAWEI-NAP-MIB • HUAWEI-NTPV3-MIB • HUAWEI-PERFORMANCE-MIB • HUAWEI-PORT-MIB • HUAWEI-PORTAL-MIB • HUAWEI-QINQ-MIB • HUAWEI-RIPv2-EXT-MIB • HUAWEI-RM-EXT-MIB • HUAWEI-RRPP-MIB • HUAWEI-SECURITY-MIB • HUAWEI-SEP-MIB • HUAWEI-SNMP-EXT-MIB • HUAWEI-SSH-MIB • HUAWEI-STACK-MIB • HUAWEI-SWITCH-L2MAM-EXT-MIB • HUAWEI-SWITCH-SRV-TRAP-MIB • HUAWEI-SYS-MAN-MIB • HUAWEI-TCP-MIB • HUAWEI-TFTPC-MIB • HUAWEI-TRNG-MIB • HUAWEI-XQOS-MIB

NOTE

For more information about MIBs supported by the CloudEngine S6730-H-V2 series, visit:
https://support.huawei.com/enterprise/en/switches/s6700-pid-6691593?category=reference-guides

Standards Compliance

The following table lists the standards that the CloudEngine S6730-H-V2 series complies with.

Standard Organization	Standard or Protocol
IETF	• RFC 768 User Datagram Protocol (UDP) • RFC 792 Internet Control Message Protocol (ICMP) • RFC 793 Transmission Control Protocol (TCP) • RFC 826 Ethernet Address Resolution Protocol (ARP) • RFC 854 Telnet Protocol Specification • RFC 951 Bootstrap Protocol (BOOTP) • RFC 959 File Transfer Protocol (FTP) • RFC 1058 Routing Information Protocol (RIP) • RFC 1112 Host extensions for IP multicasting • RFC 1157 A Simple Network Management Protocol (SNMP) • RFC 1256 ICMP Router Discovery • RFC 1305 Network Time Protocol Version 3 (NTP) • RFC 5905 Network Time Protocol Version 4 (NTP) • RFC 1349 Internet Protocol (IP) • RFC 1493 Definitions of Managed Objects for Bridges

Standard Organization	Standard or Protocol
	• RFC 1542 Clarifications and Extensions for the Bootstrap Protocol • RFC 1643 Ethernet Interface MIB • RFC 1757 Remote Network Monitoring (RMON) • RFC 1901 Introduction to Community-based SNMPv2 • RFC 1902-1907 SNMP v2 • RFC 2574 SNMP v3 • RFC 1981 Path MTU Discovery for IP version 6 • RFC 2131 Dynamic Host Configuration Protocol (DHCP) • RFC 2328 OSPF Version 2 • RFC 2453 RIP Version 2 • RFC 2460 Internet Protocol, Version 6 Specification (IPv6) • RFC 2461 Neighbor Discovery for IP Version 6 (IPv6) • RFC 2462 IPv6 Stateless Address Auto configuration • RFC 2463 Internet Control Message Protocol for IPv6 (ICMPv6) • RFC 2474 Differentiated Services Field (DS Field) • RFC 2740 OSPF for IPv6 (OSPFv3) • RFC 2863 The Interfaces Group MIB • RFC 2597 Assured Forwarding PHB Group • RFC 2598 An Expedited Forwarding PHB • RFC 2571 SNMP Management Frameworks • RFC 2865 Remote Authentication Dial In User Service (RADIUS) • RFC 3046 DHCP Option82/Relay • RFC 3376 Internet Group Management Protocol, Version 3 (IGMPv3) • RFC 3513 IP Version 6 Addressing Architecture • RFC 3579 RADIUS Support For EAP • RFC 4271 A Border Gateway Protocol 4 (BGP-4) • RFC 4760 Multiprotocol Extensions for BGP-4 • draft-grant-tacacs-02 TACACS+ • RFC 6241 Network Configuration Protocol (NETCONF) • RFC 6020 YANG - A Data Modeling Language for the Network Configuration Protocol (NETCONF) • RFC 6242 - Using the NETCONF Protocol over Secure Shell (SSH) • RFC 6244 - An Architecture for Network Management Using NETCONF and YANG
IEEE	• IEEE 802.1D Media Access Control (MAC) Bridges • IEEE 802.1p Traffic Class Expediting and Dynamic Multicast Filtering • IEEE 802.1Q Virtual Bridged Local Area Networks • IEEE 802.1ad Provider Bridges • IEEE 802.2 Logical Link Control • IEEE Std 802.3 CSMA/CD • IEEE Std 802.3ab 1000BASE-T specification • IEEE Std 802.3ad Aggregation of Multiple Link Segments • IEEE Std 802.3ae 10GE WEN/LAN Standard • IEEE Std 802.3x Full Duplex and flow control • IEEE Std 802.3z Gigabit Ethernet Standard

Standard Organization	Standard or Protocol
	• IEEE Std 802.3u Fast Ethernet Standard • IEEE 802.1ax/IEEE802.3ad Link Aggregation • IEEE 802.3ah Ethernet in the First Mile. • IEEE 802.1ag Connectivity Fault Management • IEEE 802.1ab Link Layer Discovery Protocol • IEEE 802.1D Spanning Tree Protocol • IEEE 802.1w Rapid Spanning Tree Protocol • IEEE 802.1s Multiple Spanning Tree Protocol • IEEE 802.1x Port based network access control protocol • IEEE 802.3az Automatic power adjustment on Ethernet interfaces • IEEE 802.3ba 40Gbit/s and 100Gbit/s Ethernet Standard • IEEE 802.3ac VLAN tagging • IEEE 802.3ax Link Aggregation Task Force
ITU	• ITU SG13 Y.17ethoam • ITU SG13 QoS control Ethernet-Based IP Access • ITU-T Y.1731 ETH OAM performance monitor
ISO	• ISO 10589 IS-IS Routing Protocol
MEF	• MEF 2 Requirements and Framework for Ethernet Service Protection • MEF 9 Abstract Test Suite for Ethernet Services at the UNI • MEF 10.2 Ethernet Services Attributes Phase 2 • MEF 11 UNI Requirements and Framework • MEF 13 UNI Type 1 Implementation Agreement • MEF 15 Requirements for Management of Metro Ethernet Phase 1 Network Elements • MEF 17 Service OAM Framework and Requirements • MEF 20 UNI Type 2 Implementation Agreement • MEF 23 Class of Service Phase 1 Implementation Agreement • Xmodem XMODEM/YMODEM Protocol Reference

NOTE

The listed standards and protocols are fully or partially supported by Huawei switches. For details, visit http://e.huawei.com/en or contact your local Huawei sales office.

Ordering Information

The following table lists ordering information of the CloudEngine S6730-H-V2 series.

Model	Product Description
CloudEngine S6730-H48X6CZ-V2	CloudEngine S6730-H48X6C-V2 bundle (48*10GE SFP+ ports, 6*100GE QSFP28 ports, with license, 1*expansion slot, without power module)
CloudEngine S6730-H48X6CZ-V2	CloudEngine S6730-H48X6C-V2 (48*10GE SFP+ ports, 6*40GE QSFP ports, optional license for upgrade to 6*100GE QSFP28 ports, 1*expansion slot, without power module)
CloudEngine S6730-H48X6CZ-TV2	CloudEngine S6730-H48X6C-TV2 bundle (48*10GE SFP+ ports, 6*100GE QSFP28 ports, with license, 1*expansion slot, HTM, without power module)

Model	Product Description
CloudEngine S6730-H48X6CZ-TV2	CloudEngine S6730-H48X6C-TV2 (48*10GE SFP+ ports, 6*40GE QSFP ports, optional license for upgrade to 6*100GE QSFP28 ports, 1*expansion slot, HTM, without power module)
CloudEngine S6730-H28X6CZ-V2	CloudEngine S6730-H28X6C-V2 bundle (28*10GE SFP+ ports, 6*100GE QSFP28 ports, with license, 1*expansion slot, without power module)
CloudEngine S6730-H28X6CZ-V2	CloudEngine S6730-H28X6C-V2 (28*10GE SFP+ ports, 6*40GE QSFP28 ports, optional license for upgrade to 6*100GE QSFP28 ports, 1*expansion slot, without power module)
CloudEngine S6730-H28X6CZ-TV2	CloudEngine S6730-H28X6C-TV2 bundle (28*10GE SFP+ ports, 6*100GE QSFP28 ports, with license, 1*expansion slot, HTM, without power module)
CloudEngine S6730-H28X6CZ-TV2	CloudEngine S6730-H28X6C-TV2 (28*10GE SFP+ ports, 6*40GE QSFP ports, optional license for upgrade to 6*100GE QSFP28 ports, 1*expansion slot, HTM, without power module)
CloudEngine S6730-H48X6C-V2	CloudEngine S6730-H48X6C-V2 (48*10GE SFP+ ports, 6*40GE QSFP28 ports, optional license for upgrade to 6*100GE QSFP28, without power module)
CloudEngine S6730-H24X6C-V2	CloudEngine S6730-H24X6C -V2(24*10GE SFP+ ports, 6*40GE QSFP28 ports, optional license for upgrade to 6*100GE QSFP28, without power module)
CloudEngine S6730-H48X6C-TV2	CloudEngine S6730-H48X6C-TV2 (48*10GE SFP+ ports, 6*40GE QSFP28 ports, optional license for upgrade to 6*100GE QSFP28,HTM,without power module)
CloudEngine S6730-H24X6C-TV2	CloudEngine S6730-H24X6C-TV2(24*10GE SFP+ ports, 6*40GE QSFP28 ports, optional license for upgrade to 6*100GE QSFP28,HTM,without power module)
PAC600S12-CB	600W AC power module (for S6730-H48X6C-V2/TV2 & S6730-H24X6C-V2/TV2 series models)
PAC600S12-EB	600W AC power module (for S6730-H48X6C-V2/TV2 & S6730-H24X6C-V2/TV2 series models)
PAC600S12-DB	600W AC power module (for S6730-H48X6C-V2/TV2 & S6730-H24X6C-V2/TV2 series models)
PDC1000S12-DB	1000W DC power module (for S6730-H48X6C-V2/TV2 & S6730-H24X6C-V2/TV2 series models)
PAC600S12-PB	600W AC power module (for S6730-H48X6CZ-V2/TV2 & S6730-H28X6CZ-V2/TV2 series models)
PAC1K2S12-PB	1200W AC power module (for S6730-H48X6CZ-V2/TV2 & S6730-H28X6CZ-V2/TV2 series models)
PDC1K2S12-CE	1200W DC power module (for S6730-H48X6CZ-V2/TV2 & S6730-H28X6CZ-V2/TV2 series models)
FAN-031A-B	Fan Module

License	Product Description
L-100GEUPG-S67H	S67XX-H Series,40GE to 100GE Electronic RTU License,Per Device
L-MLIC-S67H	S67XX-H Series Basic SW,Per Device
N1-S67H-M-Lic	S67XX-H Series Basic SW,Per Device
N1-S67H-M-SnS1Y	S67XX-H Series Basic SW,SnS,Per Device,1Year
N1-S67H-F-Lic	N1-CloudCampus,Foundation,S67XX-H Series,Per Device
N1-S67H-F-SnS	N1-CloudCampus,Foundation,S67XX-H Series,SnS,Per Device
N1-S67H-A-Lic	N1-CloudCampus,Advanced,S67XX-H Series,Per Device
N1-S67H-A-SnS	N1-CloudCampus,Advanced,S67XX-H Series,SnS,Per Device

License	Product Description
N1-S67H-FToA-Lic	N1-Upgrade-Foundation to Advanced,S67XX-H,Per Device
N1-S67H-FToA-SnS	N1-Upgrade-Foundation to Advanced,S67XX-H,SnS,Per Device

More Information

For more information about the Huawei Campus Switches, visit http://e.huawei.com or contact us in the following ways:

- Global service hotline: http://e.huawei.com/en/service-hotline
- Logging in to the Huawei Enterprise technical support website: http://support.huawei.com/enterprise/
- Sending an email to the customer service mailbox: support_e@huawei. com

Huawei Technologies Co., Ltd.

Address:Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China

Website:e.huawei.com

Huawei CloudEngine S5735-S-V2 Series Switches Datasheet

Huawei CloudEngine S5735-S-V2 series are standard gigabit Ethernet switches that provide GE electrical & 2.5GE optical downlink ports, 10GE uplink ports and one extended slot.

Introduction

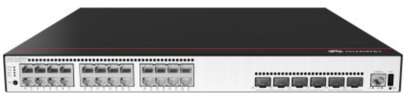
CloudEngine S5735-S-V2 series switches are developed based on next-generation high-performing hardware and software platform. CloudEngine S5735-S-V2 switches support simplified operations and maintenance (O&M), and flexible Ethernet networking. It also provides enhanced Layer 3 features and mature IPv6 features. CloudEngine S5735-S-V2 switches can be used in various scenarios. For example, it can be used as an access or aggregation switch on a campus network or as an access switch for Metropolitan Area Network. Huawei S5735-S-V2 series switches provide high-speed forwarding at line-rate on all ports in both Layer2 and Layer3 modes.

Product Overview

Models and Appearances

The following models are available in the CloudEngine S5735-S-V2 series.

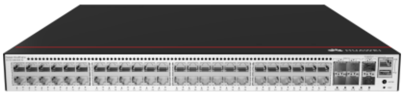
Models and appearances of the CloudEngine S5735-S-V2 series

Models and Appearances	Description
 CloudEngine S5735-S24T4XE-V2	• 24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • 1+1 power supply backup • Forwarding performance: 132 Mpps • Switching capacity*: 176 Gbps/520 Gbps
 CloudEngine S5735-S24T4XEZ-V2	• 24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • One extended slot • 1+1 power supply backup • Forwarding performance: 132 Mpps • Switching capacity*: 176 Gbps/520 Gbps
 CloudEngine S5735-S24P4XE-V2	• 24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • 3 power supplies, N+1 power supply backup • PoE+ • Forwarding performance: 132 Mpps • Switching capacity*: 176 Gbps/520 Gbps

15 - S5735-S24P4XE-V2 (24*10/100/1000BASE-T ports(PoE+), 4*10GE SFP+ ports, 2*12GE stack ports, without power module, Specially protected models)

Models and Appearances	Description
 CloudEngine S5735-S24P4XEZ-V2	• 24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • One extended slot • 3 power supplies, N+1 power supply backup • PoE+ • Forwarding performance: 132 Mpps • Switching capacity*: 176 Gbps/520 Gbps
 CloudEngine S5735-S24U4XE-V2	• 24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • 3 power supplies, N+1 power supply backup • PoE++(90W) • Forwarding performance: 132 Mpps • Switching capacity*: 176 Gbps/520 Gbps
 CloudEngine S5735-S24T8J4XE-XA-V2	• 24 x 10/100/1000Base-T ports, 8 x 2.5 GE SFP+ ports(or 2*10GE SFP+ ports), 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • Built-in dual AC power, 1+1 power supply backup • Forwarding performance: 162 Mpps • Switching capacity*: 216 Gbps/520 Gbps
 CloudEngine S5735-S24T8J4XEZ-V2	• 24 x 10/100/1000Base-T ports, 8 x 2.5 GE SFP+ ports(or 2*10GE SFP+ ports), 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • One extended slot • 1+1 power supply backup • Forwarding performance: 162 Mpps • Switching capacity*: 216 Gbps/520 Gbps
 CloudEngine S5735-S24P8J4XEZ-V2	• 24 x 10/100/1000Base-T ports, 8 x 2.5 GE SFP+ ports(or 2*10GE SFP+ ports), 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • One extended slot • 3 power supplies, N+1 power supply backup • Forwarding performance: 162 Mpps • Switching capacity*: 216 Gbps/520 Gbps
 CloudEngine S5735-S48T4XE-V2	• 48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • 1+1 power supply backup • Forwarding performance: 168 Mpps • Switching capacity*: 224 Gbps/520 Gbps
 CloudEngine S5735-S48T4XE-XA-V2	• 48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • Built-in dual AC power, 1+1 power supply backup • Forwarding performance: 168 Mpps • Switching capacity*: 224 Gbps/520 Gbps
 CloudEngine S5735-S48T4XEZ-V2	• 48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • One extended slot • 1+1 power supply backup • Forwarding performance: 168 Mpps • Switching capacity*: 224 Gbps/520 Gbps

19 - S5735-S48P4XE-V2 (48*10/100/1000BASE-T ports(PoE+), 4*10GE SFP+ ports, 2*12GE stack ports, without power module, Specially protected models)

Models and Appearances	Description
 CloudEngine S5735-S48P4XE-V2	• 48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • 3 power supplies, N+1 power supply backup • PoE+ • Forwarding performance: 168 Mpps • Switching capacity*: 224 Gbps/520 Gbps
 CloudEngine S5735-S48P4XEZ-V2	• 48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • One extended slot • 3 power supplies, N+1 power supply backup • PoE+ • Forwarding performance: 168 Mpps • Switching capacity*: 224 Gbps/520 Gbps
 CloudEngine S5735-S48U4XE-V2	• 48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports • 3 power supplies, N+1 power supply backup • PoE++(90W) • Forwarding performance: 168 Mpps • Switching capacity*: 224 Gbps/520 Gbps

*Note：The value before the slash (/) refers to the device's switching capability, while the value after the slash (/) means the system's switching capability.

Power Supply

Technical specifications of the power supplies applicable to the CloudEngine S5735-S-V2 series

Power Module	Technical Specifications	Applied Switch Model
 PAC180S12-CN	• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm • Weight: 0.8 kg • Rated input voltage range: − 100 V AC to 240 V AC, 50/60 Hz − 240 V DC • Maximum input voltage range: − 90 V AC to 290 V AC, 45 Hz to 66 Hz − 190 V DC to 290 V DC • Maximum input current: 3 A • Rated output current: 15 A • Rated output voltage: 12 V • Rated output power: 180 W • Hot swap: Supported	• CloudEngine S5735-S24T4XE-V2 • CloudEngine S5735-S24T4XEZ-V2 • CloudEngine S5735-S24T8J4XEZ-V2 • CloudEngine S5735-S48T4XE-V2 • CloudEngine S5735-S48T4XEZ-V2
 PDC1K2S12-CE	• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm • Weight: 1.5 kg • Rated input voltage range: − +48 V DC − -48V DC to -60 V DC • Maximum input voltage range:	• CloudEngine S5735-S24T4XE-V2 • CloudEngine S5735-S48T4XE-V2

Power Module	Technical Specifications	Applied Switch Model
	– +40 V DC to +57 V DC – -38.4V DC to -72 V DC • Maximum Input current: 38 A • Rated output voltage: 12 V • Rated output power: 1200W • Hot swap: Supported	
 PAC80S12-CN	• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm (1.56 in. x 2.6 in. x 8.46 in.) • Weight: 1.2 kg (2.64 lb) • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 66 Hz • Maximum input current: 2 A • Rated output current: 6.67 A • Rated output voltage: 12 V • Rated output power: 80 W • Hot swap: Supported	• CloudEngine S5735-S24T4XE-V2 • CloudEngine S5735-S24T4XEZ-V2 • CloudEngine S5735-S24T8J4XEZ-V2 • CloudEngine S5735-S48T4XE-V2 • CloudEngine S5735-S48T4XEZ-V2
 PAC600S12-PB	• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm (1.56 in. x 2.6 in. x 8.46 in.) • Weight: 1 kg (2.2 lb) • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 66 Hz – 190 V DC to 290 V DC • Maximum input current: 100 V AC to 240 V AC: 8 A 240 V DC: 4 A • Rated output current: 50 A • Rated output voltage: 12 V • Rated output power: 600 W • Hot swap: Supported	• CloudEngine S5735-S24T4XE-V2 • CloudEngine S5735-S48T4XE-V2
 PDC240S12-CN	• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm (1.56 in. x 2.6 in. x 8.46 in.) • Weight: 1.5 kg • Rated input voltage range: – +48 V DC – -48 V~-60 V DC • Maximum input voltage range: – +40 V~+57 V DC – -38.4 V~-72 V DC • Maximum input current: 7 A	• CloudEngine S5735-S24T4XE-V2 • CloudEngine S5735-S24T4XEZ-V2 • CloudEngine S5735-S24T8J4XEZ-V2 • CloudEngine S5735-S48T4XE-V2 • CloudEngine S5735-S48T4XEZ-V2

Power Module	Technical Specifications	Applied Switch Model
	• Rated output voltage: 20 V • Rated output power: 240 W • Hot swap: Supported	
 PAC600S56-EB	• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm • Weight: 0.9 kg • Rated input voltage range: – 100 V AC to 130 V AC; 50/60 Hz – 200 V AC to 240 V AC, 50/60 Hz – 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 66 Hz – 190 V DC to 290 V DC • Maximum input current: – 100 V AC to 130 V AC: 8 A – 200 V AC to 240 V AC: 8 A – 240 V DC: 4 A • Rated output current: – 53.5 V: 11.21 A – 55.5 V: 10.81 A • Rated output voltage: 53.5 V or 55.5 V • Rated output power: – 100 V AC to 130 V AC input: 300W – 200 V AC to 240 V AC & 240 V DC input: 600W • Hot swap: Supported	• CloudEngine S5735-S24P4XE-V2 • CloudEngine S5735-S24P4XEZ-V2 • CloudEngine S5735-S24U4XE-V2 • CloudEngine S5735-S24P8J4XEZ-V2 • CloudEngine S5735-S48P4XE-V2 • CloudEngine S5735-S48P4XEZ-V2 • CloudEngine S5735-S48U4XE-V2
 PAC1000S56-EB 17 & 21 - 1000W AC&240V DC Power Module(66mm Width Case,Back to Front, Power panel side exhaust)	• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm • Weight: 1.1 kg (2.43 lb) • Rated input voltage range: – 100 V AC to 130 V AC, 50/60 Hz – 200 V AC to 240 V AC, 50/60 Hz – 125V DC to 240 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45 Hz to 66 Hz – 100 V DC to 290 V DC • Input current: – 100 V AC to 130 V AC: 12 A – 200 V AC to 240 V AC: 8 A – 240 V DC: 8 A • Rated output current: – 18.69A @53.5V 18.02A @55.5V • Maximum output power: – 400W (100 V DC to 190 V DC)	• CloudEngine S5735-S24P4XE-V2 • CloudEngine S5735-S24P4XEZ-V2 • CloudEngine S5735-S24U4XE-V2 • CloudEngine S5735-S24P8J4XEZ-V2 • CloudEngine S5735-S48P4XE-V2 • CloudEngine S5735-S48P4XEZ-V2 • CloudEngine S5735-S48U4XE-V2

Power Module	Technical Specifications	Applied Switch Model
	– 900 W (100 V AC to 130 V AC input) – 1000 W (200 V AC to 240 V AC input and 190 v DC to 240 V DC input) • Hot swap: Supported	
 PDC1000S56-EB	• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm • Weight: 1 kg • Rated input voltage range: -48 V DC to -60 V DC • Maximum input voltage range: -38.4 V DC to -72 V DC • Maximum input current: 30 A • Rated output voltage: 53.5 V or 55.5 V • Rated output current: – 53.5 V: 18.69 A – 55.5 V: 18.02 A • Rated output power: 1000 W • Hot swap: Supported	• CloudEngine S5735-S24P4XE-V2 • CloudEngine S5735-S24P4XEZ-V2 • CloudEngine S5735-S24U4XE-V2 • CloudEngine S5735-S24P8J4XEZ-V2 • CloudEngine S5735-S48P4XE-V2 • CloudEngine S5735-S48P4XEZ-V2 • CloudEngine S5735-S48U4XE-V2

The following table lists its power supply configurations.

Power supply configurations of CloudEngine S5735-S-V2

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
CloudEngine S5735-S24P4XE-V2	1000 W AC (220 V)	–	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V)	–	–	774 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W DC	–	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	–	–	532 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17
	600 W AC (110 V)	–	–	247 W	• 802.3af (15.4 W per port): 16 • 802.3at (30 W per port): 8
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V)	600 W AC (220 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
	1000 W DC				
	1000 W DC	600 W AC (110 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V)	600 W AC (110 V)	-	774 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (110 V)	600 W AC (110 V)	–	532 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	600 W AC (110 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W DC	600 W AC (110 V)	600 W AC (110 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	774 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	817 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
CloudEngine S5735-S24U4XE-V2	1000 W AC (220 V)	–	–	846 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	1000 W AC (110 V)	–	–	756 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 12 • 802.3bt (90 W per port): 8
	1000 W DC	–	–	846 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
					• 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	600 W AC (220 V)	–	–	513 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17 • 802.3bt (60 W per port): 8 • 802.3bt (90 W per port): 5
	600 W AC (110 V)	–	–	228 W	• 802.3af (15.4 W per port): 14 • 802.3at (30 W per port): 7 • 802.3bt (60 W per port): 3 • 802.3bt (90 W per port): 2
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	1746 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 19
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	1566 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 17
	600 W AC (220 V)	600 W AC (220 V)	–	1083 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 18 • 802.3bt (90 W per port): 12
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	–	1386 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 23 • 802.3bt (90 W per port): 15
	1000 W DC	600 W AC (110 V)	–	846 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	1000 W AC (110 V)	600 W AC (110 V)	-	756 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 12 • 802.3bt (90 W per port): 8
	600 W AC (110 V)	600 W AC (110 V)	–	513 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17 • 802.3bt (60 W per port): 8 • 802.3bt (90 W per port): 5
	1000 W AC (220 V)	1000 W AC (220 V)	1000 W AC (220 V)	2268 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
	1000 W DC	1000 W DC	1000 W DC		• 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	1653 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 18
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	2268 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 24
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	1926 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 21
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	2268 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 24
	1000 W DC	1000 W DC	600 W AC (110 V)	1746 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 19
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	600 W AC (110 V)	1566 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 24 • 802.3bt (90 W per port): 17
	1000 W DC	600 W AC (110 V)	600 W AC (110 V)	846 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	1000 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	756 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 12 • 802.3bt (90 W per port): 8
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	798 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24 • 802.3bt (60 W per port): 13 • 802.3bt (90 W per port): 8
CloudEngine S5735-	1000 W AC (220 V)	–	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
S48P4XE-V2	1000 W AC (110 V)	–	–	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25
	1000 W DC	–	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28
	600 W AC (220 V)	–	–	513 W	• 802.3af (15.4 W per port): 33 • 802.3at (30 W per port): 17
	600 W AC (110 V)	–	–	228 W	• 802.3af (15.4 W per port): 14 • 802.3at (30 W per port): 7
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	1566 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	600 W AC (220 V)	600 W AC (220 V)	–	1083 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 36
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	–	1386 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 46
	1000 W DC	600 W AC (110 V)	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28
	1000 W AC (110 V)	600 W AC (110 V)	-	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25
	600 W AC (110 V)	600 W AC (110 V)	–	513 W	• 802.3af (15.4 W per port): 33 • 802.3at (30 W per port): 17
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	1653 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W DC	1000 W DC	600 W AC (110	1680 W	• 802.3af (15.4 W per port): 48

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
			V)		• 802.3at (30 W per port): 48
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	600 W AC (110 V)	1566 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W DC	600 W AC (110 V)	600 W AC (110 V)	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28
	1000 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	798 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 26
CloudEngine S5735-S48U4XE-V2	1000 W AC (220 V)	–	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	1000 W AC (110 V)	–	–	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25 • 802.3bt (60 W per port): 12 • 802.3bt (90 W per port): 8
	1000 W DC	–	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	600 W AC (220 V)	–	–	513 W	• 802.3af (15.4 W per port): 33 • 802.3at (30 W per port): 17 • 802.3bt (60 W per port): 8 • 802.3bt (90 W per port): 5
	600 W AC (110 V)	–	–	228 W	• 802.3af (15.4 W per port): 14 • 802.3at (30 W per port): 7 • 802.3bt (60 W per port): 3 • 802.3bt (90 W per port): 2
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	1746 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 29 • 802.3bt (90 W per port): 19
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	1566 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 26 • 802.3bt (90 W per port): 17
	600 W AC (220 V)	600 W AC (220 V)	–	1083 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 36

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
					• 802.3bt (60 W per port): 18 • 802.3bt (90 W per port): 12
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	–	1386 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 46 • 802.3bt (60 W per port): 23 • 802.3bt (90 W per port): 15
	1000 W DC	600 W AC (110 V)	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	1000 W AC (110 V)	600 W AC (110 V)	-	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25 • 802.3bt (60 W per port): 12 • 802.3bt (90 W per port): 8
	600 W AC (110 V)	600 W AC (110 V)	–	513 W	• 802.3af (15.4 W per port): 33 • 802.3at (30 W per port): 17 • 802.3bt (60 W per port): 8 • 802.3bt (90 W per port): 5
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	2880 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 48 • 802.3bt (90 W per port): 32
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	1653 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 27 • 802.3bt (90 W per port): 18
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	2286 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 38 • 802.3bt (90 W per port): 25
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	1926 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 32 • 802.3bt (90 W per port): 21
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	2376 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 39 • 802.3bt (90 W per port): 26
	1000 W DC	1000 W DC	600 W AC (110 V)	1746 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
					• 802.3bt (60 W per port): 29 • 802.3bt (90 W per port): 19
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	600 W AC (110 V)	1566 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48 • 802.3bt (60 W per port): 26 • 802.3bt (90 W per port): 17
	1000 W DC	600 W AC (110 V)	600 W AC (110 V)	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28 • 802.3bt (60 W per port): 14 • 802.3bt (90 W per port): 9
	1000 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25 • 802.3bt (60 W per port): 12 • 802.3bt (90 W per port): 8
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	798 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 26 • 802.3bt (60 W per port): 13 • 802.3bt (90 W per port): 8
CloudEngine S5735-S24P4XEZ-V2	1000 W AC (220 V)	–	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V)	–	–	774 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W DC	–	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	–	–	532 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17
	600 W AC (110 V)	–	–	247 W	• 802.3af (15.4 W per port): 16 • 802.3at (30 W per port): 8
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (110	600 W AC (110	–	532 W	• 802.3af (15.4 W per port): 24

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
	V)	V)			• 802.3at (30 W per port): 17
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	817 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
CloudEngine S5735-S24P8J4XEZ-V2	1000 W AC (220 V)	–	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V)	–	–	774 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W DC	–	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	–	–	532 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17
	600 W AC (110 V)	–	–	247 W	• 802.3af (15.4 W per port): 16 • 802.3at (30 W per port): 8
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	–	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (110 V)	600 W AC (110 V)	–	532 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 17

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	840 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	817 W	• 802.3af (15.4 W per port): 24 • 802.3at (30 W per port): 24
CloudEngine S5735-S48P4XEZ-V2	1000 W AC (220 V)	–	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28
	1000 W AC (110 V)	–	–	756 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 25
	1000 W DC	–	–	846 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 28
	600 W AC (220 V)	–	–	513 W	• 802.3af (15.4 W per port): 33 • 802.3at (30 W per port): 17
	600 W AC (110 V)	–	–	228 W	• 802.3af (15.4 W per port): 14 • 802.3at (30 W per port): 7
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	–	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	–	1566 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	600 W AC (220 V)	600 W AC (220 V)	–	1083 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 36
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	–	1386 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 46
	600 W AC (110 V)	600 W AC (110 V)	–	513 W	• 802.3af (15.4 W per port): 33 • 802.3at (30 W per port): 17
	1000 W AC	1000 W AC	1000 W AC	1680 W	• 802.3af (15.4 W per port): 48

Module	Power Module 1	Power Module 2	Power Module 3	Available PoE Power	Maximum Number of Ports (Fully Loaded)
	(220 V) 1000 W DC	(220 V) 1000 W DC	(220 V) 1000 W DC		• 802.3at (30 W per port): 48
	600 W AC (220 V)	600 W AC (220 V)	600 W AC (220 V)	1653 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (220 V) 1000 W DC	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (220 V) 1000 W DC	600 W AC (220 V)	600 W AC (220 V)	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V) 1000 W DC	1000 W AC (110 V)	1680 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 48
	600 W AC (110 V)	600 W AC (110 V)	600 W AC (110 V)	798 W	• 802.3af (15.4 W per port): 48 • 802.3at (30 W per port): 26

Product Features and Highlights

Powerful Service Processing Capability

- CloudEngine S5735-S-V2 supports a broad set of Layer 2/Layer 3 multicast protocols, such as PIM SM, PIM DM, PIM SSM, and IGMP snooping. This capability is ideal for high-definition video backhaul and video conferencing access.
- CloudEngine S5735-S-V2 provides multiple Layer 3 features including OSPF, IS-IS, BGP, and VRRP, meeting enterprises' access and aggregation service needs and enabling a variety of voice, video, and data applications.

Multiple Security Control Mechanisms

- CloudEngine S5735-S-V2 supports MAC address authentication, 802.1X authentication, and implements dynamic delivery of policies (VLAN, QoS, and ACL) to users.
- CloudEngine S5735-S-V2 provides a series of mechanisms to defend against DoS attacks and user-targeted attacks. DoS attacks are targeted at switches and include SYN flood, Land, Smurf, and ICMP flood attacks. User-targeted attacks include bogus DHCP server attacks, IP/MAC address spoofing, DHCP request flood, and changing of the DHCP CHADDR value.
- CloudEngine S5735-S-V2 sets up and maintains a DHCP snooping binding table, and discards the packets that do not match the table entries. The DHCP snooping trusted port feature ensures that users connect only to the authorized DHCP server.
- CloudEngine S5735-S-V2 supports strict ARP learning, which protects a network against ARP spoofing attacks to ensure that users can connect to the Internet normally.

Multiple Reliability Mechanisms

- CloudEngine S5735-S-V2 is equipped with two pluggable power modules that work in 1+1 redundancy backup mode. Mixed installation of AC and DC power modules is supported, allowing for flexible configuration of AC or DC power modules according to service requirements.
- In addition to supporting traditional Spanning Tree Protocol (STP), Rapid Spanning Tree Protocol (RSTP), and Multiple Spanning Tree Protocol (MSTP), CloudEngine S5735-S-V2 is also designed with the industry's latest Ethernet Ring Protection Switching (ERPS) technology. This protocol is reliable, easy to maintain, and implements fast protection switching within 50 ms. ERPS is defined in ITU-T G.8032, and it implements millisecond-level protection switching based on traditional Ethernet MAC and bridging functions.

- CloudEngine S5735-S-V2 supports Smart Link, which implements backup of uplinks. One CloudEngine S5735-S-V2 switch can connect to multiple aggregation switches through multiple links, significantly improving reliability of access devices.

Easy Network deployment

- CloudEngine S5735-S-V2 supports Huawei Easy Operation, a solution that provides zero-touch deployment, replacement of faulty devices without additional configuration, USB-based deployment, batch device configuration, and batch remote upgrade. The capabilities facilitate device deployment, upgrade, service provisioning, and other management and maintenance operations, and also greatly reduce O&M costs. CloudEngine S5735-S-V2 can be managed using SNMP v1/v2c/v3, CLI, web-based network management system, or SSH v2.0. Additionally, it supports RMON, multiple log hosts, port traffic statistics collection, and network quality analysis, which facilitate network optimization and reconstruction.

Mature IPv6 Technologies

- CloudEngine S5735-S-V2 uses the mature, stable VRP platform and supports IPv4/IPv6 dual stack, IPv6 RIPng.
- CloudEngine S5735-S-V2 can be deployed on a pure IPv4 network, a pure IPv6 network, or a shared IPv4/IPv6 network, helping achieve IPv4-to-IPv6 transition.

Intelligent Stack (iStack)

- CloudEngine S5735-S-V2 supports intelligent stack (iStack). This technology combines multiple switches into a logical switch. Member switches in a stack implement redundancy backup to improve device reliability and use inter-device link aggregation to improve link reliability.
- iStack provides high network scalability. You can increase ports, bandwidth, and processing capacity of a stack by simply adding member switches to the stack.
- iStack also simplifies device configuration and management. After a stack is set up, multiple physical switches are virtualized into one logical device. You can log in to any member switch in the stack to manage all the member switches in the stack. CloudEngine S5735-S-V2 support stacking through electrical ports.

Network Slicing Functions

- CloudEngine S5735-S-V2 provides a range of VLAN slicing functions to meet diversified SLA requirements of different services and customers. Service isolation and bandwidth guarantee are implemented based on QoS. Slices can be completely isolated from each other without affecting each other. Traffic is isolated at the physical layer, and network slicing is performed for services on the same physical network. The Network Slicing technology can be used at the access, aggregation, and core layers to meet differentiated SLA requirements of new services on campus networks.

PoE Function

CloudEngine S5735-S-V2 PoE models can support PoE++(up to 90W power supply), Meeting high-power power supply requirements for Wi-Fi 6 APs, IP cameras, and Video phones.

- **Perpetual PoE**: When a PoE switch is abnormal Power-off or the software version is upgraded, the power supply to PDs is not interrupted. This capability ensures that PDs are not powered off during the switch reboot.
- **Fast PoE**: PoE switches can supply power to PDs within seconds after they are powered on. This is different from common switches that generally take 1 to 3 minutes to start to supply power to PDs. When a PoE switch reboots due to a power failure, the PoE switch continues to supply power to the PDs immediately after being powered on without waiting until it finishes reboot. This greatly shortens the power failure time of PDs.

Intelligent O&M

- CloudEngine S5735-S-V2 provides telemetry technology to collect device data in real time and send the data to Huawei campus network analyzer CampusInsight. The CampusInsight analyzes network data based on the intelligent fault identification algorithm, accurately displays the real-time network status, effectively demarcates and locates faults in a timely manner, and identifies network problems that affect user experience, accurately guaranteeing user experience.
- CloudEngine S5735-S-V2 supports the WebMaster solution, featuring an embedded local management system. It automatically discovers network topology and provides full-network visibility, including topology views, network elements (NEs), and device alarms. The solution also enables unified management of switches, AR routers, and APs—without requiring an external network management system.WebMaster offers a graphical interface for intuitive operations, including guided service provisioning, one-click batch upgrades, zero-configuration device replacement, and automatic authentication and onboarding of

NEs. It can also detect loops caused by incorrect cable connections and eliminate them automatically.With its rich feature set, WebMaster enables the entire campus network to be visible, manageable, and controllable—all through a single device.

Intelligent Upgrade

- CloudEngine S5735-S-V2 supports the intelligent upgrade feature. Specifically, CloudEngine S5735-S-V2 obtains the version upgrade path and downloads the newest version for upgrade from the Huawei Online Upgrade Platform (HOUP). The entire upgrade process is highly automated and achieves one-click upgrade. In addition, preloading the version is supported, which greatly shortens the upgrade time and service interruption time.
- The intelligent upgrade feature greatly simplifies device upgrade operations and makes it possible for the customer to upgrade the version independently. This greatly reduces the customer's maintenance costs. In addition, the upgrade policies on the HOUP platform standardize the upgrade operations, which greatly reduces the risk of upgrade failures.

Cloud Management

- The Huawei cloud management platform allows users to configure, monitor, and inspect switches on the cloud, reducing on-site deployment and O&M manpower costs and decreasing network OPEX. Huawei switches support both cloud management and on-premise management modes. These two management modes can be flexibly switched as required to achieve smooth evolution while maximizing return on investment (ROI).

OPS(Open Programmability System)

- CloudEngine S5735-S-V2 supports Open Programmability System (OPS), an open programmable system based on the Python language. IT administrators can program the O&M functions of a CloudEngine S5735-S-V2 switch through Python scripts to quickly innovate functions and implement intelligent O&M.

Licensing

CloudEngine S5735-S-V2 supports both the traditional feature-based licensing mode and the latest Huawei IDN One Software (N1 mode for short) licensing mode. The N1 mode is ideal for deploying Huawei CloudCampus Solution in the on-premises scenario, as it greatly enhances the customer experiences in purchasing and upgrading software services with simplicity.

Software Package Features in N1 Mode

Switch Functions	N1 Basic Software	N1 Foundation Software Package	N1 Advanced Software Package
Basic network functions: Layer 2 functions, IPv4, IPv6 and others Note: For details, see the Service Features	√	√	√
Basic network automation based on the iMaster NCE-Campus: • NE management: Device management, topology management and discovery • User access authentication	×	√	√
Automation and intelligent O&M: Basic functions for intelligent network analysis of CampusInsight.	×	√	√
Advanced network automation and intelligent O&M: Endpoint plug&play of iMaster NCE-Campus and Application analysis function of CampusInsight.	×	×	√

Product Specifications

Functions and Features

Item	Description
MAC address table	MAC address learning and aging
	32K MAC entries (MAX)
	Static, dynamic, and blackhole MAC address entries
	Packet filtering based on source MAC addresses
	Interface-based MAC learning limiting
VLAN features	4K VLANs simultaneously
	1K VLANif interface simultaneously
	Sub-VLAN, Super-VLAN, Mux VLAN, Voice VLAN
	QinQ and enhanced selective QinQ
	VLAN Stacking, VLAN Mapping
	LNP, VCMP, GVRP
ACL	ACL Hardware Specifications: 2.3K (shared ingress/egress)
Ethernet loop protection	Smart Link tree topology and Smart Link multi-instance, providing millisecond-level protection switchover
	ERPS (G.8032)
	STP (IEEE 802.1d), RSTP (IEEE 802.1w), and MSTP (IEEE 802.1s)
	BPDU protection, root protection, and loop protection
	BPDU tunnel
	LLDP, LLDP-MED
	SEP(Smart Ethernet Protection)
Probe protocol	LBDT/ L2PT/ DLDP
	SmartLink/ MonitorLink
	802.1ag/ 802.3ah/ Y.1731
Multicast	PIM DM, PIM SM, PIM SSM
	IGMPv1/v2/v3, IGMPv1/v2/v3 snooping, MLD snooping
	Multicast load balancing among member ports of a trunk
	Interface-based multicast traffic statistics
	Multicast VLAN, Multicast Static MAC
IP routing	BGP/BGP4+/ISIS/ISISv6/RIP/RIPng/OSPFv2/OSPFv3/VRRPv4/VRRPv6
	Static route/Routing Policy/Policy-Based Routing
	Up to 8192 FIBv4 entries
	Up to 4000 FIBv6 entries

Item	Description
IPv6 features	Up to 1024 ND entries (MAX)
	Path MTU (PMTU)
	IPv6 ping, IPv6 tracert, and IPv6 Telnet
Reliability	LACP
	VRRP
	BFD
	LLDP
	Dual-Boot backup
QoS/ACL	2432rules per IPv4 ACL
	1536 rules per IPv6 ACL
	Rate limiting on packets sent and received by an interface
	Packet redirection
	Interface-based traffic policing and two-rate and three-color CAR
	Eight queues on each interface
	DRR, SP, and DRR+SP queue scheduling algorithms
	Re-marking of the 802.1p priority and DSCP priority
	Packet filtering at Layer 2 to Layer 4, filtering out invalid frames based on the source MAC address, destination MAC address, source IP address, destination IP address, TCP/UDP port number, protocol type, and VLAN ID
	Rate limiting in each queue and traffic shaping on interfaces
	Network Slicing（VLAN）
Security	Hierarchical user management and password protection
	DoS attack defense, ARP attack defense, and ICMP attack defense
	Binding of the IP address, MAC address, interface number, and VLAN ID
	Port isolation, port security, and sticky MAC
	Blackhole MAC address entries
	Limit on the number of learned MAC addresses
	IEEE 802.1x authentication and limit on the number of users on an interface
	Portal authentication
	AAA authentication, RADIUS authentication, HWTACACS authentication, and NAC
	SSH V2.0
	Hypertext Transfer Protocol Secure (HTTPS)
	CPU defense
	Blacklist and whitelist
	DHCP client, DHCP relay, DHCP server, DHCP snooping

Item	Description
	DHCPv6 client, DHCPv6 relay
Management and maintenance	iStack, stack bandwidth: 80Gbps
	Maximum number of stacked devices: 9
	Fast stack upgrade
	Cloud management based on Netconf/Yang
	Virtual Cable Test (VCT)
	Remote configuration and maintenance using Telnet
	SNMPv1/v2/v3
	RMON
	eSight and web-based NMS
	HTTPS
	LLDP/LLDP-MED
	System logs and multi-level alarms
	802.3az EEE
	IFIT
	Port Mirroring, RSPAN, Flow mirroring
	Registration Center Deployment
	GVRP
	iPCA、sFlow、NQA、Telemetry
	1588V2
	WebMaster Management
Interoperability	Supports VBST (Compatible with PVST/PVST+/RPVST)

Hardware Specifications

Hardware specifications of the CloudEngine S5735-S24T4XE-V2/S24T4XEZ-V2/-S24P4XE-V2 models

Item		CloudEngine S5735-S24T4XE-V2	CloudEngine S5735-S24T4XEZ-V2	CloudEngine S5735-S24P4XE-V2
Physical specifications	Dimensions (H x W x D, mm)	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm
	Chassis height	1 U	1 U	1 U
	Chassis weight (including packaging)	7.48 kg	7.25 kg	7.74 kg
Fixed port	GE port	24	24	24(PoE+)
	GE SFP port	NA	NA	NA
	10GE SFP+ port	4	4	4

Item		CloudEngine S5735-S24T4XE-V2	CloudEngine S5735-S24T4XEZ-V2	CloudEngine S5735-S24P4XE-V2
	Dedicated 12GE stack port	2	2	2
Extended slot		NA	1	NA
Management port	Console port (RJ45)	Supported	Supported	Supported
	USB port	USB 2.0	USB 2.0	USB 2.0
CPU	Frequency	1.1 GHz	1.1 GHz	1.1 GHz
	Cores	2	2	2
Storage	Memory (RAM)	2 GB	2 GB	2 GB
Power supply system	Power supply type	• 80 W AC(pluggable) • 180 W AC(pluggable) • 240 W DC(pluggable) • 600 W AC(pluggable) • 1200 W DC(pluggable)	• 80 W AC(pluggable) • 180 W AC(pluggable) • 240 W DC(pluggable)	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)
	Power supply redundancy	1+1	1+1	1+1+1
	Rated voltage range	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 130 V, 200 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC
	Maximum voltage range	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC
	Maximum input current	• The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.
	Maximum power consumption of the device	• 33.1 W (2 *80W AC) • 45.45 W (2 *180W AC) • 85.2 W (2* 1200W DC)	• 36.40 W (with two 80 W AC power modules) • 34.74 W (with two 240 W DC power modules)	• 49.9 W (without PD, 2*600W AC) • 1023.22 W(with PD,PD Power consumption of :840W, 3*1000W AC）
	Typical power consumption	25.2 W	34.01 W	39.43 W

Item		CloudEngine S5735-S24T4XE-V2	CloudEngine S5735-S24T4XEZ-V2	CloudEngine S5735-S24P4XE-V2
	Static power consumption	23.2 W	22.14 W	28.65 W
Heat dissipation system	Heat dissipation mode	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment
	Number of fan modules	1	1	2
	Airflow	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear
	Maximum heat dissipation of the device (BTU/hour)	• 177.43 (2 *180W AC) • 276.38 (2* 1200W DC)	• 177.43 (2 *180W AC) • 276.38 (2* 1200W DC)	• Without PDs: 272.97 • With PDs: 3156.19
Environmen t parameters	Long-term operating temperature	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.
	Short-term operating temperature[3]	NA	NA	NA
	Storage temperature	-40°C to +70°C	-40°C to +70°C	-40°C to +70°C
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	5% to 95% (non-condensing)
	Operating altitude	5000 m	5000 m	5000 m
	Noise under normal temperature (sound power)	47 dB(A)	47 dB(A)	48.8 dB(A)
	Noise under high temperature (sound power)	51 dB(A)	51 dB(A)	60.9 dB(A)
	Noise under normal temperature (sound pressure)	35 dB(A)	35 dB(A)	36.8 dB(A)
	Surge protection specification (power port)	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in

Item		CloudEngine S5735-S24T4XE-V2	CloudEngine S5735-S24T4XEZ-V2	CloudEngine S5735-S24P4XE-V2
		differential mode, ±4 kV in common mode	differential mode, ±4 kV in common mode	differential mode, ±4 kV in common mode
Reliability	MTBF (year)[2]	168.93	57.69	120.24
	MTTR (hour)	0.5	0.5	0.5
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification

Hardware specifications of the CloudEngine S5735-S24P4XEZ-V2/-S24U4XE-V2/-S24U4XE-V2 models

Item		CloudEngine S5735-S24P4XEZ-V2	CloudEngine S5735-S24U4XE-V2	CloudEngine S5735-S24T8J4XE-XA-V2
Physical specifications	Dimensions (H x W x D, mm)	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm
	Chassis height	1 U	1 U	1 U
	Chassis weight (including packaging)	7.4 kg	7.4 kg	6.85 kg
Fixed port	GE port	24(PoE+)	24(PoE++)	24
	2.5GE SFP port	NA	NA	8
	10GE SFP+ port	4	4	4
	Dedicated 12GE stack port	2	2	2
Extended slot		1	NA	NA
Management port	Console port (RJ45)	Supported	Supported	Supported
	USB port	USB 2.0	USB 2.0	USB 2.0
CPU	Frequency	1.1 GHz	1.1 GHz	1.1 GHz
	Cores	2	2	2
Storage	Memory (RAM)	2 GB	2 GB	2 GB
Power supply system	Power supply type	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)	Built-in dual AC power
	Power supply redundancy	1+1+1	1+1+1	1+1

Item		CloudEngine S5735-S24P4XEZ-V2	CloudEngine S5735-S24U4XE-V2	CloudEngine S5735-S24T8J4XE-XA-V2
	Rated voltage range	• AC input: 100 V AC to 130 V, 200 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 130 V, 200 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 110V DC to 250V DC
	Maximum voltage range	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 264 V AC, 47 Hz to 63 Hz • High-Voltage DC input: 88 V DC to 300 V DC
	Maximum input current	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.
	Maximum power consumption of the device	• Without PoE: 45.80 W (with two 600 W AC power modules) 55.70 W (with two 1000 W AC power modules) 64.68 W (with two 1000 W DC power modules) • Full PoE load: 1023.22 W (PoE: 840 W, with three 1000 W AC power modules)	• 55.2 W (without PD, 2*600W AC) • 2430.11 W(with PD,PD Power consumption of :2268W, 3*1000W AC）	55.12 W
	Typical power consumption	35.98 W	36.98 W	41.18 W
	Static power consumption	30.43 W	28.8 W	22.45 W
Heat dissipation system	Heat dissipation mode	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment
	Number of fan modules	2	2	2
	Airflow	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear
	Maximum heat dissipation of the device (BTU/hour)	- Without PoE: 156.27 (with two 600 W AC power modules) 190.05 (with two 1000 W AC power modules) 220.69 (with two 1000 W	• Without PDs: 259.32 • With PDs: 8857.81	188.07

Item		CloudEngine S5735-S24P4XEZ-V2	CloudEngine S5735-S24U4XE-V2	CloudEngine S5735-S24T8J4XE-XA-V2
		DC power modules) - Full PoE load: 3491.33 (PoE: 840 W, with three 1000 W AC power modules)		
Environment parameters	Long-term operating temperature	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.
	Short-term operating temperature[3]	NA	NA	NA
	Storage temperature	-40°C to +70°C	-40°C to +70°C	-40°C to +70°C
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	5% to 95% (non-condensing)
	Operating altitude	5000 m	5000 m	5000 m
	Noise under normal temperature (sound power)	• Three 600 W AC PoE power modules with 30% load: 49 dBA • Three 1000 W AC PoE power modules with 30% load: 48.5 dBA • Three 1000 W DC PoE power modules with 30% load: 47.3 dBA	3*600W PoE AC 30% load: 50 dB(A) 3*1000W PoE AC 30% load: 49.9 dB(A) 3*100W PoE DC 30% load: 48.5 dB(A)	41.9 dB(A)
	Noise under normal temperature (sound pressure)	• Three 600 W AC PoE power modules with 30% load: 37 dBA • Three 1000 W AC PoE power modules with 30% load: 36.9 dBA • Three 1000 W DC PoE power modules with 30% load: 35.5 dBA	3*600W PoE AC 30% load: 38 dB(A) 3*1000W PoE AC 30% load: 37.9 dB(A) 3*100W PoE DC 30% load: 36.5 dB(A)	29.9 dB(A)
	Surge protection specification (power port)	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	±6 kV in differential mode, ±6 kV in common mode
Reliability	MTBF (year)[2]	57.72	92.58	59.93
	MTTR (hour)	0.5	0.5	0.5

Item		CloudEngine S5735-S24P4XEZ-V2	CloudEngine S5735-S24U4XE-V2	CloudEngine S5735-S24T8J4XE-XA-V2
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification

Hardware specifications of the CloudEngine S5735-S24T8J4XEZ-V2/-S24P8J4XEZ-V2/-S48T4XE-V2 models

Item		CloudEngine S5735-S24T8J4XEZ-V2	CloudEngine S5735-S24P8J4XEZ-V2	CloudEngine S5735-S48T4XE-V2
Physical specifications	Dimensions (H x W x D, mm)	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm
	Chassis height	1 U	1 U	1 U
	Chassis weight (including packaging)	7.35 kg	7.59 kg	7.74 kg
Fixed port	GE port	24	24	48
	2.5GE SFP port	8	8	NA
	10GE SFP+ port	4	4	4
	Dedicated 12GE stack port	2	2	2
Extended slot		1	1	NA
Management port	Console port (RJ45)	Supported	Supported	Supported
	USB port	USB 2.0	USB 2.0	USB 2.0
CPU	Frequency	1.1 GHz	1.1 GHz	1.1 GHz
	Cores	2	2	2
Storage	Memory (RAM)	2 GB	2 GB	2 GB
Power supply system	Power supply type	• 80 W AC(pluggable) • 180 W AC(pluggable) • 240 W DC(pluggable)	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)	• 80 W AC(pluggable) • 180 W AC(pluggable) • 240 W DC(pluggable) • 600 W AC(pluggable) • 1200 W DC(pluggable)
	Power supply redundancy	1+1	1+1+1	1+1
	Rated voltage range	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to

Item		CloudEngine S5735-S24T8J4XEZ-V2	CloudEngine S5735-S24P8J4XEZ-V2	CloudEngine S5735-S48T4XE-V2
		V DC	V DC	-60 V DC
	Maximum voltage range	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC
	Maximum input current	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.
	Maximum power consumption of the device	• 58.29 W (with two 80 W AC power modules) • 57.77 W (with two 240 W DC power modules)	• Without PoE: 45.80 W (with two 600 W AC power modules) 55.70 W (with two 1000 W AC power modules) 64.68 W (with two 1000 W DC power modules) • Full PoE load: 1023.22 W (PoE: 840 W, with three 1000 W AC power modules)	• 55.12 W (2 *80W AC) • 64.51 W (2 *180W AC) • 93.26 W (2* 1200W DC)
	Typical power consumption	52.51 W	53.59 W	39.1 W
	Static power consumption	34.35 W	32.79 W	25.31 W
Heat dissipation system	Heat dissipation mode	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment
	Number of fan modules	2	2	2
	Airflow	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear
	Maximum heat dissipation of the device (BTU/hour)	• 198.89 (with two 80 W AC power modules) • 197.12 (with two 240 W DC power modules)	• Without PoE: 156.27 (with two 600 W AC power modules) 190.05 (with two 1000 W AC power modules) 220.69 (with two 1000 W DC power modules) • Full PoE load: 3491.33 (PoE: 840 W, with three 1000 W AC power modules)	• 221.79 (2 *180W AC) • 317.33 (2* 1200W DC)

Item		CloudEngine S5735-S24T8J4XEZ-V2	CloudEngine S5735-S24P8J4XEZ-V2	CloudEngine S5735-S48T4XE-V2
Environment parameters	Long-term operating temperature	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.
	Short-term operating temperature[3]	NA	NA	NA
	Storage temperature	-40°C to +70°C	-40°C to +70°C	-40°C to +70°C
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	5% to 95% (non-condensing)
	Operating altitude	5000 m	5000 m	5000 m
	Noise under normal temperature (sound power)	47 dB(A)	Three 600 W AC PoE power modules with 30% load: 49 dBA Three 1000 W AC PoE power modules with 30% load: 48.5 dBA Three 1000 W DC PoE power modules with 30% load: 47.3 dBA	41.9 dB(A)
	Noise under normal temperature (sound pressure)	35 dB(A)	Three 600 W AC PoE power modules with 30% load: 37 dBA Three 1000 W AC PoE power modules with 30% load: 36.9 dBA Three 1000 W DC PoE power modules with 30% load: 35.5 dBA	29.9 dB(A)
	Surge protection specification (power port)	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode
Reliability	MTBF (year)[2]	57.88	53.44	124.62
	MTTR (hour)	0.5	0.5	0.5
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing	• EMC certification • Safety certification • Manufacturing	• EMC certification • Safety certification • Manufacturing

Item		CloudEngine S5735-S24T8J4XEZ-V2	CloudEngine S5735-S24P8J4XEZ-V2	CloudEngine S5735-S48T4XE-V2
		certification	certification	certification

Hardware specifications of the CloudEngine S5735-S48T4XE-XA-V2/-S48T4XEZ-V2/-S48P4XE-V2 models

Item		CloudEngine S5735-S48T4XE-XA-V2	CloudEngine S5735-S48T4XEZ-V2	CloudEngine S5735-S48P4XE-V2
Physical specifications	Dimensions (H x W x D, mm)	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm
	Chassis height	1 U	1 U	1 U
	Chassis weight (including packaging)	6.82 kg	7.48 kg	8.00 kg
Fixed port	GE port	48	48	48(PoE+)
	10GE SFP+ port	4	4	4
	Dedicated 12GE stack port	2	2	2
Extended slot		NA	1	NA
Management port	Console port (RJ45)	Supported	Supported	Supported
	USB port	USB 2.0	USB 2.0	USB 2.0
CPU	Frequency	1.1 GHz	1.1 GHz	1.1 GHz
	Cores	2	2	2
Storage	Memory (RAM)	2 GB	2 GB	2 GB
Power supply system	Power supply type	Built-in dual AC power	• 80 W AC(pluggable) • 180 W AC(pluggable) • 240 W DC(pluggable)	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)
	Power supply redundancy	1+1	1+1	1+1+1
	Rated voltage range	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 110V DC to 250V DC	• AC input: 100 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 130 V, 200 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC
	Maximum voltage range	• AC input: 90 V AC to 264 V AC, 47 Hz to 63 Hz • High-Voltage DC input: 88 V DC to 300 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC

Item		CloudEngine S5735-S48T4XE-XA-V2	CloudEngine S5735-S48T4XEZ-V2	CloudEngine S5735-S48P4XE-V2
			72 V DC	• DC input: -38.4 V DC to -72 V DC
	Maximum input current	4.0 A	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.
	Maximum power consumption of the device	55.12 W	• 49.84 W (with two 80 W AC power modules) • 48.98 W (with two 240 W DC power modules)	• 71.45 W (without PD, 2*600W AC) • 1931.44 W(with PD,PD Power consumption of :1680W, 3*1000W AC）
	Typical power consumption	39.1 W	44.1 W	54.51 W
	Static power consumption	18.45 W	24.68 W	33.99 W
Heat dissipation system	Heat dissipation mode	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment	Air-cooled heat dissipation and intelligent speed adjustment
	Number of fan modules	2	2	2
	Airflow	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear
	Maximum heat dissipation of the device (BTU/hour)	188.07	• 170.06 (with two 80 W AC power modules) • 167.12 (with two 240 W DC power modules)	• Without PDs: 286.62 • With PDs: 6653.60
Environment parameters	Long-term operating temperature	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1ºC every time the altitude increases by 220 m.
	Short-term operating temperature[3]	NA	NA	NA
	Storage temperature	-40°C to +70°C	-40°C to +70°C	-40°C to +70°C
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	5% to 95% (non-condensing)
	Operating altitude	5000 m	5000 m	5000 m
	Noise under	41.9 dB(A)	41.9 dB(A)	3*600W PoE AC 30% load:

Item		CloudEngine S5735-S48T4XE-XA-V2	CloudEngine S5735-S48T4XEZ-V2	CloudEngine S5735-S48P4XE-V2
	normal temperature (sound power)			50 dB(A) 3*1000W PoE AC 30% load: 49.9 dB(A) 3*100W PoE DC 30% load: 48.5 dB(A)
	Noise under normal temperature (sound pressure)	29.9 dB(A)	29.9 dB(A)	3*600W PoE AC 30% load: 38 dB(A) 3*1000W PoE AC 30% load: 37.9 dB(A) 3*100W PoE DC 30% load: 36.5 dB(A)
	Surge protection specification (power port)	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode
Reliability	MTBF (year)[2]	43.19	41.6	79.02
	MTTR (hour)	0.5	0.5	0.5
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification

Hardware specifications of the CloudEngine S5735-S48P4XEZ-V2/-S48U4XE-V2 models

Item		CloudEngine S5735-S48P4XE-V2	CloudEngine S5735-S48U4XE-V2	
Physical specifications	Dimensions (H x W x D, mm)	43.6 mm x 442 mm x 420 mm	43.6 mm x 442 mm x 420 mm	
	Chassis height	1 U	1 U	
	Chassis weight (including packaging)	7.75 kg	12.83 kg	
Fixed port	GE port	48(PoE+)	48(PoE++)	
	10GE SFP+ port	4	4	
	Dedicated 12GE stack port	2	2	
Extended slot		1	NA	
Management port	Console port (RJ45)	Supported	Supported	
	USB port	USB 2.0	USB 2.0	

Item		CloudEngine S5735-S48P4XE-V2	CloudEngine S5735-S48U4XE-V2	
CPU	Frequency	1.1 GHz	1.1 GHz	
	Cores	2	2	
Storage	Memory (RAM)	2 GB	2 GB	
Power supply system	Power supply type	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)	• 600 W PoE AC (pluggable) • 1000 W PoE AC (pluggable) • 1000 W PoE DC (pluggable)	
	Power supply redundancy	1+1+1	1+1+1	
	Rated voltage range	• AC input: 100 V AC to 130 V, 200 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	• AC input: 100 V AC to 130 V, 200 V AC to 240 V AC, 50/60 Hz • High-Voltage DC input: 240 V DC • DC input: -48 V DC to -60 V DC	
	Maximum voltage range	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC, 45 Hz to 65 Hz • High-Voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	
	Maximum input current	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	
	Maximum power consumption of the device	• Without PoE: 62.90 W (with two 600 W AC power modules) 67.20 W (with two 1000 W AC power modules) 76.93 W (with two 1000 W DC power modules) • Full PoE load: 1931.44 W (PoE: 1680 W, with three 1000 W AC power modules)	• Without PoE: 70.15 W (with two 600 W AC power modules) 74.22 W (with two 1000 W AC power modules) 83.66 W (with two 1000 W DC power modules) • Full PoE load: 3117.00 W (PoE: 2880 W, with three 1000 W AC power modules)	
	Typical power consumption	52.87 W	56.23 W	
	Static power consumption	33.89 W	31.57 W	
Heat dissipation	Heat dissipation mode	Air-cooled heat dissipation and intelligent speed	Air-cooled heat dissipation and intelligent speed	

Item		CloudEngine S5735-S48P4XE-V2	CloudEngine S5735-S48U4XE-V2	
system		adjustment	adjustment	
	Number of fan modules	2	2	
	Airflow	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear	
	Maximum heat dissipation of the device (BTU/hour)	• Without PoE: 214.62 (with two 600 W AC power modules) 227.93 (with two 1000 W AC power modules) 261.47 (with two 1000 W DC power modules) • Full PoE load: 6590.27 (PoE: 1680 W, with three 1000 W AC power modules)	• Without PoE: 239.36 (with two 600 W AC power modules) 253.25 (with two 1000 W AC power modules) 285.46 (with two 1000 W DC power modules) • Full PoE load: 10635.52 (PoE: 2880 W, with three 1000 W AC power modules)	
Environment parameters	Long-term operating temperature	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1°C every time the altitude increases by 220 m.	• 0-1800 m altitude: -5°C to +50°C • 1800-5000 m altitude: The operating temperature reduces by 1°C every time the altitude increases by 220 m.	
	Short-term operating temperature[3]	NA	NA	
	Storage temperature	-40°C to +70°C	-40°C to +70°C	
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	
	Operating altitude	5000 m	5000 m	
	Noise under normal temperature (sound power)	• Three 600 W AC PoE power modules with 30% load: 50 dBA • Three 1000 W AC PoE power modules with 30% load: 49.9 dBA • Three 1000 W DC PoE power modules with 30% load: 48.5 dBA	• Three 600 W AC PoE power modules with 30% load: 50 dBA • Three 1000 W AC PoE power modules with 30% load: 49.9 dBA • Three 1000 W DC PoE power modules with 30% load: 48.5 dBA	
	Noise under normal temperature (sound pressure)	• Three 600 W AC PoE power modules with 30% load: 38 dBA • Three 1000 W AC PoE power modules with 30% load: 37.9 dBA	• Three 600 W AC PoE power modules with 30% load: 38 dBA • Three 1000 W AC PoE power modules with 30% load: 37.9 dBA	

Item		CloudEngine S5735-S48P4XE-V2	CloudEngine S5735-S48U4XE-V2	
		• Three 1000 W DC PoE power modules with 30% load: 36.5 dBA	• Three 1000 W DC PoE power modules with 30% load: 36.5 dBA	
	Surge protection specification (power port)	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	• AC power port: ±6 kV in differential mode, ±6 kV in common mode • DC power port: ±2 kV in differential mode, ±4 kV in common mode	
Reliability	MTBF (year)[2]	40.14	73.29	
	MTTR (hour)	0.5	0.5	
	Availability	> 0.99999	> 0.99999	
Certification		• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification	

NOTE

1: The power consumption under different load conditions is calculated according to the ATIS standard. Additionally, the EEE function is enabled and there is no PoE power output.

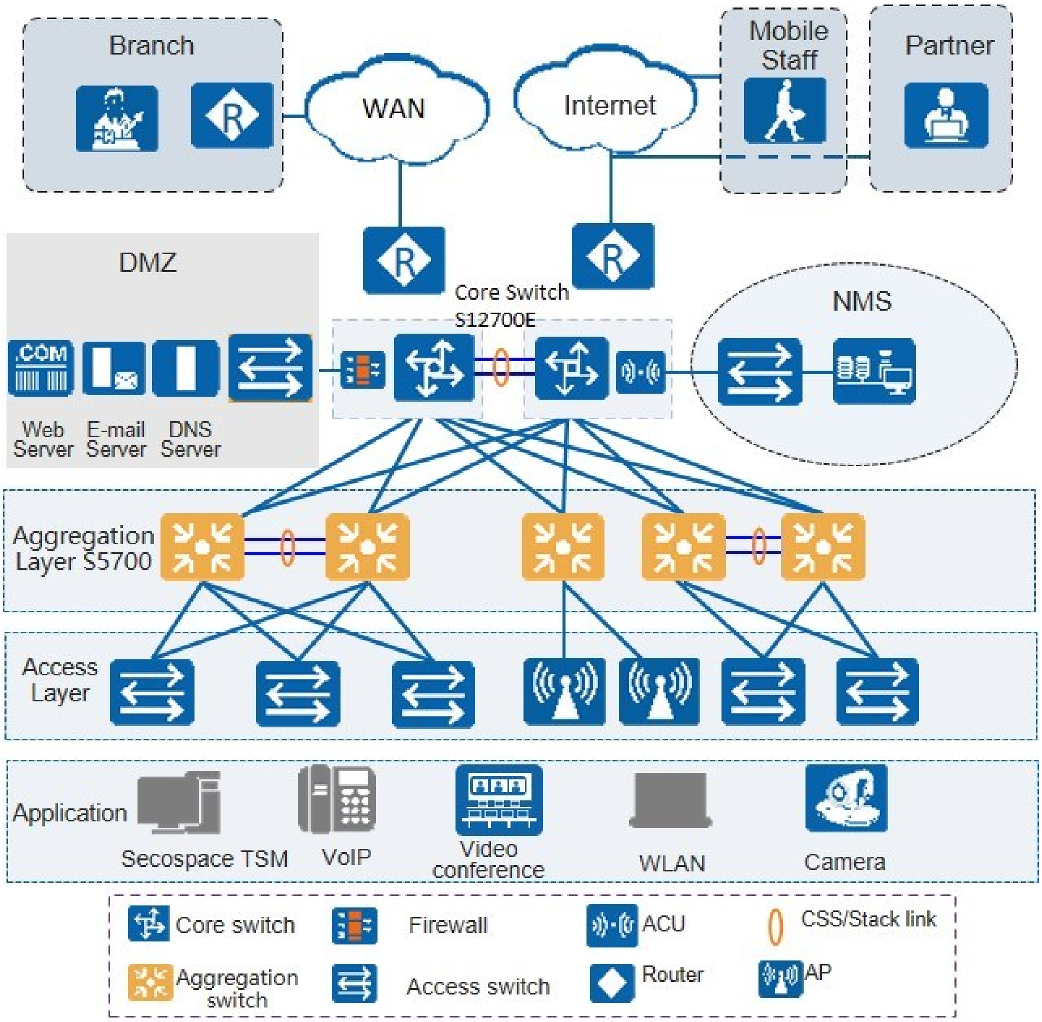
2: The reliability parameter values are calculated based on the typical configuration of the device. The parameter values vary according to the modules configured by the customer.

3: Short term indicates that the successive operating time is no more than 96 hours, the total operating time is no more than 360 hours, or the number of times the operating temperature is over 45° C is no more than 15 in a year.

Networking and Applications

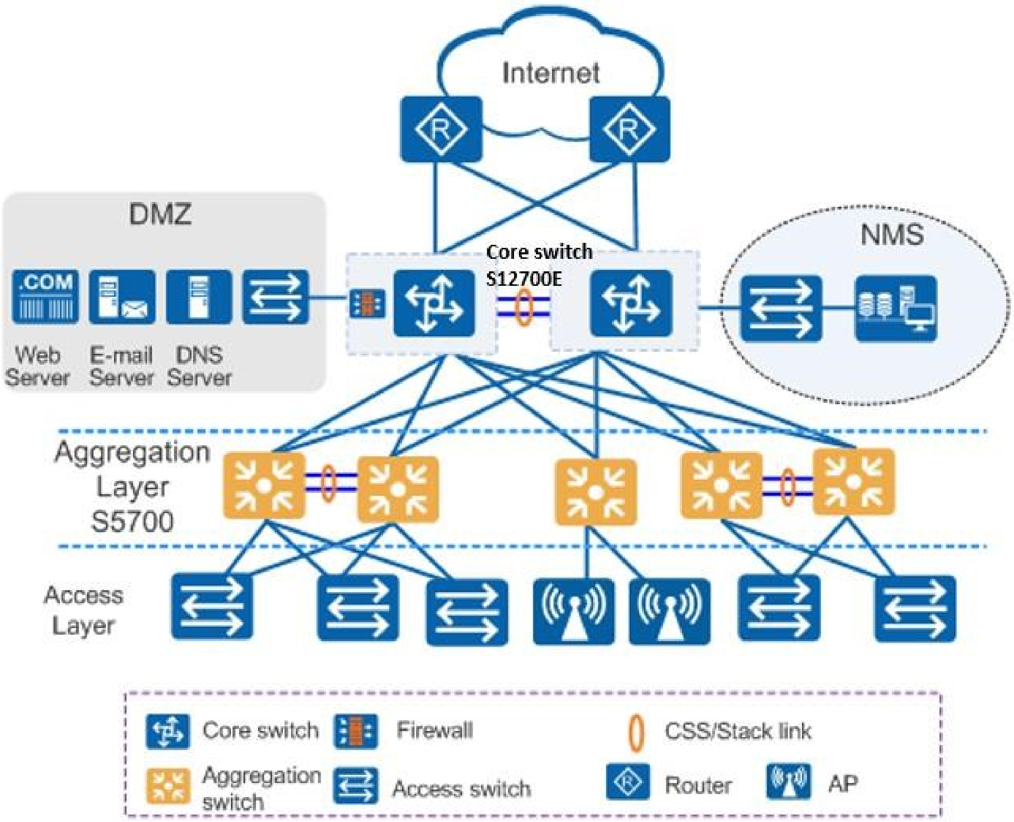
Large-Scale Enterprise Campus Network

CloudEngine S5735-S-V2 series switches can be deployed at the access layer of a campus network to build a high-performance and highly reliable enterprise network.



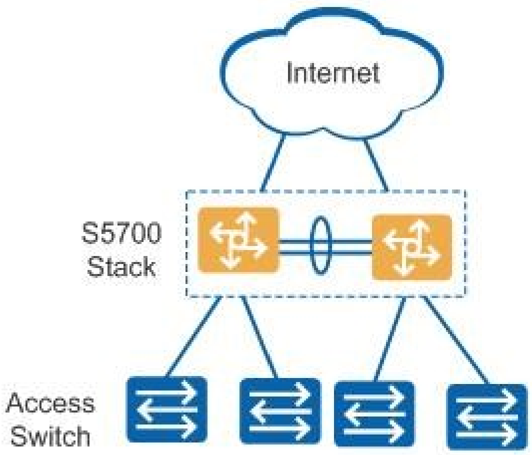
Small- or Medium-scale Enterprise Campus Network

CloudEngine S5735-S-V2 series switches can be deployed at the aggregation layer of a campus network to build a high-performance, multi-service, and highly reliable enterprise network.



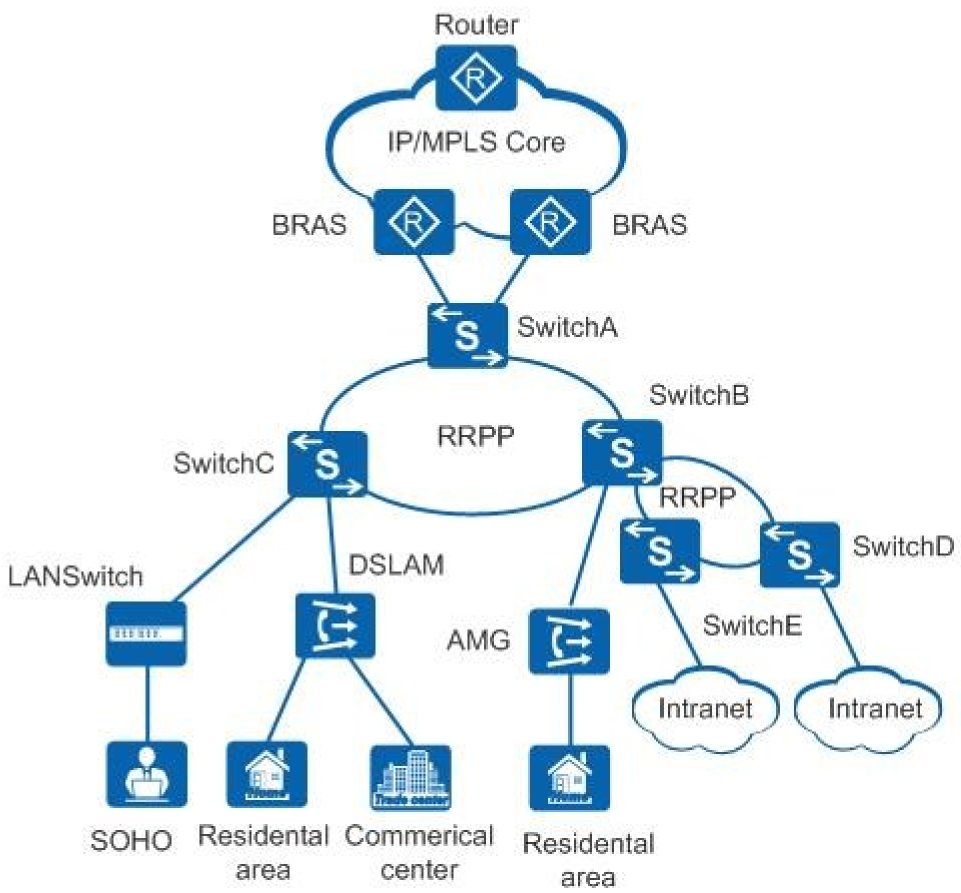
Small-scale Enterprise Campus Network

With powerful aggregation and routing capabilities of CloudEngine S5735-S-V2 series switches make them suitable for use as core switches in a small-scale enterprise network. Two or more S5735-S-V2 switches use iStack technology to ensure high reliability. They provide a variety of access control policies to achieve centralized management and simplify configuration.



Application on a MAN

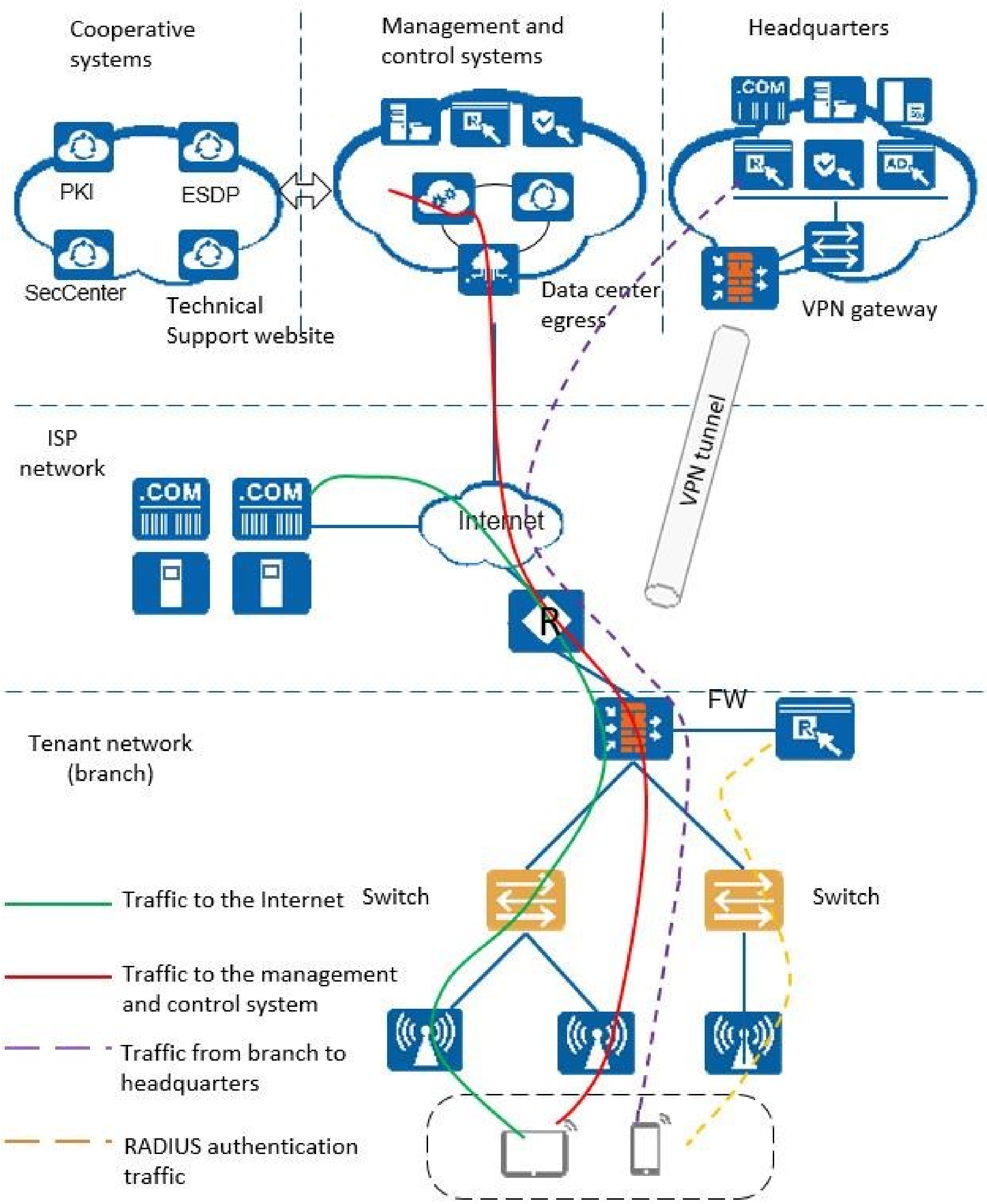
CloudEngine S5735-S-V2 series switches can be deployed at the access layer of a MAN(Metropolitan Area Network) to build a high-performance, multi-service, and highly reliable ISP MAN network.



Application in Public Cloud

CloudCampus Solution is a network solution suite based on Huawei public cloud.CloudEngine S5735-S-V2 series switches can be located at the access layer.

The switches are plug-and-play. They go online automatically after being powered on and connected with network cables, without the need for complex configurations, and use bidirectional certificate authentication to ensure management channel security. The switches provide the NETCONF and YANG interfaces, through which the management and control system delivers configurations to them. In addition, remote maintenance and fault diagnosis can be performed on the management and control system.



Safety and Regulatory Compliance

Certification Category	Description
Safety	• IEC 60950-1 • EN 60950-1/A11/A12 • UL 60950-1 • CSA C22.2 No 60950-1 • AS/NZS 60950.1 • CNS 14336-1 • IEC60825-1 • IEC60825-2 • EN60825-1 • EN60825-2

Certification Category	Description
Electromagnetic Compatibility (EMC)	• CISPR22 Class A • CISPR24 • EN55022 Class A • EN55024 • ETSI EN 300 386 Class A • CFR 47 FCC Part 15 Class A • ICES 003 Class A • AS/NZS CISPR22 Class A • VCCI Class A • IEC61000-4-2 • ITU-T K 20 • ITU-T K 21 • ITU-T K 44 • CNS13438
Environment	• RoHS • REACH • WEEE

NOTE

- EMC: electromagnetic compatibility
- CISPR: International Special Committee on Radio Interference
- EN: European Standard
- ETSI: European Telecommunications Standards Institute
- CFR: Code of Federal Regulations
- FCC: Federal Communication Commission
- IEC: International Electrotechnical Commission
- AS/NZS: Australian/New Zealand Standard
- VCCI: Voluntary Control Council for Interference
- UL: Underwriters Laboratories
- CSA: Canadian Standards Association
- IEEE: Institute of Electrical and Electronics Engineers
- RoHS: restriction of the use of certain hazardous substances
- REACH: Registration Evaluation Authorization and Restriction of Chemicals
- WEEE: Waste Electrical and Electronic Equipment

MIB and Standards Compliance

Supported MIBs

Category	MIB
Public MIB	• BRIDGE-MIB • DISMAN-NSLOOKUP-MIB • DISMAN-PING-MIB • DISMAN-TRACEROUTE-MIB • ENTITY-MIB

Category	MIB
	• EtherLike-MIB • IF-MIB • IP-FORWARD-MIB • IPv6-MIB • LAG-MIB • LLDP-EXT-DOT1-MIB • LLDP-EXT-DOT3-MIB • LLDP-MIB • NOTIFICATION-LOG-MIB • NQA-MIB • OSPF-TRAP-MIB • P-BRIDGE-MIB • Q-BRIDGE-MIB • RFC1213-MIB • RIPv2-MIB • RMON-MIB • SAVI-MIB • SNMP-FRAMEWORK-MIB • SNMP-MPD-MIB • SNMP-NOTIFICATION-MIB • SNMP-TARGET-MIB • SNMP-USER-BASED-SM-MIB • SNMPv2-MIB • TCP-MIB • UDP-MIB
Huawei-proprietary MIB	• HUAWEI-AAA-MIB • HUAWEI-ACL-MIB • HUAWEI-ALARM-MIB • HUAWEI-ALARM-RELIABILITY-MIB • HUAWEI-BASE-TRAP-MIB • HUAWEI-BRAS-RADIUS-MIB • HUAWEI-BRAS-SRVCFG-EAP-MIB • HUAWEI-BRAS-SRVCFG-STATICUSER-MIB • HUAWEI-CBQOS-MIB • HUAWEI-CDP-COMPLIANCE-MIB • HUAWEI-CONFIG-MAN-MIB • HUAWEI-CPU-MIB • HUAWEI-DAD-TRAP-MIB • HUAWEI-DC-MIB • HUAWEI-DATASYNC-MIB • HUAWEI-DEVICE-MIB • HUAWEI-DHCPR-MIB • HUAWEI-DHCPS-MIB • HUAWEI-DHCP-SNOOPING-MIB

Category	MIB
	• HUAWEI-DIE-MIB • HUAWEI-DNS-MIB • HUAWEI-DLDP-MIB • HUAWEI-ELMI-MIB • HUAWEI-ERPS-MIB • HUAWEI-ERRORDOWN-MIB • HUAWEI-ENERGYMNGT-MIB • HUAWEI-EASY-OPERATION-MIB • HUAWEI-ENTITY-EXTENT-MIB • HUAWEI-ENTITY-TRAP-MIB • HUAWEI-ETHARP-MIB • HUAWEI-ETHOAM-MIB • HUAWEI-FLASH-MAN-MIB • HUAWEI-FWD-RES-TRAP-MIB • HUAWEI-GARP-APP-MIB • HUAWEI-GTSM-MIB • HUAWEI-HGMP-MIB • HUAWEI-HWTACACS-MIB • HUAWEI-IF-EXT-MIB • HUAWEI-INFOCENTER-MIB • HUAWEI-IPPOOL-MIB • HUAWEI-IPV6-MIB • HUAWEI-ISOLATE-MIB • HUAWEI-L2IF-MIB • HUAWEI-L2MAM-MIB • HUAWEI-L2VLAN-MIB • HUAWEI_LDT-MIB • HUAWEI-LLDP-MIB • HUAWEI-MAC-AUTHEN-MIB • HUAWEI-MEMORY-MIB • HUAWEI-MFF-MIB • HUAWEI-MFLP-MIB • HUAWEI-MSTP-MIB • HUAWEI-MULTICAST-MIB • HUAWEI-NAP-MIB • HUAWEI-NTPV3-MIB • HUAWEI-PERFORMANCE-MIB • HUAWEI-PORT-MIB • HUAWEI-PORTAL-MIB • HUAWEI-QINQ-MIB • HUAWEI-RIPv2-EXT-MIB • HUAWEI-RM-EXT-MIB • HUAWEI-RRPP-MIB • HUAWEI-SECURITY-MIB • HUAWEI-SEP-MIB

Category	MIB
	• HUAWEI-SNMP-EXT-MIB • HUAWEI-SSH-MIB • HUAWEI-STACK-MIB • HUAWEI-SWITCH-L2MAM-EXT-MIB • HUAWEI-SWITCH-SRV-TRAP-MIB • HUAWEI-SYS-MAN-MIB • HUAWEI-TCP-MIB • HUAWEI-TFTPC-MIB • HUAWEI-TRNG-MIB • HUAWEI-XQOS-MIB

NOTE

For more detailed information of MIBs supported by the CloudEngine S5735-S series, visit https://support.huawei.com/enterprise/en/switches/s5700-pid-6691579?category=reference-guides&subcategory=mib-reference.

Standard Compliance

Standard Organization	Standard or Protocol
IETF	• RFC 768 User Datagram Protocol (UDP) • RFC 792 Internet Control Message Protocol (ICMP) • RFC 793 Transmission Control Protocol (TCP) • RFC 826 Ethernet Address Resolution Protocol (ARP) • RFC 854 Telnet Protocol Specification • RFC 951 Bootstrap Protocol (BOOTP) • RFC 959 File Transfer Protocol (FTP) • RFC 1058 Routing Information Protocol (RIP) • RFC 1112 Host extensions for IP multicasting • RFC 1157 A Simple Network Management Protocol (SNMP) • RFC 1256 ICMP Router Discovery • RFC 1305 Network Time Protocol Version 3 (NTP) • RFC 1349 Internet Protocol (IP) • RFC 1493 Definitions of Managed Objects for Bridges • RFC 1542 Clarifications and Extensions for the Bootstrap Protocol • RFC 1643 Ethernet Interface MIB • RFC 1757 Remote Network Monitoring (RMON) • RFC 1901 Introduction to Community-based SNMPv2 • RFC 1902-1907 SNMP v2 • RFC 1981 Path MTU Discovery for IP version 6 • RFC 2131 Dynamic Host Configuration Protocol (DHCP) • RFC 2328 OSPF Version 2 • RFC 2453 RIP Version 2 • RFC 2460 Internet Protocol, Version 6 Specification (IPv6) • RFC 2461 Neighbor Discovery for IP Version 6 (IPv6) • RFC 2462 IPv6 Stateless Address Auto configuration • RFC 2463 Internet Control Message Protocol for IPv6 (ICMPv6)

Standard Organization	Standard or Protocol
	• RFC 2474 Differentiated Services Field (DS Field) • RFC 2740 OSPF for IPv6 (OSPFv3) • RFC 2863 The Interfaces Group MIB • RFC 2597 Assured Forwarding PHB Group • RFC 2598 An Expedited Forwarding PHB • RFC 2571 SNMP Management Frameworks • RFC 2865 Remote Authentication Dial In User Service (RADIUS) • RFC 3046 DHCP Option82 • RFC 3376 Internet Group Management Protocol, Version 3 (IGMPv3) • RFC 3513 IP Version 6 Addressing Architecture • RFC 3579 RADIUS Support For EAP • RFC 4271 A Border Gateway Protocol 4 (BGP-4) • RFC 4760 Multiprotocol Extensions for BGP-4 • draft-grant-tacacs-02 TACACS+ • RFC 6241 Network Configuration Protocol (NETCONF)
IEEE	• IEEE 802.1D Media Access Control (MAC) Bridges • IEEE 802.1p Traffic Class Expediting and Dynamic Multicast Filtering • IEEE 802.1Q Virtual Bridged Local Area Networks • IEEE 802.1ad Provider Bridges • IEEE 802.2 Logical Link Control • IEEE Std 802.3 CSMA/CD • IEEE Std 802.3i 10BASE-T specification • IEEE Std 802.3u 100BASE-T specification • IEEE Std 802.3ab 1000BASE-T specification • IEEE Std 802.3ab RJ45 con Auto-MDIX • IEEE Std 802.3ad Aggregation of Multiple Link Segments • IEEE Std 802.3ae 10GE WEN/LAN Standard • IEEE Std 802.3x Full Duplex and flow control • IEEE Std 802.3z Gigabit Ethernet Standard • IEEE802.1ax/IEEE802.3ad Link Aggregation • IEEE 802.1ab Link Layer Discovery Protocol • IEEE 802.1D Spanning Tree Protocol • IEEE 802.1w Rapid Spanning Tree Protocol • IEEE 802.1s Multiple Spanning Tree Protocol • IEEE 802.1x Port based network access control protocol • IEEE 802.3af DTE Power via MIDI • IEEE 802.3at DTE Power via the MDI Enhancements • IEEE 802.3ac VLAN tagging • IEEE 802.3ax Link Aggregation Task Force
ITU	• ITU SG13 Y.17ethoam • ITU SG13 QoS control Ethernet-Based IP Access • ITU-T Y.1731 ETH OAM performance monitor
ISO	• ISO 10589 IS-IS Routing Protocol

Standard Organization	Standard or Protocol
MEF	• MEF 2 Requirements and Framework for Ethernet Service Protection • MEF 9 Abstract Test Suite for Ethernet Services at the UNI • MEF 10.2 Ethernet Services Attributes Phase 2 • MEF 11 UNI Requirements and Framework • MEF 13 UNI Type 1 Implementation Agreement • MEF 15 Requirements for Management of Metro Ethernet Phase 1 Network Elements • MEF 17 Service OAM Framework and Requirements • MEF 20 UNI Type 2 Implementation Agreement • MEF 23 Class of Service Phase 1 Implementation Agreement • XMODEM/YMODEM Protocol Reference

Ordering Information

The following table lists ordering information of the CloudEngine S5735-S-V2 series switches.

Model	Product Description
CloudEngine S5735-S24T4XE-V2	CloudEngine S5735-S24T4XE-V2 (24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports, without power module)
CloudEngine S5735-S24T4XEZ-V2	CloudEngine S5735-S24T4XEZ-V2 (24*10/100/1000BASE-T ports, 4*10GE SFP+ ports, 2*12GE stack ports, expansion card slot, without power module)
CloudEngine S5735-S24P4XE-V2	CloudEngine S5735-S24P4XE-V2 (24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports, PoE+, without power module)
CloudEngine S5735-S24P4XEZ-V2	CloudEngine S5735-S24P4XEZ-V2 (24*10/100/1000BASE-T ports (PoE+), 4*10GE SFP+ ports, 2*12GE stack ports, expansion card slot, without power module)
CloudEngine S5735-S24U4XE-V2	CloudEngine S5735-S24U4XE-V2 (24 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x12GE stack ports, PoE++, without power module)
CloudEngine S5735-S24T8J4XE-XA-V2	CloudEngine S5735-S24T8J4XE-XA-V2 (24*10/100/1000BASE-T ports, 8*2.5GE SFP ports (or 2*10GE SFP+ ports), 4*10GE SFP+ ports, 2*12GE stack ports, dual built-in AC power)
CloudEngine S5735-S24T8J4XEZ-V2	CloudEngine S5735-S24T8J4XEZ-V2 (24*10/100/1000BASE-T ports, 8*2.5GE SFP ports (or 2*10GE SFP+ ports), 4*10GE SFP+ ports, 2*12GE stack ports, expansion card slot, without power module)
CloudEngine S5735-S24P8J4XEZ-V2	CloudEngine S5735-S24P8J4XEZ-V2 (24*10/100/1000BASE-T ports (PoE+), 8*2.5GE SFP ports (or 2*10GE SFP+ ports), 4*10GE SFP+ ports, 2*12GE stack ports, expansion card slot, without power module)
CloudEngine S5735-S48T4XE-V2	CloudEngine S5735-S48T4XE (48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports, without power module)
CloudEngine S5735-S48T4XE-XA-V2	CloudEngine S5735-S48T4XE-XA-V2 (48*10/100/1000BASE-T ports, 4*10GE SFP+ ports, 2*12GE stack ports, dual built-in AC power)
CloudEngine S5735-S48T4XEZ-V2	CloudEngine S5735-S48T4XEZ-V2 (48*10/100/1000BASE-T ports, 4*10GE SFP+ ports, 2*12GE stack ports, expansion card slot, without power module)
CloudEngine S5735-S48P4XE-V2	CloudEngine S5735-S48P4XE-V2 (48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GE stack ports, PoE+, without power module)
CloudEngine S5735-	CloudEngine S5735-S48P4XEZ-V2 (48*10/100/1000BASE-T ports (PoE+), 4*10GE SFP+ ports,

Model	Product Description
S48P4XEZ-V2	2*12GE stack ports, expansion card slot, without power module)
CloudEngine S5735-S48U4XE-V2	CloudEngine S5735-S48U4XE-V2 (48 x 10/100/1000Base-T ports, 4 x 10 GE SFP+ ports, 2 x 12GEstack ports, PoE++, without power module)
PAC80S12-CN	80 W AC power module
PAC180S12-CN	180 W AC power module
PDC240S12-CN	240 W DC power module
PAC600S12-PB	600W AC &240 V DC Power Module
PDC1K2S12-CE	1200 W DC power module
PAC600S56-EB	600 W AC PoE power module
PAC1000S56-EB	1000 W AC PoE power module
PDC1000S56-EB	1000 W DC PoE power module
N1-S57S-M-Lic	S57XX-S Series Basic SW,Per Device
N1-S57S-M-SnS1Y	S57XX-S Series Basic SW,SnS,Per Device,1Year
N1-S57S-F-Lic	N1-CloudCampus,Foundation,S57XX-S Series,Per Device
N1-S57S-F-SnS1Y	N1-CloudCampus,Foundation,S57XX-S Series,SnS,Per Device,1Year
N1-S57S-A-Lic	N1-CloudCampus,Advanced,S57XX-S Series,Per Device
N1-S57S-A-SnS1Y	N1-CloudCampus,Advanced,S57XX-S Series,SnS,Per Device,1Year
N1-S57S-FToA-Lic	N1-Upgrade-Foundation to Advanced,S57XX-S,Per Device
N1-S57S-FToA-SnS1Y	N1-Upgrade-Foundation to Advanced,S57XX-S,SnS,Per Device,1Year

More Information

For more information about Huawei Campus Switches, visit http://e.huawei.com or contact us in the following ways:

- Global service hotline: http://e.huawei.com/en/service-hotline
- Logging in to the Huawei Enterprise technical support website: http://support.huawei.com/enterprise/
- Sending an email to the customer service mailbox: support_e@huawei.com

Huawei Technologies Co., Ltd.

Address:Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China

Website:e.huawei.com

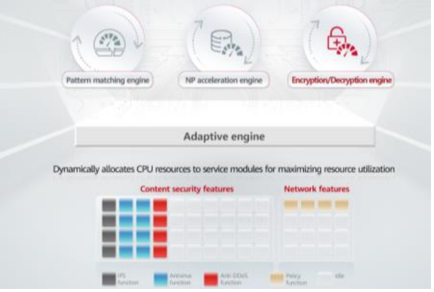
HUAWEI HiSecEngine USG6500F Series AI Firewalls

As digitalization is sweeping the world, extensive connections, explosive growth of data, and booming intelligent applications are profoundly changing the way we live and work. Enterprise services are going digital and moving to the cloud, which promotes the transformation of enterprise networks while bringing greater challenges to network security. As threats increase, unknown threats are ever-changing and highly covert. As users' requirements for security services increase, performance and latency become bottlenecks. With mass numbers of security policies and logs, threat handling and O&M are extremely time-consuming. As the "first gate" on network borders, firewalls are the first choice for enterprise security protection. However, traditional firewalls can only analyze and block threats based on signatures and therefore are unable to effectively handle unknown threats. In addition, the effectiveness of threats depends on the professional experience of O&M personnel. The single-point, reactive, and in-event defense method cannot effectively defend against unknown threat attacks, let alone threats hidden in encrypted traffic.

With new hardware and software architectures, The HiSecEngine USG6500F series AI firewalls, in either desktop or 1U models, are next-generation AI firewalls that feature intelligent defense, outstanding performance, and simplified O&M, effectively addressing the preceding challenges. The HiSecEngine USG6500F series AI firewalls use intelligence technologies to enable border defense to accurately block known and unknown threats. Equipped with multiple built-in security-dedicated acceleration engines, The HiSecEngine USG6500F series AI firewalls support enhanced forwarding, content security detection, and IPsec service processing acceleration. The security O&M platform implements unified management and O&M of multiple types of security products, such as firewalls, Anti-DDoS devices, reducing security O&M OPEX. In addition, the HiSecEngine USG6500F-DL series AI firewalls support the LTE function, which can be used to implement flexible, efficient, and fast network deployment in remote areas or mobile office scenarios.

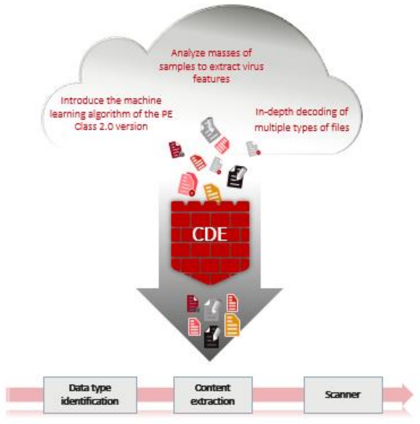
01 Product Highlights

Excellent performance



By leveraging fresh-new hardware and software architectures of forwarding and control separation, the HiSecEngine USG6500F series AI firewalls dynamically allocate resources to service modules through the adaptive security engine (ASE), maximizing resource utilization and improving overall service performance. For core services, the HiSecEngine USG6500F series AI firewalls also support network processor (NP), pattern matching, and encryption/decryption engines. These engines greatly improve short-packet forwarding, reduce the forwarding latency, and enhance application identification, intrusion prevention detection, and IPSec service performance.

Intelligent defense



- ✓ Signature:Malicious file family = **1:N**
- ✓ Detection speeds are **equivalent to the signature detection performance**
- ✓ Industry-leading **unknown threat** detection capabilities

The HiSecEngine USG6500F series AI firewalls provide content security functions, such as application identification, IPS, antivirus, and URL filtering to protect intranet servers and users against threats. The HiSecEngine USG6000F series AI firewalls also support to detect unknown threats by interworking with sandbox.

Traditional IPS signatures are manually produced through analysis, resulting in low productivity. Also, the accuracy of the signatures depends heavily on expert experience. Huawei innovatively enables the IPS signature production on the intelligent cloud by adopting intelligence technologies and utilizing expert experience. Such an intelligent mode helps increase the signature productivity by 30 times compared with manual production, reduce errors caused by manual analysis, and continuously improve the accuracy of intrusion detection.

The built-in antivirus content-based detection engine (CDE) powered by intelligence technologies can detect unknown threats and provide in-depth data analysis. With these capabilities, the CDE-boosted firewall is able to gain insight into threat activities and quickly detect malicious files, effectively improving the threat detection rate.

USG supports to detect and defend malware spreading and network attacks, like Worm, Virus, Trojan-horse, Spyware, etc. malware spreading and botnet, DoS/DDoS, SQL injection, cross site attack, ransomware, etc.

Simplified O&M

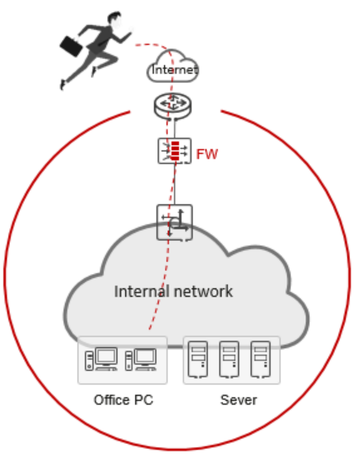
The HiSecEngine USG6500F series AI firewalls provides a brand-new web UI, which intuitively visualizes threats as well as displays key information such as device status, alarms, traffic, and threat events. With multi-dimensional data drilling, the web UI offers optimal user experience, enhanced usability, and simplified O&M.

The HiSecEngine USG6500F series AI firewalls can be centrally managed by the security management platform SecoManager, implementing a shift from single-point defense to collaborative network protection. The SecoManager provides policy tuning and intelligent O&M capabilities. It can also manage security products, such as anti-DDoS devices to quickly eliminate network threats and improve security handling effectiveness.

The HiSecEngine USG6500F series AI firewalls can also be managed by NCE-Campus, and NCE-Campus can also support to manage switch, AR, POL device at the same time, even third party devices.

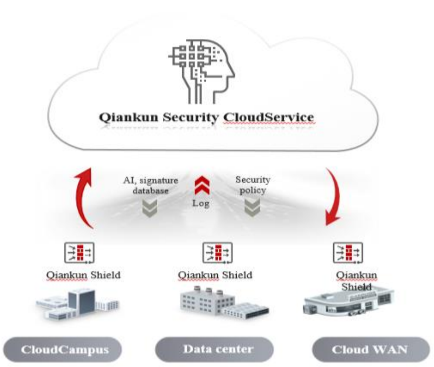
A wide range of network features

Huawei HiSecEngine USG6500F series AI firewalls provide various network features such as VPN, IPv6, and intelligent traffic steering.



- Provides various VPN features such as IPsec VPN and SSL VPN, and supports multiple encryption algorithms, such as DES, 3DES, AES, and SHA, ensuring secure and reliable data transmission.
- Provides secure and rich IPv6 network switchover, policy control, security protection, and service visualization capabilities, helping government, media, carrier, Internet, and finance sectors implement IPv6 reconstruction.
- Provides dynamic and static intelligent traffic steering based on multi-egress links, selects the outbound interface based on the specified link bandwidth, weight, or priority, forwards traffic to each link based on the specified traffic steering mode, and dynamically tunes the link selection result in real time to maximize the usage of link resources and improve user experience.
- Most threats and attacks come from network traffic. Firewalls are deployed at the egress of the local network to interwork with Huawei Qiankun security cloud service to implement automatic threat analysis and handling. This ensures the interconnection between the intranet and extranet, effectively intercepts traffic attacks, and automatically handles external attack sources. Protects enterprise network resources.

Collaboration with Huawei Qiankun Security Cloud Service



- Most threats and attacks come from network traffic. Firewalls are deployed at the egress of the local network to interwork with Huawei Qiankun security cloud service to implement automatic threat analysis and handling. This ensures the interconnection between the intranet and extranet, effectively intercepts traffic attacks, and automatically handles external attack sources. Protects enterprise network resources.
- By associating with Huawei Qiankun security cloud service, the firewall can obtain security services such as border protection and response on demand. Lightweight deployment and unified cloud O&M effectively reduce hardware stacking and greatly reduce enterprise security investment and O&M difficulties.

02 Deployment

Small data center border protection

- Firewalls are deployed at egresses of data centers, and functions and system resources can be virtualized. The firewall has multiple types of interfaces, such as 10G (SFP+), GE (RJ45) and GE(SFP) interfaces. Services can be flexibly expanded without extra interface cards.
- The intrusion prevention capability effectively blocks a variety of malicious attacks and delivers differentiated defense based on virtual environment requirements to guarantee data security.
- VPN tunnels can be set up between firewalls and mobile workers and between firewalls and branch offices for secure and low-cost remote access and mobile working.

Enterprise border protection

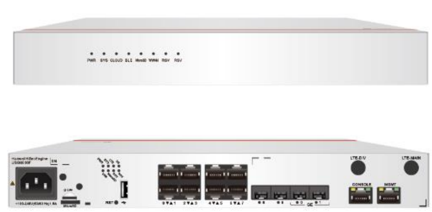
- Firewalls are deployed at the network border. The built-in traffic probe can extract packets of encrypted traffic to monitor threats in encrypted traffic in real time.
- The policy control and data filtering functions of the firewalls are used to monitor social network applications to prevent data breach and protect enterprise networks.

03 Product Appearance

- **Rich access capability :** Ethernet, LTE/5G RU and GPON.
- **Figure**

HiSecEngine USG6510F-D/USG6530F-D

HiSecEngine USG6510F-DL/USG6530F-DL

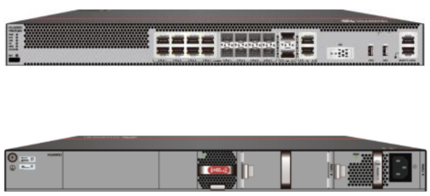


HiSecEngine USG6510F-DPL/ USG6530F-DPL

HiSecEngine USG6560F-D

23 - USG6555F(HTM)-AC Host(2*GE RJ45 + 8*GE COMBO + 2*10GE SFP+,1 AC power)

HiSecEngine USG6525F/USG6555F/USG6565F/USG6585F



HiSecEngine USG6585F-B

04 Software Features

Feature	Description
Integrated protection	Integrates firewall, VPN, intrusion prevention, antivirus, bandwidth management, Anti-DDoS, and URL filtering functions, and provides a global configuration view and integrated policy management.
Application identification and control	Application identification based on signatures, correlation, and behaviors instead of ports; 6000+ preset applications, which can be further classified; support for user-defined applications; 50+ categories and 20+ risk labels for access control based on categories and labels; automatic update of the application identification signature database.
Security policy management	Supports traffic management and control based on the VLAN ID, 5-tuple, security zone, region, application, and time range, and implements integrated content security inspection. Provides predefined templates for common attack defense scenarios to facilitate security policy deployment. Supports interworking with third-party policy management software (FireMon and Algosec) to facilitate security O&M.
Bandwidth management	Manages per-IP bandwidth based on service application identification to guarantee the network experience of key services and users. The management and control can be implemented by limiting the maximum bandwidth, guaranteeing the minimum bandwidth, and changing the application forwarding priority.

Intrusion prevention	Obtains the latest threat information in a timely manner and accurately detects and prevents vulnerability exploits; covers tens of thousands of CVE vulnerabilities; prevents the exploit of vulnerabilities (such as those in Windows and Unix/Linux operating systems, databases, Apache, IIS, and Tomcat as well as middleware), web attacks (such as SQL injection, XSS, and RCE), botnets, remote control, and Trojan horses; supports brute force cracking detection based on user behavior; provides 13,000+ predefined signatures and supports user-defined signatures and automatic signature database update; supports attack forensics collection, full-flow packet obtaining (including three-way handshake information), and attack fragment display to facilitate O&M; supports X-Forwarded-For (XFF) field extraction. The USG6500F-D series supports a maximum of 10000 IPS signatures.
Antivirus	Detects malware in files transmitted through protocols like HTTP, FTP, SMTP, POP3, IMAP4, NFS, and SMB; detects Trojan horses, worms, spyware, vulnerability exploits, adware, hacker tools, Rootkit, backdoors, grayware, botnet programs, ransomware, phishing software, cryptojacking software, and web shell programs; supports virus detection for Office files, executable files (Windows/Linux/MacOS), script files, flash files, PDF files, RTF files, web pages, and images; supports attack forensics collection; supports the inspection of archive files of up to 100 nested compression levels in multiple compression formats, such as tar, gzip, zip, rar, and 7z, and supports multiple actions, such as alert, block, add declaration, and attachment deletion.
Advanced malware prevention	The heuristic antivirus engine uses detection technologies such as AI, semantic analysis, and Emulator, coupled with threat and reputation information, to detect packed malware, script morphing, and malware embedded in compound documents. It can detect billions of malware variants and supports automatic update of the signature database. In addition, it can send suspicious files to the local or cloud sandbox for further inspection to detect zero-day malware.
Web security	The URL category database on the cloud contains 560 million URLs in over 130 categories, such as news, games, gambling, drugs, and malicious web pages. URLs cover over 100 languages, and key categories of URLs cover over 20 languages. The URL category query servers are deployed in multiple countries/regions to provide high-speed and low-latency category query services. User-defined URL/host whitelist and blacklist are supported. HTTPS traffic can be filtered without decryption. TLS/SSL traffic can be decrypted before filtering. HTTP/2 and QUIC traffic can be filtered, and URL categories can be imported in batches. Supports Safe Search enforcement across five major search engines: YouTube, Bing, Google, Yahoo, and Yandex, with mandatory filtering of illegal or inappropriate content in search results. URL access can be controlled based on users/user groups, time ranges, and security zones to precisely manage users' online behaviors.
DNS security	Based on massive threat information, technologies such as AI and knowledge graph are used to detect malicious DNS requests, including C&C domain names, DGA-generated domain names, compromised sites, and malicious domain names such as cryptojacking, ransomware, and phishing domain names. The local malicious domain name database supports a maximum of 500,000 malicious domain names. DNS category-based filtering, DNS safe search, and DNS redirection (sinkhole) are also supported.
Anti-botnet/spyware	Supports the detection and prevention of viruses and advanced malware, such as botnets, Trojan horses, worms, remote control tools, and spyware, and prevents the download of malware; quickly detects malicious traffic like C&C based on signatures, IP addresses, and domain reputation information; displays the roles of communication parties in botnet attack logs.
Threat information	Huawei Intelligent Security Center leverages multiple AI algorithms and expert analysis to generate massive threat information about IP addresses, domain names, URLs, and files on a daily basis. The threat information is automatically synchronized to devices for threat detection to quickly block emerging attacks. In addition, it can interconnect with third-party threat information sources to enrich inspection rules.

OT/IoT security	Supports common industrial control protocols such as Modbus, S7, Profinet, and OPC, identification and control of IoT devices such as cameras, and IoT asset risk assessment. Supports vulnerability detection for IoT devices like cameras and industrial control software and protocols like ICS/SCADA. The traffic probe function, coupled with HiSec Insight situational awareness system, can learn the traffic behavior baseline of IoT assets and detect and evaluate IoT asset risks. 1.For details about the list of supported OT protocols, see https://isecurity.huawei.com/security/wiki/application (Business Systems > Industrial). 2. The USG6500F-D series does not support industrial control protocols. 3. Firewalls are deployed at Level 3.5 or above of the Purdue model.
Flow probe	Collects and parses metadata and attack forensics information, including network-layer metadata (such as IP addresses, ports, and packet characteristics) and application-layer metadata (such as field information obtained after in-depth parsing of protocols like HTTP, DNS, TLS, and SSH), and sends the data to the HiSec Insight security situational awareness platform for further analysis by using algorithms, such as deep learning and machine learning algorithms, to detect potential, unknown, and advanced threats in network traffic. The USG6500F-D series does not support Flow probe.
Anti-DDoS	Uses technologies such as source IP address detection, fingerprint detection, and dynamic traffic limiting to defend against over 10 common DDoS attacks and over 20 single-packet attacks, such as SYN flood, UDP flood, ICMP flood, HTTP flood, HTTPS flood, DNS flood, and SIP flood attacks, and supports traffic baseline learning and IP reputation-based filtering. The USG6500F-D series supports only single-packet attack defense.
Mail filtering	Supports mail address filtering (covering the sender and recipient addresses) and SMTP mail sending rate limiting.
DLP	Supports identification of 100+ real file types, user-defined file name extensions, and file type-based upload/download control; supports keyword filtering for Office documents, web pages, code, and TXT files; supports user-defined keywords, regular expressions, and weight configuration.
SaaS access control	Supports SaaS application identification and access control based on signatures, DNS, IP addresses (IP address database of the top 50 SaaS applications), and first packets, and supports traffic steering based on SaaS applications, ensuring good SaaS application experience.
Behavior audit	Audits and regulates common user online behaviors, including FTP operations (upload, download, and command), HTTP operations (posting, search, and browsing), DNS, Telnet, SNMP, and email sending and receiving operations.
Intelligent uplink selection	Supports service-specific PBR and intelligently selects the optimal link based on multiple types of load balancing criteria (such as the bandwidth ratio and link health status) in multi-ISP scenarios.
VPN encryption	Supports various highly reliable VPN features, such as IPsec VPN, SSL VPN, and GRE, and multiple encryption algorithms, such as DES, 3DES, AES, SHA, SM2, SM3, and SM4.
SSL-encrypted traffic inspection	Detects and defends against threats hidden in TLS/SSL-encrypted traffic, performs application-layer protection, such as intrusion prevention, antivirus, data filtering, and URL filtering, on decrypted TLS/SSL traffic, and supports URL category whitelist.
SSL offloading	Replaces the server to implement SSL encryption and decryption, reducing the server load and implementing load balancing of HTTP traffic.
Diversified reports	Provides visualized and multi-dimensional reports by IP address, application, time, traffic, or threat.
Security virtualization	Supports virtualization of multiple types of security services, including firewall, intrusion prevention, antivirus, and VPN services; allows users to separately conduct personalized management on the same physical device.
Routing	Supports multiple types of IPv4/IPv6 routing protocols, such as RIP, OSPF, BGP, IS-IS, RIPng, OSPFv3, BGP4+, and IPv6 IS-IS.

IP multicast	Supports IPv4 Layer 3 multicast protocols, such as IGMP, MSDP, and PIM, and provides point-to-multipoint services to reduce bandwidth consumption.
Deployment and reliability	Supports transparent (Layer 2), routing (Layer 3), tap, and hybrid working modes and high availability (HA), including the Active/Active and Active/Standby modes.
Server load balancing	Supports IPv6, Layer 4/Layer 7 server load balancing, and multiple session persistence methods such as source IP address-based and HTTP cookie-based session persistence; supports SSL offloading and encryption; combines services and security policies to improve service security; supports health check based on multiple protocols such as TCP, RADIUS, DNS, and HTTP to detect server status changes promptly.
Security center	The built-in asset identification module can identify assets such as Windows, Linux, Android, and iOS assets and cameras, perform correlation analysis on threat logs and assets, and display asset risk assessment results and the entire kill chain.
SRv6	Supports IS-IS for SRv6, BGP for SRv6, SRv6 BE, SRv6 TE policy, SRv6 midpoint protection, SRv6 microloop avoidance, SRv6 OAM, SRv6 SRH compression, SRv6 TI-LFA FRR, and EVPN L3VPN.
Secure SD-WAN	Provides a built-in secure SD-WAN solution for low-cost and business-level Internet links. Supports zero-touch provisioning (ZTP) through email to complete device provisioning in minutes without requiring technical skills. Supports forward error correction (FEC) to prevent pixelated display and video freezing at a 30% packet loss rate; supports real-time link switching based on link quality ensure key application experience. Supports multi-link routing and dual-CPE flexible networking to ensure uninterrupted connections for site services; supports E2E IPsec encryption to ensure secure service transmission.
User authentication	Supports multiple authentication modes for Internet access users, including local Portal authentication and single sign-on (SSO). In local Portal authentication, the built-in Portal page of the device can be pushed to users, and the account and password entered on the Portal page by a user can be sent to the local database or RADIUS, HWTACACS, AD, or LDAP authentication server for authentication. SSO includes RADIUS SSO and Agile Controller (NCE-Campus) SSO.
O&M capability	Supports telemetry to automatically read information from hardware, such as fans, power modules, optical modules, Ethernet ports, temperature sensors, and drivers, and sends interface traffic statistics, CPU usage, and memory usage to the collector.
PPPoE	Functions as a PPPoE client to provide Internet access services, including user authentication and authorization and dynamic IP address allocation.
Posture compliance check	Supports Operating System Version Check, Operating System Patch Check, Antivirus Software Check, Firewall Check, Running Process Check, File Security Check, Registry Check, Port Check, Anti-Screenshot, Anti Double Redirect, and Prevents Nested Remote Desktop Connections.

Firewall Interworking with Huawei Qiankun Security CloudService

Service	Description
Border protection and response service	Intrusion detection and prevention: Leverages signatures to block application-layer attacks, terminates the transfer of phishing emails and malware (such as viruses and Trojan horses), detects compromised hosts on the internal network, and disconnects these hosts from the Internet to protect the security and stability of customer services. Automatic event analysis: Combines intelligent analysis and manual analysis by experts to analyze security events to ensure the accuracy of attack blocking and security alarms and optimize attack identification rules. Whitelist and blacklist: Supports whitelist and blacklist configuration on the Portal or app to protect services from threats. Periodic security reports: Generates weekly and monthly reports on security protection events and sends the reports to users by email. Emergency notification: Identifies emergencies from security events and sends notifications to users via SMS and email.

05 Specifications

- **System Performance and Capacity**

Model	USG6510F-D	USG6530F-D	USG6560F-D	USG6510F-DL USG6510F-DPL	USG6530F-DL USG6530F-DPL
IPv4 Firewall Throughput[1] (1518/512/64-byte, UDP) [10]	6/6/3.6 Gbps	12/12/3.6 Gbps	12/12/3.6 Gbps	6/6/3.6 Gbps	12/12/3.6 Gbps
IPv6 Firewall Throughput[1] (1518/512/84-byte, UDP) [10]	6/6/3.6 Gbps	12/12/3.6 Gbps	12/12/3.6 Gbps	6/6/3.6 Gbps	12/12/3.6 Gbps
Secure SD-WAN Throughput[9] (1400/512 byte,UDP)[10]	5/5 Gbps	6/6 Gbps	6.6/6.6 Gbps	5/5 Gbps	6/6 Gbps
SD-WAN EVPN max tunnels	200	200	200	200	200
Firewall Throughput (Packets Per Second)	5.4 Mpps	5.4 Mpps	5.4 Mpps	5.4 Mpps	5.4 Mpps
Firewall Latency (64-byte, UDP)	18 µs	18 µs	18 µs	18 µs	18 µs
FW + SA* Throughput[2]	1.8 Gbps	2.2 Gbps	3 Gbps	1.8 Gbps	2.2 Gbps
NGFW Throughput (HTTP 100K)[3]	1.6 Gbps	1.8 Gbps	2.2Gbps	1.6 Gbps	1.8 Gbps
NGFW Throughput (Enterprise Mix)[4]	1 Gbps	1.2 Gbps	1.3 Gbps	1 Gbps	1.2 Gbps
Threat Protection Throughput (HTTP 100K)[7]	1.3 Gbps	1.5 Gbps	2 Gbps	1.3 Gbps	1.5 Gbps
Threat Protection Throughput (Enterprise Mix)[5]	800 Mbps	1 Gbps	1.2 Gbps	800 Mbps	1 Gbps
Concurrent Sessions	800,000	1,000,000	1,000,000	800,000	1,000,000
IPv6 Concurrent Sessions[1]	200,000	500,000	500,000	200,000	500,000

Model	USG6510F-D	USG6530F-D	USG6560F-D	USG6510F-DL USG6510F-DPL	USG6530F-DL USG6530F-DPL
New Sessions/Second (HTTP1.1)[1]	40,000/s	50,000	50,000	40,000/s	50,000
IPv6 New Sessions/Second (HTTP1.1)[1]	8000/s	30,000	30,000	8,000	30,000
IPsec VPN Throughput[1] (AES-256 + SHA256, 1420-byte)	2 Gbps	3.7 Gbps	3.7 Gbps	2 Gbps	3.7 Gbps
Maximum IPSec VPN Tunnels (GW to GW)	1,000	2,000	2,000	1,000	2,000
Maximum IPsec VPN Tunnels (Client to GW)	1,000	2,000	2,000	1,000	2,000
SSL Inspection Throughput[8]	400 Mbps	400 Mbps	550 Mbps	400 Mbps	400 Mbps
SSL VPN Throughput[6]	200 Mbps	300 Mbps	300 Mbps	200 Mbps	300 Mbps
Concurrent SSL VPN Users (Default/Maximum)	100/300	100/1000	100/1000	100/300	100/1000
Firewall Policies (Maximum)	3,000	3,000	3,000	3,000	3,000
Virtual Firewalls	10	20	20	10	20
URL Filtering: Categories	More than 130.				
URL Filtering: URLs	A database of over 560 million URLs in the cloud.				
Automated IPS Signature Updates	Yes, an industry-leading security center from Huawei (https://isecurity.huawei.com/security/service/ips).				
Third-Party and Open-Source Ecosystem	Open API for integration with third-party products, providing NETCONF interfaces. Other third-party management software based on SNMP, SSH, and Syslog.				
VLANs (Maximum)	4094				
VLANIF Interfaces (Maximum)	4094				

Model	USG6525F	USG6555F	USG6565F	USG6585F	USG6585F-B
IPv4 Firewall Throughput[1] (1518/512/64-byte, UDP)[10]	2.5/2.5/2.5 Gbps	5/5/3.6 Gbps	7/7/3.6 Gbps	9/9/4 Gbps	20/18/5 Gbps
IPv6 Firewall Throughput[1] (1518/512/84-byte, UDP) [10]	2.5/2.5/2.5 Gbps	5/5/3.6 Gbps	7/7/3.6 Gbps	9/9/4 Gbps	20/18/5 Gbps
Secure SD-WAN Throughput[9] (1400/512 byte,UDP) [10]	2.5/2.5 Gbps	5/5 Gbps	6/6 Gbps	9/6.6 Gbps	10/6.8 Gbps
SD-WAN EVPN max tunnels	200	200	200	200	200
Firewall Throughput (Packets Per Second)	3.75 Mpps	5.4 Mpps	5.4 Mpps	6 Mpps	7.5Mpps
Firewall Latency (64-byte, UDP)	18 μs	18 μs	18 μs	18 μs	15 μs
FW + SA* Throughput[2]	2.2 Gbps	3 Gbps	3 Gbps	3 Gbps	4.5Gbps
NGFW Throughput (HTTP 100K)[3]	1.8 Gbps	2.1 Gbps	2.2 Gbps	2.2 Gbps	3.3Gbps
NGFW Throughput (Enterprise Mix) [4]	1.2 Gbps	1.2 Gbps	1.2 Gbps	1.3 Gbps	2Gbps
Threat Protection Throughput (HTTP 100K)[7]	1.5 Gbps	1.8 Gbps	2 Gbps	2 Gbps	3Gbps
Threat Protection Throughput (Enterprise Mix)[5]	1 Gbps	1 Gbps	1.1 Gbps	1.2 Gbps	1.8Gbps
Concurrent Sessions	3,000,000	4,000,000	4,000,000	4,000,000	4,000,000
IPv6 Concurrent Sessions[1]	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000
New Sessions/Second (HTTP1.1)[1]	80,000	80,000	80,000	80,000	120,000
IPv6 New Sessions/Second (HTTP1.1)[1]	80,000	80,000	80,000	80,000	120,000
IPsec VPN Throughput[1] (AES-256 + SHA256, 1420-byte)	2.5 Gbps	3.7 Gbps	3.7 Gbps	3.7 Gbps	5.6 Gbps
Maximum IPSec VPN Tunnels (GW to GW)	4,000	4,000	4,000	4,000	4,000
Maximum IPsec VPN Tunnels (Client to GW)	4,000	4,000	4,000	4,000	4,000
SSL Inspection Throughput[8]	550 Mbps	550 Mbps	550 Mbps	550 Mbps	830 Mbps

Model	USG6525F	USG6555F	USG6565F	USG6585F	USG6585F-B
SSL VPN Throughput[6]	300 Mbps	500 Mbps	500 Mbps	500 Mbps	750 Mbps
Concurrent SSL VPN Users (Default/Maximum)	100/1000	100/2000	100/2000	100/2000	100/2000
Firewall Policies (Maximum)	15,000				
Virtual Firewalls	100				
URL Filtering: Categories	More than 130.				
URL Filtering: URLs	A database of over 560 million URLs in the cloud.				
Automated IPS Signature Updates	Yes, an industry-leading security center from Huawei (https://isecurity.huawei.com/security/service/ips).				
Third-Party and Open-Source Ecosystem	Open API for integration with third-party products, providing NETCONF interfaces. Other third-party management software based on SNMP, SSH, and Syslog.				
VLANs (Maximum)	4094				
VLANIF Interfaces (Maximum)	4094				

1. Performance is tested under ideal conditions based on RFC2544, 3511. The actual result may vary with deployment environments.
2. SA performances are measured using 100 KB HTTP files.
3. NGFW throughput is measured with Firewall, SA, and IPS enabled; the performance is measured using 100 KB HTTP files.
4. NGFW throughput is measured with Firewall, SA, and IPS enabled; the performance is measured using the Enterprise Mix Traffic Model.
5. The threat protection throughput is measured with Firewall, SA, IPS, and AV enabled; the performance is measured using the Enterprise Mix Traffic Model.
6. SSL VPN throughput is measured using TLS v1.2 with AES128-SHA.
7. The threat protection throughput is measured with Firewall, SA, IPS, and AV enabled, the performances are measured using 100 KB HTTP files.
8. SSL inspection throughput is measured with IPS-enabled and HTTPS traffic using TLS v1.2 with TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256.
9. The SD-WAN tunnel is packed with GRE over IPSec.
10. The USG6500F-D/DL/DPL series, as well as the USG6585F-B host bundle, include performance enhancement licenses. Activating the license enables the specified performance metrics.

*SA: indicates service awareness.

06 Hardware Specifications

Model	USG6510F-D	USG6530F-D	USG6510F-DL
Chassis Height	Desktop		
Dimensions (W x D x H) mm	250 x 210 x 43.6		320 x 220 x 43.6
Fixed Interface	10*GE RJ45 + 2*GE SFP	10*GE RJ45 + 2*10GE SFP+	4*GE SFP + 8*GE RJ45 + LTE
USB Port	1 x USB 2.0		
Weight	1.55 kg		2.34 kg
Hardware	Optional, 64 GB microSD card available for purchase		
Power Supply(AC)	100 V to 240 V, 50 Hz/60 Hz		
Maximum power consumption of the machine	23.67 W	22.5 W	34.1 W
Power Supplies	Single power supply		
Operating Environment (Temperature/Humidity)	Temperature: 0° C to 45° C Humidity: 5% to 95%, non-condensing		
Non-operating Environment	Temperature: –40° C to +70° C Humidity: 5% to 95%, non-condensing		

Model	USG6510F-DPL	USG6530F-DL	USG6530F-DPL	USG6560F-D
Chassis Height	Desktop			
Dimensions (W x D x H) mm	320 x 220 x 43.6			
Fixed Interface	4*GE SFP + 8*GE RJ45 + LTE（Support 4*PoE）	2*10GE SFP+ + 2*GE SFP + 8*GE RJ45 + LTE	2*10GE SFP+ + 2*GE SFP + 8*GE RJ45 + LTE（Support 4*PoE）	2*10GE SFP+ + 2*GE SFP+8*GE RJ45
USB Port	1 x USB 2.0			
Weight	2.6 kg	2.34 kg	2.6 kg	2.29 kg
Hardware	Optional, 64 GB microSD card available for purchase			
Power Supply(AC)	100 V to 240 V, 50 Hz/60 Hz			
Maximum power consumption of the machine	35.7 W	34.1 W	35.7 W	29.4W
Power Supplies	Single power supply			
Operating Environment (Temperature/Humidity)	Temperature: 0° C to 45° C Humidity: 5% to 95%, non-condensing			
Non-operating Environment	Temperature: –40° C to +70° C Humidity: 5% to 95%, non-condensing			

Model	USG6525F	USG6555F	USG6565F	USG6585F	USG6585F-B
Chassis Height	1 U				
Dimensions (W x D x H) mm	442 × 420 × 43.6				
Fixed Interface	2*GE RJ45 + 8*GE COMBO + 2*10GE SFP+				16*GE RJ45 + 8*GE COMBO + 2*10GE SFP+
USB Port	2 x USB 2.0				1 × USB 2.0
Weight	5.46 kg				5.816 kg
Hardware	Optional, M.2 SSD (64 GB/240 GB/960GB), hot-swappable				
Power Supply	100 V to 240 V, 50 Hz/60 Hz				
Maximum power consumption of the machine	36.8 W				53.2 W
Power Supplies	Optional dual power modules for 1+1 redundancy				
Operating Environment	Temperature: 0° C to 45° C Humidity: 5% to 95%, non-condensing				
Storage environment	Temperature: –40° C to +70° C Humidity: 5% to 95%, non-condensing				

07 Ordering Information

- **Note：**
  - **The ordering information of USG6510F-DL/USG6530F-D/USG6530F-DL/ USG6560F-D is the same as USG6510F-D;**
  - **The ordering information of USG6530F-DPL is the same as USG6530F-DPL;**
  - **The license information of USG6555F/USG6565F/USG6585F/USG6585F-B is the same as USG6525F;**
  - **The four models USG6510F-D/USG6530F-D/USG6555F/USG6585F support Qiankun Security Cloud Service;**
  - **Some parts of this table list the sales strategies in different regions. For more information, please contact your Huawei representative.**

Product	Model	Description
USG6510F-D	USG6510F-D-AC	USG6510F-D AC host (10*GE RJ45+2*GE SFP,1*Adapter)
USG6510F-DPL	USG6510F-DPL-AC	USG6510F-DPL AC Host(4*GE SFP+8*GE RJ45+ LTE, 1*Adapter, include SSL VPN 100 users); Supports PoE/PoE+/PoE++, Maximum power supply capability:150W.
USG6525F	USG6525F-AC	USG6525F AC host (2*GE RJ45 + 8*GE COMBO + 2*10GE SFP+, 1 AC power)
	USG6525F-DC	USG6525F DC host (2*GE RJ45 + 8*GE COMBO + 2*10GE SFP+)
Function License		
SSL VPN	LIC-USG6KF-SSLVPN-100	Quantity of SSL VPN Concurrent Users (100 Users)
	LIC-USG6KF-SSLVPN-200	Quantity of SSL VPN Concurrent Users (200 Users)
	LIC-USG6KF-SSLVPN-500	Quantity of SSL VPN Concurrent Users (500 Users)
	LIC-USG6KF-SSLVPN-1000	Quantity of SSL VPN Concurrent Users (1000 Users)
	LIC-USG6KF-SSLVPN-2000	Quantity of SSL VPN Concurrent Users (2000 Users)
	LIC-USG6KF-SSLVPN-5000	Quantity of SSL VPN Concurrent Users (5000 Users)
NGFW License		
IPS Update Service	LIC-USG6525F-IPS-1Y	IPS Update Service Subscribe Per Year (Applies to USG6525F)
URL Filtering Update Service	LIC-USG6525F-URL-1Y	URL Update Service Subscribe Per Year (Applies to USG6525F)
Antivirus Update Service	LIC-USG6525F-AV-1Y	AV Update Service Subscribe Per Year (Applies to USG6525F)
Threat Protection Bundle (IPS, AV, URL)	LIC- USG6510F-D -TP-1Y	Threat Protection Subscription Per Year (Applies to USG6510F-D Overseas)
	LIC-USG6510F-DPL-TPU-1Y	Threat Protection Database Upgrade Service (Applies to USG6510F-D), Per Device, Per Year
	LIC-USG6525F-TP-1Y	Threat Protection Subscription Per Year (Applies to USG6525F Overseas)
IPv6+	LIC-6500F-IPv6+-LIC	IPv6+ Feature (includes SRv6, channel subinterface, iFit) (Applies to USG6500F)
Industrial Protocol	LIC-USG6525F-ICS-1Y	Industrial Control Security Service Subscribe Per Year (Applies to USG6525F)
N1 License		
USG6510F-D	N1- USG6510F-D -F-Lic	N1-USG6510F-D Foundation, Per Device
	N1- USG6510F-D -F-SnS1Y	N1-USG6510F-D Foundation, SnS, Per Device, Per Year
USG6510F-DPL	N1-USG6510F-DPL-F-Lic	N1-USG6510F-DPL Foundation, Per Device
	N1-USG6510F-DPL-A-Lic	N1-USG6510F-DPL Advanced, Per Device
	N1-USG6510F-DPL-F-SnS1AY	N1-USG6510F-DPL Foundation, SnS, Per Device, Per Year
	N1-USG6510F-DPL-A-SnS1AY	N1-USG6510F-DPL Advanced, SnS, Per Device, Per Year
USG6525F	N1-USG6525F-F-Lic	N1-USG6525F Foundation, Per Device
	N1-USG6525F-F-SnS1Y	N1-USG6525F Foundation, SnS, Per Device, Per Year
	N1-USG6525F-A-Lic	N1-USG6525F Advanced, Per Device
	N1-USG6525F-A-SnS1Y	N1-USG6525F Advanced, SnS, Per Device, Per Year
QianKun Cloud Deployment License		
USG6510F-D	N1-C-USG6510F-D-F-Lic	Cloud Deployment Model Foundation, Per Device,1 Year
	LIC-USG6510F-D-BA-1Y	Border Protection and Response - Threat automatic blocking (Applies to USG6510F-D), Per Device, 1 Year
	LIC-USG6510F-D-TPU-1Y	Threat Protection Database Upgrade Service (Applies to USG6510F-D), Per Device, 1 Year
USG6510F-DPL	N1-C-USG6510F-DPL-F-Lic	Cloud Deployment Model Foundation, Per Device,1 Year
	LIC-USG6510F-DPL-BA-1Y	Border Protection and Response - Threat automatic blocking (Applies to USG6525F), Per Device, 1 Year
	LIC-USG6510F-DPL-TPU-1Y	Threat Protection Database Upgrade Service (Applies to USG6525F), Per Device, 1 Year

QianKun Cloud Deployment License		
USG6525F	N1-C-USG6525F-F-Lic	Cloud Deployment Model Foundation, Per Device,1 Year
	LIC-USG6525F-BA-1Y	Border Protection and Response - Threat automatic blocking (Applies to USG6525F), Per Device, 1 Year
	LIC-USG6525F-TPU-1Y	Threat Protection Database Upgrade Service (Applies to USG6525F), Per Device, 1 Year
Qiankun OP mode		
USG6510F-D	LIC-USG6510F-D-TPU-1Y	Border Protection and Response - Threat automatic blocking (Applies to USG6510F-D), Per Device, 1 Year
USG6510F-DPL	LIC-USG6510F-DPL-TPU-1Y	Threat Protection Database Upgrade Service (Applies to USG6510F-DPL), Per Device, 1 Year
USG6525F	LIC-USG6525F-TPU-1Y	Threat Protection Database Upgrade Service (Applies to USG6525F), Per Device, 1 Year
N1 SASE Branch Interconnection license		
USG6510F-D	N1-USG6510F-D-S-S-Lic	N1 SASE Branch Interconnection Standard Package (Package for USG6510F-D)
	N1-USG6510F-D-S-S-S1Y	N1 SASE Branch Interconnection Standard Package (Package for USG6510F-D),Per Device,1 Year
USG6510F-DPL	N1-USG6510F-DPL-S-S-Lic	N1 SASE Branch Interconnection Standard Package(Package for USG6510F-DPL)
	N1-USG6510F-DPL-S-S-S1Y	N1 SASE Branch Interconnection Standard Package(Package for USG6510F-DPL),Per Device,1 Year
USG6525F	N1-USG6525F-S-S-Lic	N1 SASE Branch Interconnection Standard Package(Package for USG6525F)
	N1-USG6525F-S-S-S1Y	N1 SASE Branch Interconnection Standard Package(Package for USG6525F),Per Device,1 Year

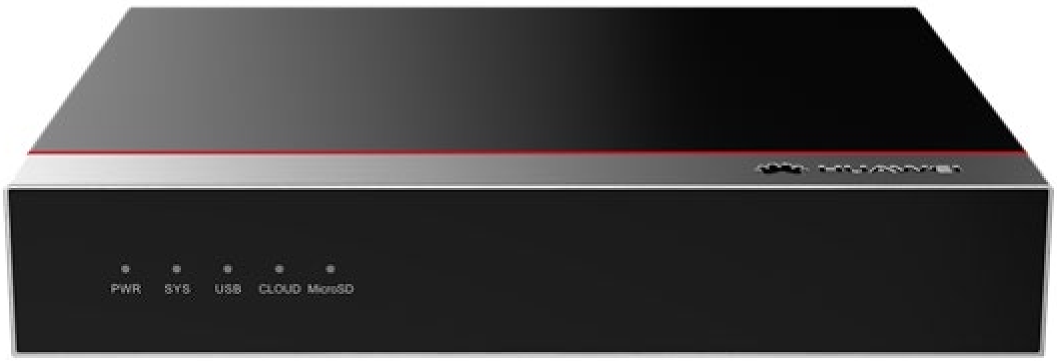
30 - AC6508 mainframe (10*GE ports, 2*10GE SFP+ ports, with the AC/DC adapter)

Huawei AC6508 Wireless Access Controller Datasheet

Product Overview

The AC6508 is a small-capacity box wireless access controller (AC) for small and medium enterprises. It can manage up to 512 access points (APs) and integrates the GE Ethernet switch function, achieving integrated access for wired and wireless users. The WLAN AC features high scalability and offers users considerable flexibility in configuring the number of managed APs. When used with Huawei's full series 802.11ax, 802.11ac,802.11n and 802.11be APs, the AC6508 can be used to construct small and medium campus networks, enterprise office networks, wireless Metropolitan Area Networks (MANs), and hotspot coverage networks.

Huawei AC6508 wireless access controller



Product Features

Large-capacity and high-performance design

- The AC6508 can manage up to 512 APs, meeting requirements of small and medium campuses.
- Provides 2 x 10GE optical interfaces and 10 x GE electrical interfaces, supporting up to 10 Gbit/s forwarding performance.

SmartRadio for air interface optimization

- Load balancing during smart roaming: The load balancing algorithm can work during smart roaming for load balancing detection among APs on the network after STA roaming to adjust the STA load on each AP, improving network stability.
- Intelligent DFA technology: The dynamic frequency assignment (DFA) algorithm is used to automatically detect adjacent-channel and co-channel interference, and identify any 2.4 GHz redundant radio. Through automatic inter-AP negotiation, the redundant radio is automatically switched to another mode (dual-5G AP models support 2.4G-to-5G switchover) or is disabled to reduce 2.4 GHz co-channel interference and increase the system capacity.
- Intelligent conflict optimization technology: The dynamic enhanced distributed channel access (EDCA) and airtime scheduling algorithms are used to schedule the channel occupation time and service priority of each user. This ensures that each user is assigned relatively equal time for using channel resources and user services are scheduled in an orderly manner, improving service processing efficiency and user experience.

Various roles

- The AC6508 has a built-in Portal/AAA server and can provide Portal/802.1X authentication for users, reducing customer investment.

Flexible networking

- The WLAN AC can be deployed in inline, bypass, bridge, and Mesh network modes, and supports both centralized and local forwarding.
- The WLAN AC and APs can be connected across a Layer 2 or Layer 3 network. In addition, NAT can be deployed when APs are deployed on the private network and the WLAN AC is deployed on the public network.
- The WLAN AC is compatible with Huawei full-series 802.11n, 802.11ac, 802.11ax and 802.11be APs and supports hybrid networking of 802.11n, 802.11ac, 802.11ax and 802.11be APs for simple scalability.

Built-in application identification server

- Supports Layer 4 to Layer 7 application identification and can identify over 6000 applications, including common office applications and P2P download applications, such as Lync, FaceTime, YouTube, and Facebook.
- Supports application-based policy control technologies, including traffic blocking, traffic limit, and priority adjustment policies.
- Supports automatic application expansion in the application signature database.

Comprehensive reliability design

- Supports AC 1+1 HSB, and N+1 backup, ensuring uninterrupted services.
- Supports port backup based on the Link Aggregation Control Protocol (LACP) or Multiple Spanning Tree Protocol (MSTP).
- Supports WAN authentication escape between APs and WLAN ACs. In local forwarding mode, this feature retains the online state of existing STAs and allows access of new STAs when APs are disconnected from WLAN ACs, ensuring service continuity.

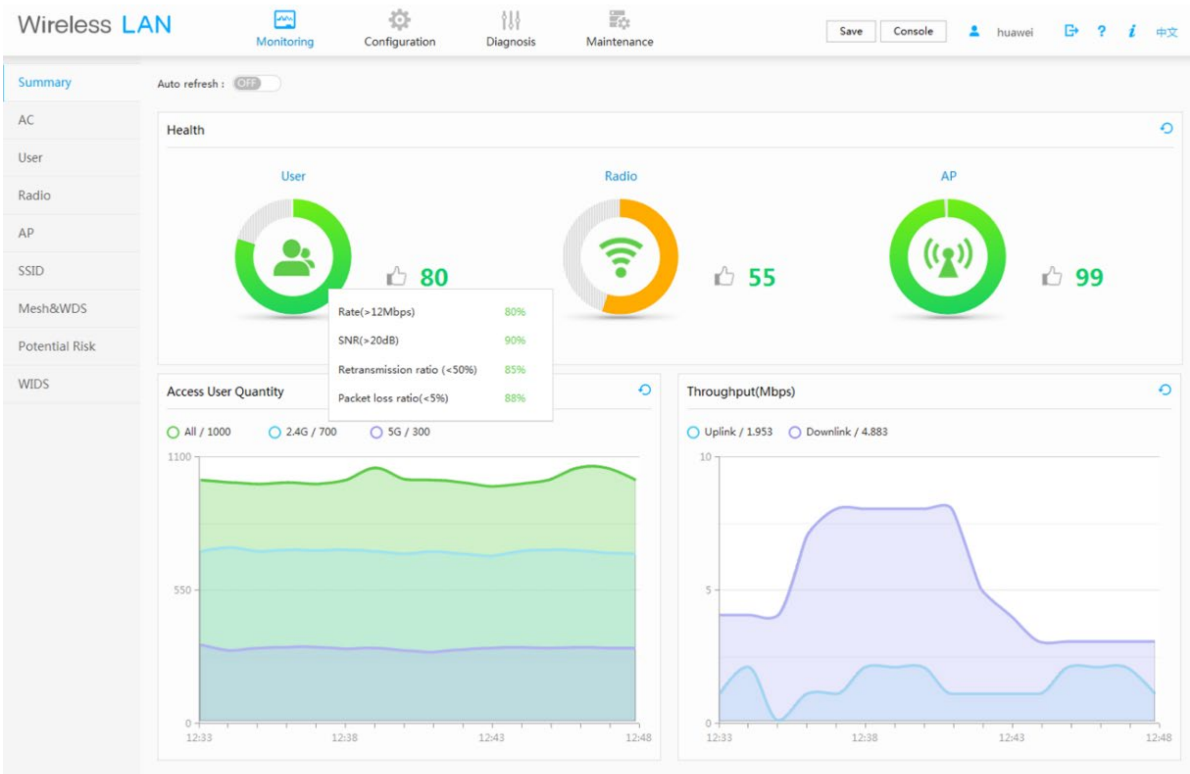
Built-in visualized network management platform

The AC6508 has a built-in web system that is easy to configure and provides comprehensive monitoring and intelligent diagnosis.

Health-centric one-page monitoring, visualized KPIs

- One page integrates the summary and real-time statistics. KPIs are displayed in graphs, including user performance, radio performance, and AP performance, enabling users to extract useful information from the massive amounts of monitored data, while also knowing the device and network status instantly.

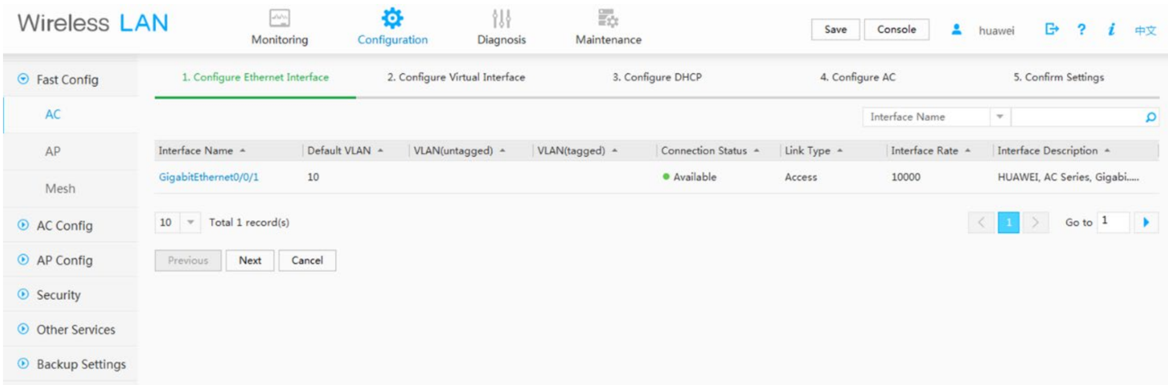
Monitoring interface



Profile-based configuration by AP group simplifies configuration procedure and improves efficiency.

- The web system supports AP group-centric configuration and automatically selects the common parameters for users, meaning that users do not need to pre-configure the common parameters, simplifying the configuration procedure.
- If two AP groups have small configuration differences, users can copy the configurations of one AP group to the other. This improves configuration efficiency because users only need to modify the original configurations, not create entirely new ones each time.

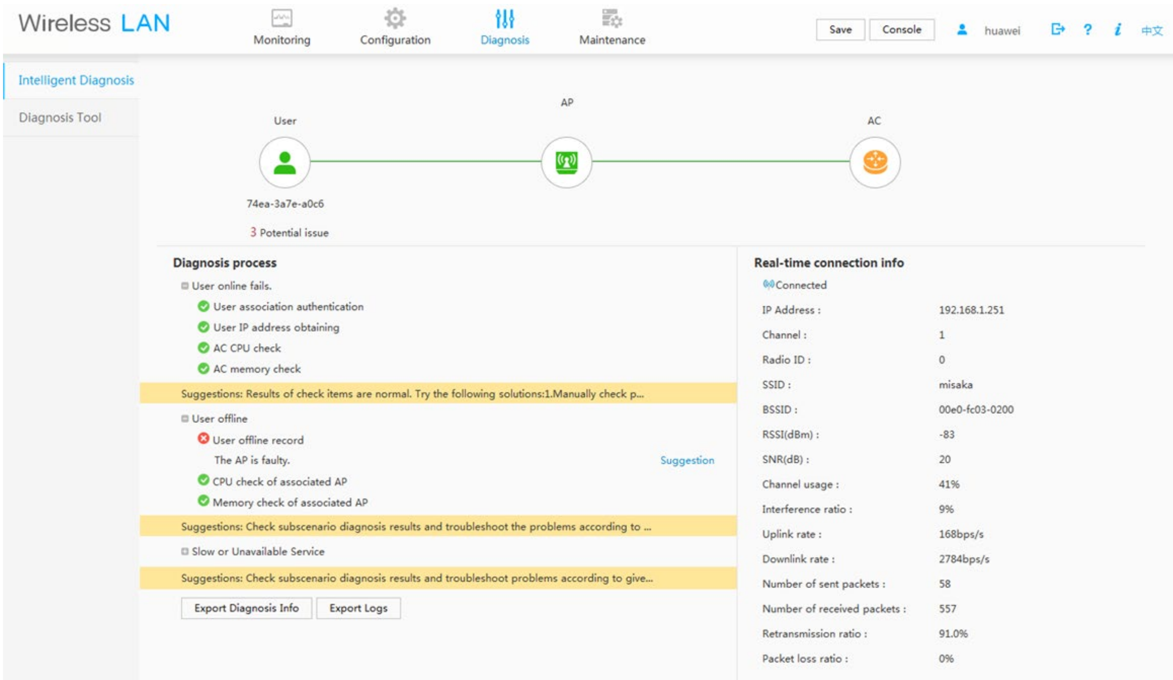
Configuration interface



One-click diagnosis solves 80% of common network problems.

- The web system supports real-time and periodic one-click intelligent diagnosis from the dimensions of users, APs, and WLAN ACs, and provides feasible suggestions for troubleshooting the faults.

Intelligent diagnosis



AC6508 Features

Switching and forwarding features

Feature		Description
Ethernet features	Ethernet	Operating modes of full duplex, half duplex, and auto-negotiation Rates of an Ethernet interface: 10 Mbit/s, 100 Mbit/s, 1000 Mbit/s, and auto-negotiation • Flow control on interfaces • Jumbo frames • Link aggregation • Load balancing among links of a trunk • Interface isolation and forwarding restriction • Broadcast storm suppression

Feature		Description
		• 802.3az Energy Efficient Ethernet (EEE)
	VLAN	Access modes of access, trunk, and hybrid Default VLAN VLAN pool
	MAC	Automatic learning and aging of MAC addresses Static, dynamic, and blackhole MAC address entries Packet filtering based on source MAC addresses Interface-based MAC learning limiting
	ARP	Static and dynamic ARP entries ARP in a VLAN Aging of ARP entries
	LLDP	LLDP
Ethernet loop protection	MSTP	STP RSTP MSTP BPDU protection, root protection, and loop protection Partitioned STP
IPv4 forwarding	IPv4 features	ARP and RARP ARP proxy Auto-detection NAT Bonjour protocol
	Unicast routing features	Static route RIP-1 and RIP-2 OSPF BGP IS-IS Routing policies and policy-based routing URPF check DHCP server and relay DHCP snooping
	Multicast routing features	IGMPv1, IGMPv2, and IGMPv3 PIM-SM Multicast routing policies RPF
IPv6 forwarding	IPv6 features	ND protocol
	Unicast routing features	Static route RIPng OSPFv3 BGP4+ IS-IS IPv6 DHCPv6

Feature		Description
		DHCPv6 snooping
	Multicast routing features	MLD MLD snooping
Device reliability	BFD	BFD
Layer 2 multicast features	Layer 2 multicast	IGMP snooping Prompt leave Multicast traffic control Inter-VLAN multicast replication
Ethernet OAM	EFM OAM	Neighbor discovery Link monitoring Fault notification Remote loopback
QoS features	Traffic classification	Traffic classification based on the combination of the L2 protocol header, IP 5-tuple, and 802.1p priority
	Action	Access control after traffic classification Traffic policing based on traffic classification Re-marking packets based on traffic classifiers Class-based packet queuing Associating traffic classifiers with traffic behaviors
	Queue scheduling	PQ scheduling DRR scheduling PQ+DRR scheduling WRR scheduling PQ+WRR scheduling
	Congestion avoidance	SRED WRED
	Application control	Smart Application Control (SAC)
Configuration and maintenance	Terminal service	Configurations using command lines Error message and help information in English Login through console and Telnet terminals Send function and data communications between terminal users
	File system	File systems Directory and file management File uploading and downloading using FTP and TFTP
	Debugging and maintenance	Unified management over logs, alarms, and debugging information Electronic labels User operation logs Detailed debugging information for network fault diagnosis Network test tools such as traceroute and ping commands Intelligent diagnosis Interface mirroring and flow mirroring

Feature		Description
	Version upgrade	Device software loading and online software loading BIOS online upgrade In-service patching
Security and management	Network management	ICMP-based ping and traceroute SNMPv1, SNMPv2c, and SNMPv3 Standard MIB RMON NetStream Packet Conservation Algorithm for Internet 2.0 (iPCA 2.0) ,flow detection based on applications, 5-tuple, and flows
	System security	Different user levels for commands, preventing unauthorized users from accessing device SSHv2.0 RADIUS and HWTACACS authentication for login users ACL filtering DHCP packet filtering (with the Option 82 field) Local attack defense function that can protect the CPU and ensure that the CPU can process services Defense against control packet attacks Defenses against attacks such as source address spoofing, Land, SYN flood (TCP SYN), Smurf, ping flood (ICMP echo), Teardrop, broadcast flood, and Ping of Death attacks IPSec URL filtering Antivirus Intrusion prevention

Wireless networking capabilities

Feature	Description
Networking between APs and WLAN ACs	APs and WLAN ACs can be connected through a Layer 2 or Layer 3 network. APs can be directly connected to a WLAN AC. APs are deployed on a private network, while WLAN ACs are deployed on the public network to implement NAT traversal. WLAN ACs can be used for Layer 2 bridge forwarding or Layer 3 routing. WAN authentication escape is supported between APs and WLAN ACs. In local forwarding mode, this feature retains the online state of existing STAs and allows access of new STAs when APs are disconnected from WLAN ACs, ensuring service continuity.
Forwarding mode	Direct forwarding (distributed forwarding or local forwarding) Tunnel forwarding (centralized forwarding) Centralized authentication and distributed forwarding In direct forwarding mode, user authentication packets support tunnel forwarding. Soft GRE forwarding.

Feature	Description
	Tunnel forwarding + EoGRE tunnel
WLAN AC discovery	An AP can obtain the device's IP address in any of the following ways: • Static configuration • DHCP • DNS The WLAN AC uses DHCP or DHCPv6 to allocate IP addresses to APs. DHCP or DHCPv6 relay is supported. On a Layer 2 network, APs can discover the WLAN AC by sending broadcast CAPWAP packets.
Wireless networking mode	WDS bridging: • Point-to-point (P2P) wireless bridging • Point-to-multipoint (P2MP) wireless bridging • Automatic topology detection and loop prevention (STP) Wireless mesh network • Access authentication for mesh devices • Mesh routing algorithm • Go-online without configuration • Mesh network with multiple MPPs • Vehicle-ground fast link handover • Mesh client mode
CAPWAP tunnel	Centralized CAPWAP CAPWAP control tunnel and data tunnel (optional) CAPWAP tunnel forwarding and direct forwarding in an extended service set (ESS) Datagram Transport Layer Security (DTLS) encryption, which is enabled by default for the CAPWAP control tunnel Heartbeat detection and tunnel reconnection
Active and standby WLAN ACs	Enables and disables the switchback function. Supports load balancing. Supports 1+1 hot backup. **NOTE** *In 1+1 VRRP HSB mode, WLAN ACs share one virtual IP address, simplifying the network topology.* Supports N+1 backup. Supports wireless configuration synchronization between WLAN ACs.

AP management

Feature	Description
AP access control	Displays MAC addresses or SNs of APs in the whitelist. Adds a single AP or multiple APs (by specifying a range of MAC addresses or SNs) to the whitelist. Automatically discovering and manually confirming APs. Automatically discovering APs without manually confirming them.

Feature	Description
AP profile management	Specifies the default AP profile that is applied to automatically discovered APs.
AP group management	The AP group function is used to configure multiple APs in batches. When multiple APs managed by a WLAN AC require the same configurations, you can add these APs to one AP group and configure the AP group to complete AP configuration.
AP region management	Supports three AP region deployment modes: • Distributed deployment: APs are deployed independently. An AP is equivalent to a region and does not interfere with other APs. APs work at the maximum power and do not perform radio calibration. • Common deployment: APs are loosely deployed. The transmit power of each radio is less than 50% of the maximum transmit power. • Centralized deployment: APs are densely deployed. The transmit power of each radio is less than 25% of the maximum transmit power. Specifies the default region to which automatically discovered APs are added.
AP type management	Manages AP attributes including the number of interfaces, AP types, number of radios, radio types, maximum number of virtual access points (VAPs), maximum number of associated users, and radio gain (for APs deployed indoors). Provides default AP types.
Network topology management	Supports LLDP topology detection.
AP working mode management	Supports AP working mode switchover. The AP working mode can be switched to the Fat or cloud mode on the AC.

Radio management

Feature	Description
Radio profile management	The following parameters can be configured in a radio profile: • Radio working mode and rate • Automatic or manual channel and power adjustment mode • Radio calibration interval • The radio type can be set to 802.11b, 802.11b/g, 802.11b/g/n, 802.11g, 802.11n, 802.11g/n, 802.11a, 802.11a/n, 802.11ac, 802.11ax, or 802.11be. You can bind a radio to a specified radio profile. Supports MU-MIMO.
Unified static configuration of parameters	Radio parameters such as the channel and power of each radio are configured on the WLAN AC and then delivered to APs.
Dynamic management	APs can automatically select working channels and power when they go online. In an AP region, APs automatically adjust working channels and power in the event of signal interference: • Partial calibration: The optimal working channel and power of a specified AP can be adjusted. • Global calibration: The optimal working channels and power of all the APs in a specified region can be adjusted. When an AP is removed or goes offline, the WLAN AC increases the power of neighboring APs to compensate for the coverage hole. Automatic selection and calibration of radio parameters in AP regions are supported.
Enhanced service capabilities	Band steering: Enables terminals to preferentially access the 5G frequency band, achieving

Feature	Description
	load balancing between the 2.4G and 5G frequency bands. Smart roaming: Enables sticky terminals to roam to APs with better signals. • 802.11k and 802.11v smart roaming • 802.11r fast roaming (≤ 50 ms)

WLAN service management

Feature	Description
ESS management	Allows you to enable SSID broadcast, set the maximum number of access users, and set the association aging time in an ESS. Isolates APs at Layer 2 in an ESS. Maps an ESS to a service VLAN. Associates an ESS with a security profile or a QoS profile. Enables IGMP for APs in an ESS. Supports Chinese SSIDs.
VAP-based service management	Adds multiple VAPs at a time by binding radios to ESSs. Displays information about a single VAP, VAPs with a specified ESS, or all VAPs. Supports configuration of offline APs. Creates VAPs according to batch delivered service provisioning rules in automatic AP discovery mode.
Service provisioning management	Supports service provisioning rules configured for a specified radio of a specified AP type. Adds automatically discovered APs to the default AP region. The default AP region is configurable. Applies a service provisioning rule to a region to enable APs in the region to go online.
Multicast service management	Supports IGMP snooping. Supports IGMP proxy.

Load balancing	Performs load balancing among radios in a load balancing group.
	• Supports two load balancing modes:
	  – Based on the number of STAs connected to each radio
	  – Based on the traffic volume on each radio

Bring Your Own Device (BYOD)	• Identifies device types according to the OUI in the MAC address. • Identifies device types according to the user agent (UA) field in an HTTP packet. • Identifies device types according to DHCP Option information. • Carries device type information in RADIUS authentication and accounting packets.
Location services	Locates AeroScout and Ekahau tags. Locates Wi-Fi terminals. Locates Bluetooth terminals. Locates Bluetooth tags.
Spectrum analysis	Identifies the following interference sources: Bluetooth, microwave ovens, cordless phones, ZigBee, game controller, 2.4 GHz/5 GHz wireless audio and video devices, and baby monitors. Works with the eSight to display spectrums of interference sources.
Rogue device monitoring	Supports WIDS/WIPS attack detection to monitor, identify, prevent, and take

Feature	Description
	countermeasures against rogue devices and implement refined management and control.
Hotspot2.0	Supports a Hotspot2.0 network.
Internet of Things (IoT)	Supports IoT cards on the AP to converge the WLAN and IoT.
Navi WLAN AC	Supports remote STA access on the Navi WLAN AC.
Centralized license control	Supports a license server as the centralized AP license control point. Allows a license server to manage license clients. Supports license synchronization between a license server and clients.

WLAN user management

Feature	Description
Address allocation of wireless users	Functions as a DHCP server to assign IP addresses to wireless users.
WLAN user management	Supports user blacklist and whitelist. Controls the number of access users: • Based on APs • Based on SSIDs Logs out users in any of the following ways: • Using RADIUS DM messages • Using commands Supports various methods to view information: • Allows you to view the user status by specifying the user MAC address, AP ID, radio ID, or WLAN ID. • Displays the number of online users in an ESS, AP, or radio. • Collects packet statistics on air interface based on user.
WLAN user roaming	Supports intra-AC Layer 2 roaming. **NOTE** *Users can roam between APs connected to different physical ports on a WLAN AC.* Supports inter-VLAN Layer 3 roaming on a WLAN AC. Supports roaming between WLAN ACs. Supports fast key negotiation in 802.1X authentication. Authenticates users who request to reassociate with the WLAN AC and rejects the requests of unauthorized users. Delays clearing user information after a user goes offline so that the user can rapidly go online again.
User group management	Supports ACLs. Supports user isolation: • Inter-group isolation • Intra-group isolation

WLAN security

Feature	Description

Feature	Description
WLAN security profile management	Manages authentication and encryption modes using WLAN security profiles.
Authentication modes	Open system authentication with no encryption WEP authentication/encryption WPA/WPA2/WPA3 authentication and encryption: • WPA/WPA2-PSK+TKIP • WPA/WPA2-PSK+CCMP • WPA/WPA2-802.1X+TKIP • WPA/WPA2-802.1X+CCMP • WPA3-802.1X+GCMP512 • WPA/WPA2-PSK+TKIP-CCMP • WPA/WPA2-802.1X+TKIP-CCMP WPA/WPA2-PPSK authentication and encryption WPA3-SAE+CCMP authentication and encryption WAPI authentication and encryption: • Supports centralized WAPI authentication. • Supports three-certificate WAPI authentication, which is compatible with traditional two-certificate authentication. • Issues a certificate file together with a private key. Allows users to use MAC addresses as accounts for authentication by the RADIUS server. Portal authentication: • Authentication through an external Portal server • Built-in Portal authentication and authentication page customization 802.1X authentication: • Authentication through an external 802.1X server. • Built-in 802.1X authentication.
Combined authentication	Combined MAC authentication: • PSK+MAC authentication MAC+portal authentication: • MAC authentication is used first. When MAC authentication fails, portal authentication is used.
EAP types	EAP-TLS, EAP-TTLS, EAP-PEAP, EAP-CHAP, EAP-SIM, EAP-AKA, EAP-GTC, EAP-FAST, EAP-PEAP, EAP-MD5, EAP-MSCHAPv2, PEAPv0, PEAPv1
AAA	Local authentication/local accounts (MAC addresses and accounts) RADIUS authentication Multiple authentication servers: • Supports backup authentication servers. • Specifies authentication servers based on the account. • Configures authentication servers based on the account. • Binds user accounts to SSIDs.
Security isolation	Port-based isolation User group-based isolation
Security standards	802.11i, Wi-Fi Protected Access 2 (WPA2), WPA,802.1X

Feature	Description
	Advanced Encryption Standards (AES), Temporal Key Integrity Protocol (TKIP), and Extensible Authentication • Protocol (EAP) types: • EAP-Transport Layer Security (TLS) • EAP-Tunneled TLS (TTLS) or Microsoft Challenge Handshake Authentication Protocol Version 2 (MSCHAPv2) • Protected EAP (PEAP) v0 or EAP-MSCHAPv2 • EAP-Flexible Authentication via Secure Tunneling (FAST) • PEAP v1 or EAP-Generic Token Card (GTC) EAP-Subscriber Identity Module (SIM)
WIDS	• Rogue device scan, identification, defense, and countermeasures, which includes dynamic blacklist configuration and detection of rogue APs, STAs, and network attacks.
Authority control	• ACL limit based on the following: • Port • User group • User
Other security features	• SSID hiding • IP source guard: • Configures IP and MAC binding entries statically. • Generates IP and MAC binding entries dynamically.

WLAN QoS

Feature	Description
WMM profile management	Enables or disables Wi-Fi Multimedia (WMM). Allows a WMM profile to be applied to radios of multiple APs.
Traffic profile management	Manages traffic from APs and maps packet priorities according to traffic profiles. Applies a QoS policy to each ESS by binding a traffic profile to each ESS.
AC traffic control	Manages QoS profiles. Uses ACLs to perform traffic classification. Limits incoming and outgoing traffic rates for each user based on inbound and outbound CAR parameters. Limits the traffic rate based on ESSs or VAPs.
AP traffic control	Controls traffic of multiple users and allows users to share bandwidth. Limits the rate of a specified VAP.
Packet priority configuration	Sets the QoS priority (IP precedence or DSCP priority) for CAPWAP control channels. Sets the QoS priority for CAPWAP data channels: • Allows you to specify the CAPWAP header priority. • Maps 802.1p priorities of user packets to ToS priorities of tunnel packets.
Airtime fair scheduling	Allocates equal time to users for occupying the channel, which improves users' Internet access experience.

Physical Specifications

Feature	Description
Dimensions (H x W x D)	43.6 mm x 250 mm x210 mm
Interface type	2 x 10G (SFP+) + 10 x GE
Maximum power consumption	25.06 W
Weight	1.47 kg
Operating temperature and altitude	-60 m to +1800 m: 0°C to 45°C 1800 m to 5000 m: Temperature decreases by 1°C every time the altitude increases 220 m.
Relative humidity	5% RH to 95% RH, noncondensing
Power modules	AC/DC adapter

Performance Specifications

Feature	Description
Number of managed APs	512 **NOTE** *The RUs managed by the WLAN AC do not occupy the AC's license resources. However, the total number of managed common APs and RUs cannot exceed the upper limit allowed by the AC.*
Number of access users	4096 **NOTE** *The maximum number of access users varies depending on the authentication mode.*
Number of MAC address entries	8192
Forwarding capability	10 Gbit/s **NOTE** ● *The value is the maximum forwarding capability supported by the device. The actual performance varies with the enabled functions of the device and the network environment. For details, see the product specifications.* ● *Packet length: 1514 bytes.*
Number of VLANs	4096
Number of routing entries	• IPv4: 8192 • IPv6: 2048
Number of ARP entries	6144
Number of multicast forwarding entries	2048
Number of DHCP IP address pools	128 IP address pools, each of which contains a maximum of 8192 IP addresses
Number of local accounts	1000
Number of ACLs	4096

Standards compliance

Item	Description
Safety standards	IEC60950-1 UL60950-1 CSA C22.2#60950-1 EN60950-1 AS/NZS 60950.1 GB 4943
EMC standards	FCC Part15B ETSI EN 300 386 IEC61000-4-11 IEC 61000-4-4 IEC61000-4-2 IEC61000-4-3 IEC61000-4-5 IEC61000-4-6 IEC 61000-3-2 IEC 61000-3-3 AS/NZS CISPR 32 EN55032/EN55024 ICES-003 GB9254
RoHS	Directive 2002/95/EC & 2011/65/EU
Reach	Regulation 1907/2006/EC
WEEE	Directive 2002/96/EC & 2012/19/EU

Ordering Information

Part Number	Description
02354FRJ-001	AC6508 mainframe (10*GE ports, 2*10GE SFP+ ports, with the AC/DC adapter)
88034UVY	Access Controller AP Resource License(1 AP)
88034UWA	Access Controller AP Resource License(8 AP)
88034UWB	Access Controller AP Resource License(16 AP)
88034UWC	Access Controller AP Resource License(32 AP)
88034UWD	Access Controller AP Resource License(64 AP)
88034UWE	Access Controller AP Resource License(128 AP)
88039VXM	Wireless Access Controller AP Wi-Fi Shield Resource License(1 AP)

More Information

For more information about Huawei WLAN products, visit http://e.huawei.com or contact us in the following ways:

- Global service hotline: http://e.huawei.com/en/service-hotline
- Logging in to the Huawei Enterprise technical support web: http://support.huawei.com/enterprise/
- Sending an email to the customer service mailbox: support_e@huawei.com

Huawei Technologies Co., Ltd.

Address:Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China

Website:www.huawei.com

HUAWEI

38 - AirEngine5773-21(11be indoor,2+2 dual bands,smart antenna,USB,BLE)

Huawei AirEngine 5773-21 Access Point Datasheet

Product Overview

Huawei AirEngine 5773-21 is a next-generation indoor access point (AP) in compliance with Wi-Fi 7 (802.11be). It can simultaneously provide services on 2.4 GHz (2x2 MIMO) and 5 GHz (2x2 MIMO) frequency bands, supporting a total of 4 spatial streams and achieving a device rate of up to 3.57 Gbps. The AP is empowered by brand-new Wi-Fi 7 technologies and is equipped with built-in smart antennas to enable always-on Wi-Fi signals for users, significantly enhancing users' wireless network experience. Additionally, it has a compact size, facilitating flexible deployment and saving customer investment. These strengths make the AirEngine 5773-21 ideal for indoor coverage scenarios such as SMB workplaces, hospitals, and shopping malls and supermarkets.

AirEngine 5773-21

- Provides services simultaneously on both the 2.4 GHz (2x2) and 5 GHz (2x2) frequency bands, at a rate of up to 689 Mbps at 2.4 GHz, 2.88 Gbps at 5 GHz, and 3.57 Gbps for the device.
- Has built-in smart antennas that automatically adjust the coverage direction and signal strength based on the intelligent switchover algorithm. Such capability enables the AP to flexibly adapt to the application environment changes, providing accurate and stable coverage as STAs move.
- USB port can be used for external IoT expansion (supporting protocols such as ZigBee, and RFID).
- Supports Bluetooth serial port-based O&M through built-in Bluetooth and CloudCampus APP.
- Supports Fit and cloud management working modes, and enables Huawei cloud management platform to manage and operate APs and services on the APs, reducing network O&M costs.

NOTE

The feature description and specification are based on the version of V600R25C00.

Feature Descriptions

Wi-Fi 7 (802.11be) standard

Wi-Fi 7 (802.11be) is the Wi-Fi standard, also known as IEEE 802.11be or extremely high throughput (EHT). Based on Wi-Fi 6, Wi-Fi 7 introduces technologies such as 4096-quadrature amplitude modulation (QAM), multi-resource unit (MRU), multi-link operation (MLO), enhanced multi-user multiple-input multiple-output (MU-MIMO). Drawing on these cutting-edge technologies, Wi-Fi 7 delivers a higher data transmission rate and lower latency than Wi-Fi 6.

New Features in Wi-Fi 7

Wi-Fi 7 aims to increase the WLAN throughput and provide low-latency access assurance. To achieve this goal, the Wi-Fi 7 standard defines modifications to both the physical layer (PHY) and MAC layer. Compared with Wi-Fi 6, Wi-Fi 7 brings the following technical innovations:

Multi-RU

In Wi-Fi 6, each user can send or receive frames only on the RUs allocated to them, which greatly limits the flexibility of spectrum resource scheduling. To resolve this problem and further improve spectrum efficiency, Wi-Fi 7 defines a mechanism for allocating multiple RUs to a single user. To balance the implementation complexity and spectrum utilization, the standard specifications impose certain restrictions on RU combination. That is, small RUs (containing fewer than 242 tones) can be combined only with small RUs, and large RUs (containing greater than or equal to 242 tones) can be combined only with large RUs. Small RUs and large RUs can be combined together.

Higher-Order 4096-QAM

The highest order modulation supported by Wi-Fi 6 is 1024-QAM, which allows each modulation symbol to carry up to 10 bits. To further improve the rate, Wi-Fi 7 introduces 4096-QAM so that each modulation symbol can carry 12 bits. With the same coding, 4096-QAM in Wi-Fi 7 can achieve a 20% rate increase compared with 1024-QAM in Wi-Fi 6.

Multi-Link Mechanism

To efficiently utilize all available spectrum resources, the industry is in urgent need to introduce new spectrum management, coordination, and transmission mechanisms on the 2.4 GHz, 5 GHz, and 6 GHz frequency bands. The TGbe defines multi-link aggregation technologies, including the MAC architecture of enhanced multi-link aggregation, multi-link channel access, and multi-link transmission.

There are two modes as for MLO:

- High-concurrency mode, multiple links send different data to improve bandwidth.
- High-reliability mode, multiple links send the same data, improving reliability.

Wi-Fi CSI Sensing

Wi-Fi CSI sensing is a cutting-edge technology for implementing sensing by using channel state information (Channel State Information, CSI) generated during radio signal propagation. Based on the Wi-Fi 7 standard, Huawei innovatively introduces Wi-Fi CSI to sense the presence of personnel, so that Wi-Fi signals can be sensed wherever they are. Compared with cameras, it protects user privacy and applies to scenarios such as energy saving, health care, and smart security.

Leader AP

The leader AP integrates some WLAN AC functions and can be used to manage Fit APs in small- and medium-sized enterprises and stores, implementing WLAN AC-free access not requiring licenses and saving customer investment.

Basic Specifications

Fit AP mode

Item	Description
WLAN features	Compliance with IEEE 802.11be and compatibility with IEEE 802.11a/b/g/n/ac/ax Maximum ratio combining (MRC) Space time block code (STBC) Cyclic Delay Diversity (CDD)/Cyclic Shift Diversity (CSD) Beamforming Multi-user multiple-input multiple-output (MU-MIMO) Orthogonal frequency division multiple access (OFDMA) Preamble puncturing BSS Color TxBF TWT Compliance with 4096-quadrature amplitude modulation (QAM) and compatibility with 1024-QAM, 256-QAM, 64-QAM, 16-QAM, 8-QAM, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK) Low-density parity-check (LDPC) Frame aggregation, including A-MPDU (Tx/Rx) and A-MSDU (Tx/Rx) 802.11 dynamic frequency selection (DFS) Short guard interval (GI) in 20 MHz, 40 MHz, 80 MHz, 160 MHz modes Wi-Fi multimedia (WMM) for priority-based data processing and forwarding

Item	Description
	WLAN channel management and channel rate adjustment **NOTE** For detailed management channels, see the Country Codes & Channels Compliance. Automatic channel scanning and interference avoidance Service set identifier (SSID) hiding configuration for each AP, supporting Chinese SSIDs Signal sustain technology (SST) Unscheduled automatic power save delivery (U-APSD) Multi-user call admission control (CAC) Advanced cellular coexistence (ACC), minimizing the impact of interference from cellular networks 802.11k and 802.11v smart roaming 802.11r fast roaming (≤ 50 ms) Terminal location（Single-AP access location）
Network features	Compliance with IEEE 802.3ab Auto-negotiation of the rate and duplex mode, and automatic switchover between Media Dependent Interface (MDI) and Media Dependent Interface Crossover (MDI-X) Compatibility with IEEE 802.1Q SSID-based VLAN assignment Eth-Trunk function Management channel of the AP's uplink port in tagged and untagged modes DHCP client, obtaining IP addresses through DHCP Tunnel data forwarding and direct data forwarding STA isolation in the same VLAN IPv4/IPv6 access control list (ACL) Link Layer Discovery Protocol (LLDP) Service holding when CAPWAP link disconnection in direct data forwarding mode Unified authentication on the AC AC dual-link backup Telemetry, quickly collecting AP status and application experience parameters MESH IPv6 SAVI
QoS features	WMM power save Priority mapping for upstream packets and flow-based mapping for downstream packets Queue mapping and scheduling User-based bandwidth limiting Adaptive bandwidth management (automatic bandwidth adjustment based on the user quantity and radio environment) to improve user experience Application identification and QoS classification to improve voice quality for popular applications, such as Zoom, QQ, and WeChat Airtime scheduling Air interface HQoS scheduling
Security features	Open system authentication WPA2-PSK authentication and encryption (WPA2-Personal) WPA2-802.1X authentication and encryption (WPA2-Enterprise) WPA3-SAE authentication and encryption (WPA3-Personal) WPA3-802.1X authentication and encryption (WPA3-Enterprise)

Item	Description
	WPA-WPA2 hybrid authentication WPA2-WPA3 hybrid authentication WPA/WPA2/WPA2-PPSK authentication and encryption WPA/WPA2/WPA2-DPSK authentication and encryption WAPI authentication and encryption Wireless intrusion detection system (WIDS) and wireless intrusion prevention system (WIPS), including rogue device detection and containment, attack detection and dynamic blacklist, and STA/AP blacklist and whitelist 802.1X authentication, MAC address authentication, and Portal authentication DHCP snooping 802.11w Protected Management Frames (PMF) CAPWAP DTLS data encryption and decryption URL filtering Secure boot
EAP types	EAP-TLS, EAP-TTLS, EAP-PEAP, EAP-CHAP, EAP-SIM, EAP-AKA, EAP-GTC, EAP-FAST, EAP-PEAP, EAP-MD5, EAP-MSCHAPv2, PEAPv0, PEAPv1
Maintenance features	Unified AP management and maintenance on the AC Automatic AP onboarding, automatic configuration loading, and plug-and-play (PnP) Automatic batch upgrade STelnet using SSHv2 SFTP using SSHv2 Remote wireless O&M through the Bluetooth serial port System status alarm Unified AP management on WebMaster
Sensing	Wi-Fi CSI Sensing

Fat AP mode

Item	Description
WLAN features	Compliance with IEEE 802.11be and compatibility with IEEE 802.11a/b/g/n/ac/ax Maximum ratio combining (MRC) Space time block code (STBC) Cyclic Delay Diversity (CDD)/Cyclic Shift Diversity (CSD) Beamforming Multi-user multiple-input multiple-output (MU-MIMO) Orthogonal frequency division multiple access (OFDMA) Preamble puncturing BSS Color TxBF TWT Compliance with 4096-quadrature amplitude modulation (QAM) and compatibility with 1024-QAM, 256-QAM, 64-QAM, 16-QAM, 8-QAM, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK) Low-density parity-check (LDPC) Frame aggregation, including A-MPDU (Tx/Rx) and A-MSDU (Tx/Rx)

Item	Description
	802.11 dynamic frequency selection (DFS) Short guard interval (GI) in 20 MHz, 40 MHz, 80 MHz, 160 MHz modes Wi-Fi multimedia (WMM) for priority-based data processing and forwarding WLAN channel management and channel rate adjustment **NOTE** For detailed management channels, see the Country Codes & Channels Compliance. Automatic channel scanning and interference avoidance Service set identifier (SSID) hiding configuration for each AP, supporting Chinese SSIDs Signal sustain technology (SST) Unscheduled automatic power save delivery (U-APSD) Advanced cellular coexistence (ACC), minimizing the impact of interference from cellular networks 802.11k and 802.11v smart roaming 802.11r fast roaming (≤ 50 ms)
Network features	Compliance with IEEE 802.3ab Auto-negotiation of the rate and duplex mode, and automatic switchover between Media Dependent Interface (MDI) and Media Dependent Interface Crossover (MDI-X) Compatibility with IEEE 802.1Q SSID-based VLAN assignment DHCP client, obtaining IP addresses through DHCP Tunnel data forwarding and direct data forwarding STA isolation in the same VLAN IPv4 access control list (ACL) Link Layer Discovery Protocol (LLDP) Leader AP NAT
QoS features	WMM power save Priority mapping for upstream packets and flow-based mapping for downstream packets Queue mapping and scheduling User-based bandwidth limiting Airtime scheduling
Security features	Open system authentication WPA2-PSK authentication and encryption (WPA2-Personal) WPA3-SAE authentication and encryption (WPA3-Personal) WPA-WPA2 hybrid authentication WPA2-WPA3 hybrid authentication MAC address authentication, and Portal authentication DHCP snooping 802.11w Protected Management Frames (PMF) URL filtering Secure boot
EAP types	EAP-TLS, EAP-TTLS, EAP-PEAP, EAP-CHAP, EAP-SIM, EAP-AKA, EAP-GTC, EAP-FAST, EAP-PEAP, EAP-MD5, EAP-MSCHAPv2, PEAPv0, PEAPv1
Maintenance features	STelnet using SSHv2 SFTP using SSHv2

Item	Description
	Remote wireless O&M through the Bluetooth serial port System status alarm

Cloud-Managed AP mode

Item	Description
WLAN features	Compliance with IEEE 802.11be and compatibility with IEEE 802.11a/b/g/n/ac/ax Maximum ratio combining (MRC) Space time block code (STBC) Cyclic Delay Diversity (CDD)/Cyclic Shift Diversity (CSD) Beamforming Multi-user multiple-input multiple-output (MU-MIMO) Orthogonal frequency division multiple access (OFDMA) Preamble puncturing BSS Color TxBF TWT Compliance with 4096-quadrature amplitude modulation (QAM) and compatibility with 1024-QAM, 256-QAM, 64-QAM, 16-QAM, 8-QAM, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK) Low-density parity-check (LDPC) Frame aggregation, including A-MPDU (Tx/Rx) and A-MSDU (Tx/Rx) 802.11 dynamic frequency selection (DFS) Short guard interval (GI) in 20 MHz, 40 MHz, 80 MHz, 160 MHz modes Wi-Fi multimedia (WMM) for priority-based data processing and forwarding WLAN channel management and channel rate adjustment **NOTE** For detailed management channels, see the Country Codes & Channels Compliance. Automatic channel scanning and interference avoidance Service set identifier (SSID) hiding configuration for each AP, supporting Chinese SSIDs Signal sustain technology (SST) Unscheduled automatic power save delivery (U-APSD) Automatic AP Online by NCE-Campus Multi-user call admission control (CAC) Advanced cellular coexistence (ACC), minimizing the impact of interference from cellular networks 802.11k and 802.11v smart roaming 802.11r fast roaming (≤ 50 ms)
Network features	Compliance with IEEE 802.3ab Auto-negotiation of the rate and duplex mode, and automatic switchover between Media Dependent Interface (MDI) and Media Dependent Interface Crossover (MDI-X) Compatibility with IEEE 802.1Q SSID-based VLAN assignment DHCP client, obtaining IP addresses through DHCP STA isolation in the same VLAN IPv4/IPv6 access control list (ACL)

Item	Description
	Link Layer Discovery Protocol (LLDP) Service holdover when the link to NCE-Campus is disconnected Unified authentication on the cloud management platform Network address translation (NAT) Telemetry, quickly collecting AP status and application experience parameters MESH Tunnel-AC IPv6 SAVI
QoS features	WMM power save Priority mapping for upstream packets and flow-based mapping for downstream packets Queue mapping and scheduling User-based bandwidth limiting Adaptive bandwidth management (automatic bandwidth adjustment based on the user quantity and radio environment) to improve user experience Application identification and QoS classification to improve voice quality for popular applications, such as Zoom, QQ, and WeChat Airtime scheduling Air interface HQoS scheduling
Security features	Open system authentication WPA2-PSK authentication and encryption (WPA2-Personal) WPA2-802.1X authentication and encryption (WPA2-Enterprise) WPA3-SAE authentication and encryption (WPA3-Personal) WPA3-802.1X authentication and encryption (WPA3-Enterprise) WPA-WPA2 hybrid authentication WPA2-WPA3 hybrid authentication WPA/WPA2/WPA2-PPSK authentication and encryption WPA/WPA2/WPA2-DPSK authentication and encryption 802.1X authentication, MAC address authentication, and Portal authentication Wireless intrusion detection system (WIDS) and wireless intrusion prevention system (WIPS), including rogue device detection and containment, attack detection and dynamic blacklist, and STA/AP blacklist and whitelist DHCP snooping 802.11w Protected Management Frames (PMF) CAPWAP DTLS data encryption and decryption Secure boot
EAP types	EAP-TLS, EAP-TTLS, EAP-PEAP, EAP-CHAP, EAP-SIM, EAP-AKA, EAP-GTC, EAP-FAST, EAP-PEAP, EAP-MD5, EAP-MSCHAPv2, PEAPv0, PEAPv1
Maintenance features	Unified AP management and maintenance on the cloud management platform Automatic AP onboarding, automatic configuration loading, and PnP Batch upgrade STelnet using SSHv2 SFTP using SSHv2 Remote wireless O&M through the Bluetooth serial port Real-time user configuration monitoring and fast fault locating using the NMS System status alarm

Item	Description
	Protocol (NTP)
Sensing	Wi-Fi CSI Sensing

Network Time Technical Specifications

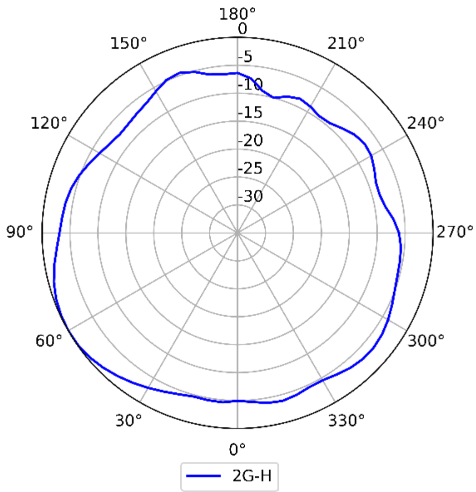
Item		Description
Technical specifications	Dimensions (Diameter × Height)	Φ180 x 45mm
	Weight	0.47 kg
	Interface type	1 x 100M/1GE/2.5GE (RJ-45) 1 x USB port **NOTE** The 2.5GE (RJ-45) supports PoE input.
	Bluetooth	BLE 5.4
	Nearlink	SLE1.0
	LED indicator	Indicates the power-on, startup, running, alarm, and fault states of the system.
Power specifications	Power input	• DC: 12 V ± 10% • PoE power supply: in compliance with 802.3at/af **NOTE** When 802.3af power is supplied, the AP will operate with restrictions, for example the USB port is unavailable, and the details refer to the Info-Finder.
	Maximum power consumption	13.3 W (excluding USB) **NOTE** • It is 13.6 W for the AP(50086767-001). • The actual maximum power consumption depends on local laws and regulations.
Environmental specifications	Operating temperature	-10°C to +50°C
	Storage temperature	-40°C to +70°C
	Operating humidity	5% to 95% (non-condensing)
	Altitude	-60 m to +5000 m
	Atmospheric pressure	53 kPa to 106 kPa
Radio specifications	Antenna type	Built-in smart antennas
	Antenna gain	2.4 GHz: 4 dBi 5 GHz: 5 dBi **NOTE** The gains above are the single-antenna peak gains.
	Maximum number of SSIDs for each radio	15
	Maximum transmit power	2.4G: 23 dBm (combined power)

Item		Description
		5G: 23 dBm (combined power) **NOTE** The actual transmit power depends on local laws and regulations.
	Frequency bands	2.400 to 2.4835 GHz ISM 5.150 to 5.250 GHz U-NII-1 5.250 to 5.350 GHz U-NII-2A 5.470 to 5.725 GHz U-NII-2C 5.725 to 5.850 GHz U-NII-3/ISM **NOTE** The available bands and channels are dependent on the configured regulatory domain (country).

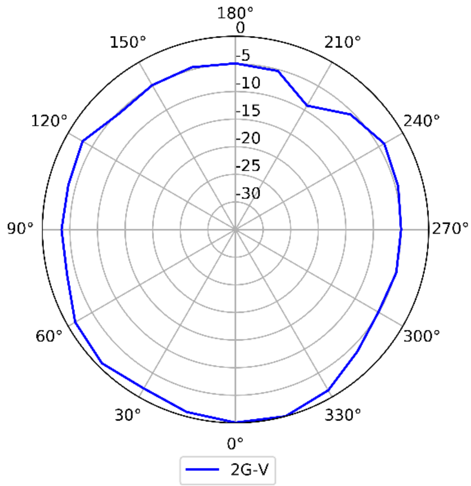
Standards Compliance

Item	Description		
Safety standards	• EN 62368-1	• IEC 62368-1	
Radio standards	• ETSI EN 300 328	• ETSI EN 301 893	• AN/NZS 4268
EMC standards	• EN 301 489-1 • EN 301 489-17 • EN 60601-1-2 • EN 55032 • EN 55035	• GB 9254 • GB 17625.2 • AS/NZS CISPR32 • CISPR 32 • CISPR 35	• IEC/EN61000-4-2 • IEC/EN 61000-4-3 • IEC/EN 61000-4-4 • IEC/EN 61000-4-5 • IEC/EN 61000-4-6 • ICES-003
IEEE standards	• IEEE 802.11a/b/g • IEEE 802.11n • IEEE 802.11ac • IEEE 802.11ax • IEEE 802.11be	• IEEE 802.11h • IEEE 802.11d • IEEE 802.11e • IEEE 802.11k	• IEEE 802.11v • IEEE 802.11w • IEEE 802.11r
Security standards	• 802.11i，Wi-Fi Protected Access (WPA)，WPA2，WPA2-Enterprise，WPA2-PSK，WPA3，WAPI • 802.1X • Advanced Encryption Standards(AES)，Temporal Key Integrity Protocol(TKIP)，WEP，Open • EAP Type(s)		
EMF	• EN 62311	• EN 50385	
RoHS	• Directive 2002/95/EC & 2011/65/EU • (EU)2015/863		
Reach	• Regulation 1907/2006/EC		
WEEE	• Directive 2002/96/EC & 2012/19/EU		

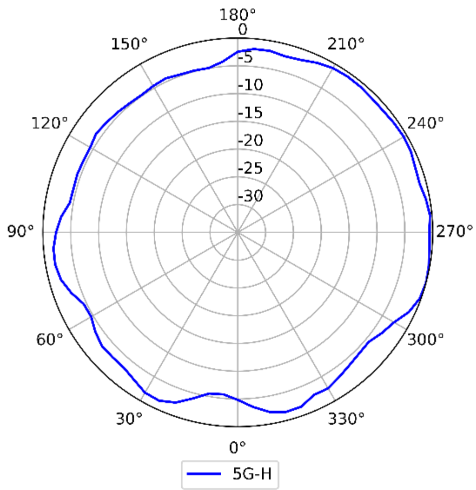
Antennas Pattern



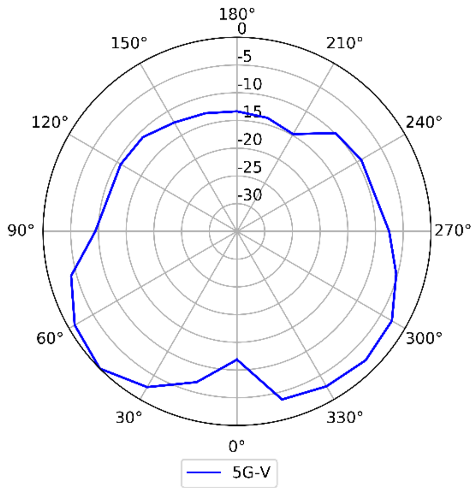
2.4GHz (Horizontal)



2.4GHz (Vertical)



5GHz (Horizontal)



5GHz (Vertical)

Huawei Technologies Co., Ltd.

Address:Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China

Website:www.huawei.com

Huawei CloudEngine S6750-S Series Switches Datasheet

Huawei CloudEngine S6750-S switches are next-generation enterprise-class access switches that provide GE/10GE downlink ports and 25GE/40GE/100GE uplink optical ports.

Introduction

Huawei CloudEngine S6750-S switches are next-generation enterprise-class core and aggregation switches that offer high performance, high reliability, cloud management, and intelligent operations and maintenance (O&M). They build on an industry-leading software and hardware platform and are purpose-built with security, IoT, and cloud in mind. With these traits, CloudEngine S6750-S can be widely used in enterprise campuses, colleges/universities and other scenarios.

CloudEngine S6750-S switches offer GE, 10GE, 40GE and 100GE port types, flexibly adapting to diversified network bandwidth requirements. They also support cloud management and implement cloud-managed network services throughout the full lifecycle from planning, deployment, monitoring, experience visibility, and fault rectification, all the way to network optimization, greatly simplifying network management.

CloudEngine S6750-S switches support free mobility, enables consistent user experience no matter the user location or IP address, fully meeting enterprises' demands for mobile offices.

CloudEngine S6750-S switches support VXLAN to implement network virtualization, achieving multi-purpose networks and multi-network convergence for greatly improved network capacity and utilization. As such, CloudEngine S6750-S switches are an ideal choice for building next-generation IoT converged networks in terms of cost, flexibility, and scalability.

Product Overview

Models and Appearances

The following models are available in CloudEngine S6750-S switches.

Appearance	Description
 CloudEngine S6750-S16X8YZ	• 16 x 1/2.5/10 GE SFP+, 8 x 1/2.5/10/25 GE SFP28 • One extended slot • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 450 Mpps • Switching capacity: 920 Gbps/2.32 Tbps*
 CloudEngine S6750-S16X10Y2CZ	• 16 x FE/1/2.5/10 GE SFP+, 10 x 1/2.5/10/25 GE SFP28, 2 x 40/100 GE QSFP28 • One extended slot • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 450 Mpps

40 - Distribution switch
40 S6750-S16X10Y2CZ
S6750-S16X10Y2CZ (16*10GE SFP+ ports, 10*25GE SFP28 ports, 2*100GE QSFP28 ports, expansion card slot, without power module)

Appearance	Description
	• Switching capacity: 1.42T bps/2.32 Tbps*
 CloudEngine S6750-S24T16X8Y2CZ	• 24 × 10/100/1000M Base-T Ethernet ports, 16 x FE/1/2.5/10 GE SFP+, 8 x 1/2.5/10/25GE SFP28, 2 x 40/100 GE QSFP28 • One extended slot • Dual pluggable power modules, 1+1 power backup • Forwarding performance: 450 Mpps • Switching capacity: 968 Gbps/2.32 Tbps*

*Note：The value before the slash (/) refers to the device's switching capability, while the value after the slash (/) means the system's switching capability.

Subcards

The following table lists the subcards applicable to the CloudEngine S6750-S.

Technical specifications of the subcards applicable to the CloudEngine S6750-S series

Subcards	Technical Specifications	Applied Switch Model
 HSIC-X08S000	• 8*10GE SFP+ or 4*25GE SFP28 (only ports 1 to 4 support 25GE) • Dimensions without packaging (H x W x D) [mm(in.)]: 39.1 mm x 100.1 mm x 220 mm (1.54 in. x 3.94 in. x 8.66 in.) • Weight without packaging [kg(lb)]: 0.44 kg (0.97 lb) • Maximum power consumption [W]: 38.5 W • Maximum heat dissipation [BTU/hour]: 131.37 BTU/hour	• CloudEngine S6750-S16X8YZ • CloudEngine S6750-S16X10Y2CZ • CloudEngine S6750-S24T16X8Y2CZ
 HSIC-Y08S000	• 8*25GE SFP28 • Dimensions without packaging (H x W x D) [mm(in.)]: 39.1 mm x 100.1 mm x 220 mm (1.54 in. x 3.94 in. x 8.66 in.) • Weight without packaging [kg(lb)]: 0.44 kg (0.97 lb) • Maximum power consumption [W]: 38.5 W • Maximum heat dissipation [BTU/hour]: 131.37 BTU/hour	• CloudEngine S6750-S16X8YZ • CloudEngine S6750-S16X10Y2CZ • CloudEngine S6750-S24T16X8Y2CZ

Fan Module

The following table lists the fan module on CloudEngine S6750-S switches.

Fan Module	Technical Specifications	Applied Switch Model
 FAN-031A-B	• Dimensions (W x D x H): 40 mm x 100.3 mm x 40 mm • Number of fans: 1 • Weight: 0.1 kg • Maximum power consumption: 21.6 W • Maximum fan speed: 24500±10% revolutions per minute (RPM) • Maximum wind rate: 31 cubic feet per minute (CFM) • Hot swap: Supported	• CloudEngine S6750-S16X8YZ • CloudEngine S6750-S16X10Y2CZ • CloudEngine S6750-S24T16X8Y2CZ

Power Supply

The following table lists the power supplies on CloudEngine S6750-S series.

<table>
<tr><th>Power Module</th><th>Technical Specifications</th><th>Applied Switch Model</th></tr>
<tr><td>
PAC180S12-CN

41- PAC180S12-CN
180W AC Power Module</td><td>• Dimensions (H x W x D): 40 mm x 66 mm x 215 mm
• Weight: 0.8 kg
• Rated input voltage range:
– 100 V AC to 130 V AC; 50/60 Hz
– 100 V AC to 240 V AC, 50/60 Hz
– 240 V DC
• Maximum input voltage range:
– 90 V AC to 290 V AC, 45 Hz to 66 Hz
– 190 V DC to 290 V DC
• Maximum input current:
– 3 A
• Rated output current:
– 15 A
• Rated output voltage: 12 V
• Rated output power:
– 180 W
• Hot swap: Supported</td><td>• CloudEngine S6750-S16X8YZ
• CloudEngine S6750-S16X10Y2CZ
• CloudEngine S6750-S24T16X8Y2CZ</td></tr>
<tr><td>
PAC600S12-PB</td><td>• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm (1.56 in. x 2.6 in. x 8.46 in.)
• Weight: 1 kg (2.2 lb)
• Rated input voltage range:
– 100 V AC to 240 V AC, 50/60 Hz
– 240 V DC
• Maximum input voltage range:
– 90 V AC to 290 V AC, 45 Hz to 66 Hz
– 190 V DC to 290 V DC
• Maximum input current:
100 V AC to 240 V AC: 8 A
240 V DC: 4 A
• Rated output current: 50 A
• Rated output voltage: 12 V
• Rated output power: 600 W
• Hot swap: Supported</td><td>• CloudEngine S6750-S16X8YZ
• CloudEngine S6750-S16X10Y2CZ
• CloudEngine S6750-S24T16X8Y2CZ</td></tr>
<tr><td>
PDC240S12-CN</td><td>• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm (1.56 in. x 2.6 in. x 8.46 in.)
• Weight: 1.5 kg (3.31 lb)
• Rated input voltage range:
– +48 V DC
– -48 V DC to -60 V DC
• Maximum input voltage range:
– +40 V DC to +57 V DC</td><td>• CloudEngine S6750-S16X8YZ
• CloudEngine S6750-S16X10Y2CZ
• CloudEngine S6750-S24T16X8Y2CZ</td></tr>
</table>

Power Module	Technical Specifications	Applied Switch Model
	− -38.4 V DC to -72 V DC • Maximum input current: 7A • Rated output voltage: 20 V • Rated output power: 240 W • Hot swap: Supported	
 PDC400S12-CB	• Dimensions (H x W x D): 39.6 mm x 66 mm x 215 mm (1.56 in. x 2.6 in. x 8.46 in.) • Weight: 0.844 kg (1.86 lb) • Rated input voltage range: − +48 V DC − -48 V DC to -60 V DC • Maximum input voltage range: − +40 V DC to +57 V DC − -38.4 V DC to -72 V DC • Maximum input current: 11A • Rated output voltage: 12 V • Rated output power: 400 W • Hot swap: Supported	

The S6750-S uses pluggable power modules. It can be configured with a single power module or double power modules for 1+1 power redundancy.

Product Features and Highlights

Enable Networks to be More Agile for Services

- Built-in high-speed and flexible processor chips, with their flexible packet processing and traffic control capabilities, CloudEngine S6750-S switches are close to services, meeting current and future challenges, and helping customers build scalable networks.
- CloudEngine S6750-S switches support fully customizing the forwarding mode, forwarding behavior, and search algorithm of traffic. New services are implemented through microcode programming. Customers do not need to replace new hardware and new services can be rolled out within six months.
- CloudEngine S6750-S switches provide open interfaces and user-defined forwarding processes to meet customized service requirements of enterprises. Enterprises can use multi-layer open interfaces to develop new protocols and functions independently. They can also hand over their requirements to vendors and jointly develop them to build an enterprise-dedicated campus network.

Delivering Abundant Services More Agilely

- With the unified user management function, the CloudEngine S6750-S switches authenticates both wired and wireless users, ensuring a consistent user experience no matter whether they are connected to the network through wired or wireless access devices. The unified user management function supports various authentication methods, including 802.1x, MAC address, and is capable of managing users based on user groups, domains, and time ranges. These functions visualize user and service management and boost the transformation from device-centric management to user experience-centric management.
- The CloudEngine S6750-S switches provide excellent quality of service(QoS) capabilities and supports queue scheduling and congestion control algorithms. Additionally, it adopts innovative priority queuing and multi-level scheduling mechanisms to implement fine-grained scheduling of data flows, meeting service quality requirements of different user terminals and services.

Fine-Grained Network Management and Visualized Fault Diagnosis

- In-situ Flow Information Telemetry (IFIT) is an in-band Operations, Administration, and Maintenance (OAM) measurement technology that uses service packets to measure real performance indicators of an IP network, such as the packet loss rate and delay. IFIT can significantly improve the timeliness and effectiveness of network O&M, thereby promoting the development of intelligent O&M.
- Three IFIT modes are available: application-level quality measurement, tunnel-level quality measurement, and native-IP IFIT measurement. Currently, CloudEngine S6750-S switches support native-IP IFIT measurement only. By providing in-band measurement capabilities, CloudEngine S6750-S switches can monitor indicators such as the delay and packet loss rate of service flows in real time. CloudEngine S6750-S switches also offer visualized O&M capabilities to centrally manage and control networks and graphically display performance data. Designed with IFIT capabilities featuring high measurement precision and easy deployment, CloudEngine S6750-S switches are ideal for constructing an intelligent O&M system and stand out with future-proof scalability.

Flexible Ethernet Networking

- In addition to traditional Spanning Tree Protocol (STP), Rapid Spanning Tree Protocol (RSTP), and Multiple Spanning Tree Protocol (MSTP), the CloudEngine S6750-S switches support Huawei-developed Smart Ethernet Protection (SEP) technology and the latest Ethernet Ring Protection Switching (ERPS) standard. SEP is a ring protection protocol specific to the Ethernet link layer, and applies to various ring network topologies, such as open ring topology, closed ring topology, and cascading ring topology. This protocol is reliable, easy to maintain, and implements fast service switching within 50 ms. ERPS is defined in ITU-T G.8032. It implements millisecond-level protection switching based on traditional Ethernet MAC and bridging functions.
- CloudEngine S6750-S switches support Smart Link and Virtual Router Redundancy Protocol (VRRP), which implement backup of uplinks. One CloudEngine S6750-S switch can connect to multiple core switches through multiple links, significantly improving reliability of aggregation devices.

Mature IPv6 Features

- The CloudEngine S6750-S series swithes are developed based on the mature, stable VRP and supports IPv4/IPv6 dual stacks, IPv6 routing protocols (RIPng, OSPFv3, BGP4+, and IS-IS for IPv6). With these IPv6 features, the CloudEngine S6750-S can be deployed on a pure IPv4 network, a pure IPv6 network, or a shared IPv4/IPv6 network, helping achieve IPv4-to-IPv6 transition.

Intelligent Stack (iStack)

- CloudEngine S6750-S switches support the iStack function that combines multiple switches into a logical switch. Member switches in a stack implement redundancy backup to improve device reliability and use inter-device link aggregation to improve link reliability. iStack provides high network scalability. You can increase a stack's ports, bandwidth, and processing capability by simply adding member switches. iStack also simplifies device configuration and management. After a stack is set up, multiple physical switches can be virtualized into one logical device. You can log in to any member switch in the stack to manage all the member switches in it.

Cloud-based Management

- The Huawei cloud management platform allows users to configure, monitor, and inspect switches on the cloud, reducing on-site deployment and O&M manpower costs and decreasing network OPEX.

VXLAN Features

- VXLAN is used to construct a Unified Virtual Fabric (UVF). As such, multiple service networks or tenant networks can be deployed on the same physical network, and service and tenant networks are isolated from each other. This capability truly achieves 'one network for multiple purposes'. The resulting benefits include enabling data transmission of different services or customers, reducing the network construction costs, and improving network resource utilization.
- This series switches are VXLAN-capable and allow centralized and distributed VXLAN gateway deployment modes. These switches also support the BGP EVPN protocol for dynamically establishing VXLAN tunnels and can be configured using NETCONF/YANG.

Link Layer Security

- CloudEngine S6750-S models all ports support MACsec. MACsec protects transmitted Ethernet data frames through identity authentication, data encryption, integrity check, and anti-replay protection, reducing the risks of information leakage and malicious network attacks. With MACsec, these switch models are able to address strict information security requirements of customers in industries such as government and finance.

Open Programmability System(OPS)

- Open Programmability System (OPS) is an open programmable system based on the Python language. IT administrators can program the O&M functions of a switch through Python scripts to quickly innovate functions and implement intelligent O&M.

Intelligent O&M

- This series switches provides telemetry technology to collect device data in real time and send the data to Huawei campus network analyzer(iMaster NCE-CampusInsight). The CampusInsight analyzes network data based on the intelligent fault identification algorithm, accurately displays the real-time network status, effectively demarcates and locates faults in a timely manner, and identifies network problems that affect user experience, accurately guaranteeing user experience.

Intelligent Upgrade

- Switches support the intelligent upgrade feature. Specifically, switches obtain the version upgrade path and download the newest version for upgrade from the Huawei Online Upgrade Platform (HOUP). The entire upgrade process is highly automated and achieves one-click upgrade. In addition, preloading the version is supported, which greatly shortens the upgrade time and service interruption time.
- The intelligent upgrade feature greatly simplifies device upgrade operations and makes it possible for the customer to upgrade the version independently. This greatly reduces the customer's maintenance costs. In addition, the upgrade policies on the HOUP platform standardize the upgrade operations, which greatly reduces the risk of upgrade failures.

Product Specifications

The following table describes the functions and features available on the CloudEngine S6750-S switches.

Functions and Features

Category	Service Features
User management	Unified user management
	802.1X, MAC, Portal, HACA authentication
	Traffic- and duration-based accounting
	User authorization based on user groups, domains, and time ranges
MAC	Automatic MAC address learning and aging
	128K MAC entries
	Static, dynamic, and blackhole MAC address entries
	Source MAC address filtering
	MAC address learning limiting based on ports and VLANs
VLAN	4K VLANs
	Access mode, Trunk mode and Hybrid mode
	Default VLAN
	QinQ and enhanced selective QinQ
	VLAN Stacking
	Dynamic VLAN assignment based on MAC addresses
ARP	ARP Snooping
DHCP	DHCPv4 Client, DHCPv4 Relay, DHCPv4 Server, DHCPv4 Snooping
	DHCPv6 Client, DHCPv6 Relay, DHCPv6 Server, DHCPv6 Snooping

Category	Service Features
IP routing	IPv4 dynamic routing protocols such as RIP, OSPF, IS-IS, and BGP
	IPv6 dynamic routing protocols such as RIPng, OSPFv3, ISISv6, and BGP4+
	Routing Policy, Policy-Based Routing
	VRF
Segment Routing	SRv6 BE (L3 EVPN)
	BGP EVPN
	SRv6 configuration through NETCONF
Multicast	IGMPv1/v2/v3 and IGMP v1/v2/v3 Snooping
	PIM-DM, PIM-SM, and PIM-SSM
	Fast-leave mechanism
	Multicast traffic control
	Multicast querier
	Multicast protocol packet suppression
MPLS	MPLS-LDP
	MPLS-L3VPN
	MPLS Qos
	MPLS TE
VXLAN	Centralized gateway
	Distributed gateway
	BGP-EVPN
	Configures VXLANs through NETCONF
QoS	Traffic classification based on Layer 2 headers, Layer 3 protocols, Layer 4 protocols, and 802.1p priority
	Actions such as ACL, Committed Access Rate (CAR), re-marking, and scheduling
	Queuing algorithms, such as PQ, DRR, WDRR, and PQ+DRR, PQ+WDRR
	Congestion avoidance mechanisms such as WRED and tail drop
	Traffic shaping
	Eight queues on each interface
	Network Slicing
Native-IP IFIT	Marks the real service packets to obtain real-time count of dropped packets and packet loss ratio
	The statistical period can be modified
	Two-way frame delay measurement
Ethernet loop protection	STP (IEEE 802.1d), RSTP (IEEE 802.1w), and MSTP (IEEE 802.1s).
	BPDU protection, root protection, and loop protection

Category	Service Features
	G.8032 Ethernet Ring Protection Switching (ERPS)
Reliability	M-LAG
	Service interface-based stacking
	Maximum number of stacked devices: 9
	Stack bandwidth (Bidirectional)
	Link Aggregation Control Protocol (LACP) and E-Trunk
	Virtual Router Redundancy Protocol (VRRP) and Bidirectional Forwarding Detection (BFD) for VRRP
	BFD for BGP/IS-IS/OSPF/static routes
	Eth-OAM 802.1ag(CFM)
	Smartlink
	LLDP
System management	Console terminal service
	Telnet/IPv6 Telnet terminal service
	SSH v1.5
	SSH v2.0
	SNMP v1/v2c/v3
	FTP、TFTP、SFTP
	BootROM upgrade and remote in-service upgrade
	Hot patch
	User operation logs
	Open Programmability System (OPS)
	Streaming Telemetry
Security and management	NAC
	RADIUS and HWTACACS authentication for login users
	MACsec
	Command line authority control based on user levels, preventing unauthorized users from using command configurations
	Defense against DoS attacks, Transmission Control Protocol (TCP) SYN Flood attacks, User Datagram Protocol (UDP) Flood attacks, broadcast storms, and heavy traffic attacks
	IPv6 RA Guard
	CPU hardware queues to implement hierarchical scheduling and protection for protocol packets on the control plane
	Remote Network Monitoring (RMON)
	Secure boot
	Netstream

Category	Service Features
	Port mirroring

NOTE

This content is applicable only to regions outside mainland China. Huawei reserves the right to interpret this content.

Hardware Specifications

The following table lists hardware specifications of the CloudEngine S6750-S switches.

Item		CloudEngine S6750-S16X8YZ	CloudEngine S6750-S16X10Y2CZ	CloudEngine S6750-S24T16X8Y2CZ
Physical specifications	Chassis dimensions (H x W x D, mm)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.40 in. x 16.54 in.)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.40 in. x 16.54 in.)	43.6 mm x 442.0 mm x 420.0 mm (1.72 in. x 17.40 in. x 16.54 in.)
	Chassis height	1U	1U	1U
	Chassis weight (full configuration weight, including weight of packaging materials)	7.67 kg	7.65 kg	7.96 kg
Fixed port	GE electrical port	NA	NA	24
	GE SFP port	NA	NA	NA
	10GE SFP+ port	16	16	16
	25GE SFP28 port	8	10	8
	40GE QSFP+ port	NA	NA	NA
	100GE QSFP28 port	NA	2	2
Management port	ETH management port	Supported	Supported	Supported
	Console port (RJ45)	Supported	Supported	Supported
	USB port	USB 2.0	USB 2.0	USB 2.0
Extended slot		• 8-port 1/10GE SFP+ interface card (1/2/3/4 support 25GE) • 8-port 1/10/25GE SFP+ interface card		
CPU	Frequency	2 GHz	2 GHz	2 GHz
	Cores	4	4	4
Memory	Memory (RAM)	4GB	4GB	4GB
	Flash	Hardware: 2 GB	Hardware: 2 GB	Hardware: 2 GB
Power supply system	Power supply type	• 180W AC Power Module • 600W AC Power Module • 240W DC Power Module • 400W DC Power Module	• 180W AC Power Module • 600W AC Power Module • 240W DC Power Module • 400W DC Power	• 180W AC Power Module • 600W AC Power Module • 240W DC Power Module • 400W DC Power

Item		CloudEngine S6750-S16X8YZ	CloudEngine S6750-S16X10Y2CZ	CloudEngine S6750-S24T16X8Y2CZ
			Module	Module
	Rated voltage range	• AC input: 100 V AC to 130 V AC, 200 V AC to 240 V AC; 50/60 Hz • High-voltage DC input: 240 V DC • DC input: –48 V DC to –60 V DC	• AC input: 100 V AC to 130 V AC, 200 V AC to 240 V AC; 50/60 Hz • High-voltage DC input: 240 V DC • DC input: –48 V DC to –60 V DC	• AC input: 100 V AC to 130 V AC, 200 V AC to 240 V AC; 50/60 Hz • High-voltage DC input: 240 V DC • DC input: –48 V DC to –60 V DC
	Maximum voltage range	• AC input: 90 V AC to 290 V AC; 45 Hz to 66 Hz • High-voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC; 45 Hz to 66 Hz • High-voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC	• AC input: 90 V AC to 290 V AC; 45 Hz to 66 Hz • High-voltage DC input: 190 V DC to 290 V DC • DC input: -38.4 V DC to -72 V DC
	Maximum input current	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.	The current specifications are related to the pluggable power module. For details, see Pluggable Power Modules.
	Typical power consumption (30% of traffic load, tested according to ATIS standard)	30% traffic under the ATIS standard and dual power modules: • 79 W (with two 180 W AC power modules) • 73 W (with two 240 W DC power modules)	30% traffic under the ATIS standard and dual power modules: • 120 W (with two 180 W AC power modules) • 109 W (with two 240 W DC power modules)	30% traffic under the ATIS standard and dual power modules: • 122 W (with two 180 W AC power modules) • 118 W (with two 240 W DC power modules)
	Maximum power consumption (100% throughput, full speed of fans)	100% traffic under the ATIS standard and dual power modules: • 85 W (with two 180 W AC power modules) • 78 W (with two 240 W DC power modules)	100% traffic under the ATIS standard and dual power modules: • 130 W (with two 180 W AC power modules) • 118 W (with two 240 W DC power modules)	100% traffic under the ATIS standard and dual power modules: • 127 W (with two 180 W AC power modules) • 123 W (with two 240 W DC power modules)
Heat dissipation system	Heat dissipation mode	Air cooling for heat dissipation, intelligent fan speed adjustment	Air cooling for heat dissipation, intelligent fan speed adjustment	Air cooling for heat dissipation, intelligent fan speed adjustment
	Number of fan modules	2	2	2
	Airflow	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear	Air intake from left, front, and right and air exhaust from rear
Environment	Long-term operating temperature	–5°C to +45°C (23°F to 113°F) at an altitude of 0 to	–5°C to +45°C (23°F to 113°F) at an altitude of 0	–5°C to +45°C (23°F to 113°F) at an altitude of 0

Item		CloudEngine S6750-S16X8YZ	CloudEngine S6750-S16X10Y2CZ	CloudEngine S6750-S24T16X8Y2CZ
parameters		1800 m (0 to 5905.44 ft.)	to 1800 m (0 to 5905.44 ft.)	to 1800 m (0 to 5905.44 ft.)
	Short-term operating temperature	-5°C to +50°C (23°F to 122°F) at an altitude of 0-1800 m (0-5905.44 ft.)	-5°C to +50°C (23°F to 122°F) at an altitude of 0-1800 m (0-5905.44 ft.)	-5°C to +50°C (23°F to 122°F) at an altitude of 0-1800 m (0-5905.44 ft.)
	Storage temperature	–40°C to +70°C (–40°F to +158°F)	–40°C to +70°C (–40°F to +158°F)	–40°C to +70°C (–40°F to +158°F)
	Relative humidity	5% to 95%, noncondensing	5% to 95%, noncondensing	5% to 95%, noncondensing
	Operating altitude	0-5000 m	0-5000 m	0-5000 m
	Noise under normal temperature (sound power)	54.2 dB(A)	54.2 dB(A)	54.2 dB(A)
	Noise under normal temperature (sound pressure)	41.2 dB(A)	41.2 dB(A)	41.2 dB(A)
	Surge protection specification (power port)	• Configured with AC power modules: ±6 kV in differential mode and ±6 kV in common mode • Configured with DC power modules: ±2 kV in differential mode and ±4 kV in common mode	• Configured with AC power modules: ±6 kV in differential mode and ±6 kV in common mode • Configured with DC power modules: ±2 kV in differential mode and ±4 kV in common mode	• Configured with AC power modules: ±6 kV in differential mode and ±6 kV in common mode • Configured with DC power modules: ±2 kV in differential mode and ±4 kV in common mode
Reliability	MTBF (year)	81.40	69.10	47.17
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification **NOTE** For details about certifications, see the section Safety and Regulatory Compliance.	• EMC certification • Safety certification • Manufacturing certification **NOTE** For details about certifications, see the section Safety and Regulatory Compliance.	• EMC certification • Safety certification • Manufacturing certification **NOTE** For details about certifications, see the section Safety and Regulatory Compliance.

NOTE

1: The power consumption under different load conditions is calculated according to the ATIS standard. Additionally.

2: The reliability parameter values are calculated based on the typical configuration of the device. The parameter values vary according to the modules configured by the customer.

Licensing

Licensing

This series switches supports both the traditional feature-based licensing mode and the latest Huawei IDN One Software (N1 mode for short) licensing mode. The N1 mode is ideal for deploying Huawei CloudCampus Solution in the on-premises scenario, as it greatly enhances the customer experiences in purchasing and upgrading software services with simplicity.

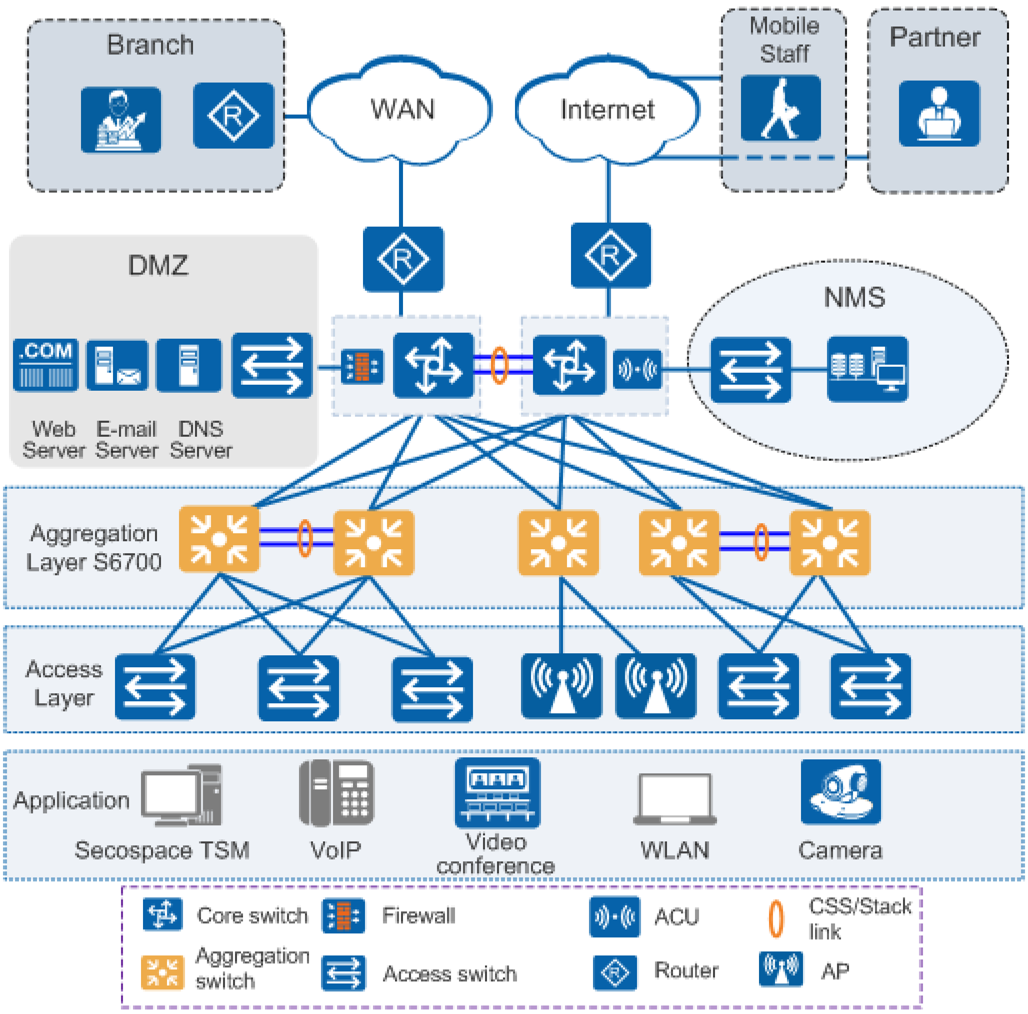
Software Package Features in N1 Mode

Switch Functions	N1 Basic Software	N1 Foundation Software Package	N1 Advanced Software Package
Basic network functions: Layer 2 functions, IPv4, IPv6, and others Note: For details, see the Service Features	√	√	√
Basic network automation based on the iMaster NCE-Campus: • NE management: Device management, topology management and discovery • User access authentication	×	√	√
Automation and intelligent O&M: Basic functions for intelligent network analysis of CampusInsight.	×	√	√
Advanced network automation and intelligent O&M: VXLAN, free mobility, Endpoint plug&play of iMaster NCE-Campus and Application analysis function of CampusInsight.	×	×	√

Networking and Applications

Large-scale Enterprise Campus Network

CloudEngine S6750-S switches can be deployed at the aggregation layer of a large-scale enterprise campus network, creating a highly reliable, scalable, and manageable enterprise campus network.



Product Accessories

Optical Modules and Fibers

10GE SFP+ ports support optical modules and cables

- GE optical module
- GE-CWDM optical module
- GE-DWDM optical module
- GE copper module
- 10GE SFP+ optical module
- 10GE-CWDM optical module
- 10GE-DWDM optical module
- 1 m, 3 m, 5 m, and 10 m SFP+ high-speed copper cables
- 3 m and 10 m SFP+ AOC cables
- 0.5 m and 1.5 m SFP+ dedicated stack cables (supported by the last 16 SFP+ ports and used only for zero-configuration stacking)

25GE SFP28 ports support optical modules and cables

- GE eSFP optical module
- GE SFP optical module
- GE-CWDM optical module

- GE-DWDM optical module
- 10GE SFP+ optical module
- 10GE-CWDM optical module
- 10GE-DWDM optical module
- 25GE SFP28 optical module
- 1 m, 3 m, 5 m, and 10 m SFP+ high-speed cables
- 3 m and 10 m SFP+ AOC cables
- 1 m, 3 m, 5 m SFP28 high-speed cables
- 3 m, 5 m, 7 m, and 10 m SFP28 AOC cables

40GE/100GE QSFP28 ports support optical modules and cables

- QSFP+ optical module
- QSFP28 optical module
- 1 m, 3 m, and 5 m QSFP+ to QSFP+ high-speed copper cables
- 10 m QSFP+ to QSFP+ AOC cable
- 1 m QSFP28 to QSFP28 high-speed copper cable
- 10 m QSFP28 to QSFP28 AOC cable

Stack Cables

CloudEngine S6750-S switches support service port stacking. The applicable stack cables are as follows:

Port Supporting Stacking	Stack Cable	Rate of a Single Port
10GE ports on the front panel	• 1 m, 3 m, and 5 m SFP+ passive high-speed cables • 10 m SFP+ active high-speed copper cables • 3 m and 10 m AOC cables • 10GE SFP+ optical module and optical fiber • 0.5 m and 1.5 m SFP+ dedicated stack cable	10 Gbit/s
40GE/100GE ports on the front panel	• 1 m QSFP28 high-speed copper cables • 10 m QSFP28 AOC cables • QSFP28 optical module and optical fiber	100Gbit/s

Safety and Regulatory Compliance

The following table lists the safety and regulatory compliance of the CloudEngine S6750-S switches.

Certification Category	Description
Safety	• IEC 60950-1 and all country deviations • EN 60950-1 • UL 60950-1 • CAN/CSA 22.2 No.60950-1 • GB 4943
Electromagnetic Compatibility (EMC)	• EMI • FCC CFR47 Part 15 Class A • EN55022 Class A • CISPR 22 Class A

Certification Category	Description
	• EN61000-3-2/IEC-1000-3-2, Power line harmonics • EN61000-4-3/IEC-1000-4-3, Radiated immunity • EN61000-4-2/IEC-1000-4-2, ESD • EN61000-4-4/IEC-1000-4-4, EFT • EN61000-4-5/IEC-1000-4-5, Surge Signal Port • EN61000-4-6/IEC-1000-4-6, Low frequency conducted immunity • EN61000-4-11/IEC-1000-4-11, Voltage dips and sags • EN61000-4-29/IEC61000-4-29, Voltage dips and sags • EMC Directive 89/336/EEC • EMC Directive 2004/108/EC • VCCI V-3 Class A • ICES-003 Class A • AS/NZS CISPR 22 Class A • CNS 13438 Class A • GB9254 Class A

NOTE

- EMC: electromagnetic compatibility
- CISPR: International Special Committee on Radio Interference
- EN: European Standard
- ETSI: European Telecommunications Standards Institute
- CFR: Code of Federal Regulations
- FCC: Federal Communication Commission
- IEC: International Electrotechnical Commission
- AS/NZS: Australian/New Zealand Standard
- VCCI: Voluntary Control Council for Interference
- UL: Underwriters Laboratories
- CSA: Canadian Standards Association
- IEEE: Institute of Electrical and Electronics Engineers

MIB and Standards Compliance

Supported MIBs

Category	MIB
Public MIB	• BRIDGE-MIB • DISMAN-NSLOOKUP-MIB • DISMAN-PING-MIB • DISMAN-TRACEROUTE-MIB • ENTITY-MIB • EtherLike-MIB • IF-MIB • IP-FORWARD-MIB • IPv6-MIB • LAG-MIB

Category	MIB
	• LLDP-EXT-DOT1-MIB • LLDP-EXT-DOT3-MIB • LLDP-MIB • NOTIFICATION-LOG-MIB • NQA-MIB • OSPF-TRAP-MIB • P-BRIDGE-MIB • Q-BRIDGE-MIB • RFC1213-MIB • RIPv2-MIB • RMON2-MIB • RMON-MIB • SAVI-MIB • SNMP-FRAMEWORK-MIB • SNMP-MPD-MIB • SNMP-NOTIFICATION-MIB • SNMP-TARGET-MIB • SNMP-USER-BASED-SM-MIB • SNMPv2-MIB • TCP-MIB • UDP-MIB
Huawei-proprietary MIB	• HUAWEI-AAA-MIB • HUAWEI-ACL-MIB • HUAWEI-ALARM-MIB • HUAWEI-ALARM-RELIABILITY-MIB • HUAWEI-BASE-TRAP-MIB • HUAWEI-BRAS-RADIUS-MIB • HUAWEI-BRAS-SRVCFG-EAP-MIB • HUAWEI-BRAS-SRVCFG-STATICUSER-MIB • HUAWEI-CBQOS-MIB • HUAWEI-CDP-COMPLIANCE-MIB • HUAWEI-CONFIG-MAN-MIB • HUAWEI-CPU-MIB • HUAWEI-DAD-TRAP-MIB • HUAWEI-DC-MIB • HUAWEI-DATASYNC-MIB • HUAWEI-DEVICE-MIB • HUAWEI-DHCPR-MIB • HUAWEI-DHCPS-MIB • HUAWEI-DHCP-SNOOPING-MIB • HUAWEI-DIE-MIB • HUAWEI-DNS-MIB • HUAWEI-DLDP-MIB • HUAWEI-ELMI-MIB

Category	MIB
	• HUAWEI-ERPS-MIB • HUAWEI-ERRORDOWN-MIB • HUAWEI-ENERGYMNGT-MIB • HUAWEI-EASY-OPERATION-MIB • HUAWEI-ENTITY-EXTENT-MIB • HUAWEI-ENTITY-TRAP-MIB • HUAWEI-ETHARP-MIB • HUAWEI-ETHOAM-MIB • HUAWEI-FLASH-MAN-MIB • HUAWEI-FWD-RES-TRAP-MIB • HUAWEI-GARP-APP-MIB • HUAWEI-GTSM-MIB • HUAWEI-HGMP-MIB • HUAWEI-HWTACACS-MIB • HUAWEI-IF-EXT-MIB • HUAWEI-INFOCENTER-MIB • HUAWEI-IPPOOL-MIB • HUAWEI-IPV6-MIB • HUAWEI-ISOLATE-MIB • HUAWEI-L2IF-MIB • HUAWEI-L2MAM-MIB • HUAWEI-L2VLAN-MIB • HUAWEI_LDT-MIB • HUAWEI-LLDP-MIB • HUAWEI-MAC-AUTHEN-MIB • HUAWEI-MEMORY-MIB • HUAWEI-MFF-MIB • HUAWEI-MFLP-MIB • HUAWEI-MSTP-MIB • HUAWEI-MULTICAST-MIB • HUAWEI-NAP-MIB • HUAWEI-NTPV3-MIB • HUAWEI-PERFORMANCE-MIB • HUAWEI-PORT-MIB • HUAWEI-PORTAL-MIB • HUAWEI-QINQ-MIB • HUAWEI-RIPv2-EXT-MIB • HUAWEI-RM-EXT-MIB • HUAWEI-RRPP-MIB • HUAWEI-SECURITY-MIB • HUAWEI-SEP-MIB • HUAWEI-SNMP-EXT-MIB • HUAWEI-SSH-MIB • HUAWEI-STACK-MIB • HUAWEI-SWITCH-L2MAM-EXT-MIB

Category	MIB
	• HUAWEI-SWITCH-SRV-TRAP-MIB • HUAWEI-SYS-MAN-MIB • HUAWEI-TCP-MIB • HUAWEI-TFTPC-MIB • HUAWEI-TRNG-MIB • HUAWEI-XQOS-MIB

NOTE

For more information about MIBs supported by CloudEngine S6750-S switches, visit: https://support.huawei.com/enterprise/en/switches/s6700-pid-6691593?category=reference-guides

Standards Compliance

The following table lists the standards that CloudEngine S6750-S switches comply with.

Standard Organization	Standard or Protocol
IETF	• RFC 768 User Datagram Protocol (UDP) • RFC 792 Internet Control Message Protocol (ICMP) • RFC 793 Transmission Control Protocol (TCP) • RFC 826 Ethernet Address Resolution Protocol (ARP) • RFC 854 Telnet Protocol Specification • RFC 951 Bootstrap Protocol (BOOTP) • RFC 959 File Transfer Protocol (FTP) • RFC 1058 Routing Information Protocol (RIP) • RFC 1112 Host extensions for IP multicasting • RFC 1157 A Simple Network Management Protocol (SNMP) • RFC 1256 ICMP Router Discovery • RFC 1305 Network Time Protocol Version 3 (NTP) • RFC 1349 Internet Protocol (IP) • RFC 1493 Definitions of Managed Objects for Bridges • RFC 1542 Clarifications and Extensions for the Bootstrap Protocol • RFC 1643 Ethernet Interface MIB • RFC 1757 Remote Network Monitoring (RMON) • RFC 1901 Introduction to Community-based SNMPv2 • RFC 1902-1907 SNMP v2 • RFC 1981 Path MTU Discovery for IP version 6 • RFC 2131 Dynamic Host Configuration Protocol (DHCP) • RFC 2328 OSPF Version 2 • RFC 2453 RIP Version 2 • RFC 2460 Internet Protocol, Version 6 Specification (IPv6) • RFC 2461 Neighbor Discovery for IP Version 6 (IPv6) • RFC 2462 IPv6 Stateless Address Auto configuration • RFC 2463 Internet Control Message Protocol for IPv6 (ICMPv6) • RFC 2474 Differentiated Services Field (DS Field) • RFC 2475 An Architecture for Differentiated Services • RFC 2740 OSPF for IPv6 (OSPFv3)

Standard Organization	Standard or Protocol
	• RFC 2863 The Interfaces Group MIB • RFC 2597 Assured Forwarding PHB Group • RFC 2598 An Expedited Forwarding PHB • RFC 2571 SNMP Management Frameworks • RFC 2865 Remote Authentication Dial In User Service (RADIUS) • RFC 3046 DHCP Option82/Relay • RFC 3376 Internet Group Management Protocol, Version 3 (IGMPv3) • RFC 3513 IP Version 6 Addressing Architecture • RFC 3579 RADIUS Support For EAP • RFC 4271 A Border Gateway Protocol 4 (BGP-4) • RFC 4760 Multiprotocol Extensions for BGP-4 • draft-grant-tacacs-02 TACACS+ • RFC5340 OSPF for IPv6 • RFC 5798 Virtual Router Redundancy Protocol (VRRP) Version 3 for IPv4 and IPv6 • RFC 6241 Network Configuration Protocol (NETCONF) • RFC 6020 YANG - A Data Modeling Language for the Network Configuration Protocol (NETCONF) • RFC 7348 Virtual eXtensible Local Area Network (VXLAN): A Framework for Overlaying Virtualized Layer 2 Networks over Layer 3 Networks • RFC 8365 A Network Virtualization Overlay Solution Using Ethernet VPN (EVPN)
IEEE	• IEEE 802.1D Media Access Control (MAC) Bridges • IEEE 802.1p Traffic Class Expediting and Dynamic Multicast Filtering • IEEE 802.1Q Virtual Bridged Local Area Networks • IEEE 802.1ad Provider Bridges • IEEE 802.2 Logical Link Control • IEEE Std 802.3 CSMA/CD • IEEE Std 802.3ab 1000BASE-T specification • IEEE Std 802.3ad Aggregation of Multiple Link Segments • IEEE Std 802.3ae 10GE WEN/LAN Standard • IEEE Std 802.3x Full Duplex and flow control • IEEE Std 802.3z Gigabit Ethernet Standard • IEEE 802.1ax/IEEE802.3ad Link Aggregation • IEEE 802.3ah Ethernet in the First Mile. • IEEE 802.1ag Connectivity Fault Management • IEEE 802.1ab Link Layer Discovery Protocol • IEEE 802.1D Spanning Tree Protocol • IEEE 802.1w Rapid Spanning Tree Protocol • IEEE 802.1s Multiple Spanning Tree Protocol • IEEE 802.1x Port based network access control protocol • IEEE 802.3az Automatic power adjustment on Ethernet interfaces
ITU	• ITU SG13 Y.17ethoam • ITU SG13 QoS control Ethernet-Based IP Access • ITU-T Y.1731 ETH OAM performance monitor

Standard Organization	Standard or Protocol
ISO	• ISO 10589 IS-IS Routing Protocol
MEF	• MEF 2 Requirements and Framework for Ethernet Service Protection • MEF 9 Abstract Test Suite for Ethernet Services at the UNI • MEF 10.2 Ethernet Services Attributes Phase 2 • MEF 11 UNI Requirements and Framework • MEF 13 UNI Type 1 Implementation Agreement • MEF 15 Requirements for Management of Metro Ethernet Phase 1 Network Elements • MEF 17 Service OAM Framework and Requirements • MEF 20 UNI Type 2 Implementation Agreement • MEF 23 Class of Service Phase 1 Implementation Agreement • Xmodem XMODEM/YMODEM Protocol Reference

NOTE

The listed standards and protocols are fully or partially supported by Huawei switches. For details, visit http://e.huawei.com/en or contact your local Huawei sales office.

Ordering Information

The following table lists ordering information of CloudEngine S6750-S switches.

Model	Product Description
CloudEngine S6750-S16X8YZ	S6750-S16X8YZ(16*10GE SFP+ ports, 8*25GE SFP28 ports, expansion card slot, without power module)
CloudEngine S6750-S24T16X8Y2CZ	S6750-S24T16X8Y2CZ(24*10/100/1000BASE-T ports, 16*10GE SFP+ ports, 8*25GE SFP28 ports, 2*100GE QSFP28 ports, expansion card slot, without power module)
CloudEngine S6750-S16X10Y2CZ	S6750-S16X10Y2CZ (16*10GE SFP+ ports, 10*25GE SFP28 ports, 2*100GE QSFP28 ports, expansion card slot, without power module)
PAC180S12-CN	180W AC power module
PAC600S12-PB	600W AC power module
PDC240S12-CN	240W AC power module
PDC400S12-CB	400W DC power module
FAN-031A-B	Fan Module
HSIC-X08S000	8-port 1/10GE SFP+ or 4-port 10/25GE interface card
HSIC-Y08S000	8-port 1/10/25GE SFP+ interface card

License	Product Description
L-VxLAN-S67	S67 Series, VxLAN License, Per Device
N1-S67S-M-Lic	S67XX-S Series Basic SW,Per Device
N1-S67S-M-SnS1Y	S67XX-S Series Basic SW,SnS,Per Device,1Year

License	Product Description
N1-S67S-F-Lic	N1-CloudCampus,Foundation,S67XX-S Series,Per Device
N1-S67S-F-SnS	N1-CloudCampus,Foundation,S67XX-S Series,SnS,Per Device(Annual fee validity period:3 years from " 90 days after PO signed ")
N1-S67S-A-SnS	N1-CloudCampus,Advanced,S67XX-S Series,SnS,Per Device(Annual fee validity period:3 years from " 90 days after PO signed ")
N1-S67S-FToA-Lic	N1-Upgrade-Foundation to Advanced,S67XX-S,Per Device
N1-S67S-FToA-SnS	N1-Upgrade-Foundation to Advanced,S67XX-S,SnS,Per Device(Annual fee validity period:3 years from " 90 days after PO signed ")
N1-S67S-A-Lic	N1-CloudCampus,Advanced,S67XX-S Series,Per Device
N1-AM-30-Lic	N1-CloudCampus, Add-On Package, Access Management, Per 30 Endpoints
N1-AM-30-SnS	N1-CloudCampus, Add-On Package, Access Management, Software Subscription and Support, Per 30 Endpoints(Annual fee validity period:3 years from " 90 days after PO signed ")
N1-EPNP-30-Lic	N1-CloudCampus, Add-On Package, Endpoints Plug and Play, Per 30 Endpoints
N1-EPNP-30-SnS	N1-CloudCampus, Add-On Package, Endpoints Plug and Play, Software Subscription and Support, Per 30 Endpoints(Annual fee validity period:3 years from " 90 days after PO signed ")
N1-APP-X7FSwitch	N1-CloudCampus, Add-On Package, Intelligent Application Analysis, X7 Series Fixed Switch, Per Device
N1-APP-X7FSwitch-SnS	N1-CloudCampus, Add-On Package, Intelligent Application Analysis, X7 Series Fixed Switch, Software Subscription and Support, Per Device(Annual fee validity period:3 years from " 90 days after PO signed ")

More Information

For more information about the Huawei Campus Switches, visit http://e.huawei.com or contact us in the following ways:

- Global service hotline: http://e.huawei.com/en/service-hotline
- Logging in to the Huawei Enterprise technical support website: http://support.huawei.com/enterprise/
- Sending an email to the customer service mailbox: support_e@huawei. com

Huawei Technologies Co., Ltd.

Address:Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China

Website:e.huawei.com

Huawei CloudEngine S5735I-S-V2 Series Industry Switches (DIN-rail) Datasheet

Huawei CloudEngine S5735I-S-V2 series industry switches are standard industry switches that provide GE downlink ports and GE, 10GE SFP+ uplink ports.

Introduction

Huawei CloudEngine S5735I-S-V2 series industry switches (S5735I-S-V2 for short) are next-generation standard Layer 3 gigabit switches that provide flexible all-gigabit access and GE/10GE uplink ports.

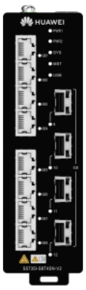
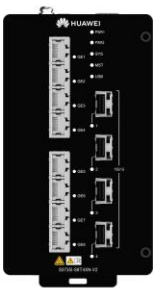
Industry switches have an industrial-grade operating temperature range as well as professional outdoor surge protection to withstand harsh outdoor environments. As such, they can be widely used in ultra-broadband operating temperature scenarios, such as smart manufacturing, smart mining, smart transportation, safe city, and electric power.

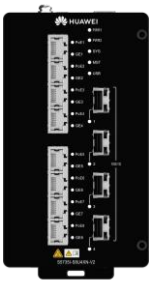
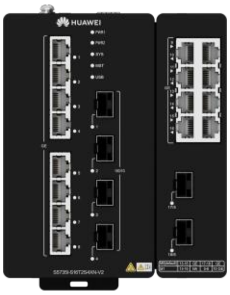
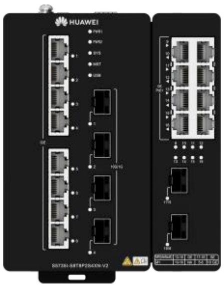
Product Overview

Models and Appearances

The following models are available in the CloudEngine S5735I-S-V2 series.

Models and appearances of the CloudEngine S5735I-S-V2 series

Models and Appearances	Description
 CloudEngine S5735I-S8T4SN-V2	• 8 x 10/100/1000Base-T Ethernet ports, 4 x GE SFP ports, 1 x DI/DO, 1 x RS485 • DC external or AC adapter • 1+1 power supply backup • Forwarding performance: 18 Mpps • Switching capacity*:24 Gbps/520 Gbps
 CloudEngine S5735I-S8T4XN-V2	• 8 x 10/100/1000Base-T Ethernet ports, 4 x 10GE SFP+ ports, 1 x DI/DO, 1 x RS485 • DC external or AC adapter • 1+1 power supply backup • Forwarding performance: 72 Mpps • Switching capacity*:96 Gbps/520 Gbps

Models and Appearances	Description
 CloudEngine S5735I-S8U4XN-V2	• 8 x 10/100/1000Base-T Ethernet ports, 4 x 10GE SFP+ ports, 1 x DI/DO, 1 x RS485 • DC external or AC adapter • 1+1 power supply backup • PoE++ • Forwarding performance: 72 Mpps • Switching capacity*:96Gbps /520 Gbps **45 - S5735I-S8U4XN-V2 (8*10/100/1000BASE-T ports, 4*10GE SFP+ ports, PoE++, DIN Rail Mounting, Dual redundant 54 to 57V DC power, Fanless)**
 CloudEngine S5735I-S8U2XN-V2	• 8 x 10/100/1000Base-T Ethernet ports, 2 x 10GE SFP+ ports • DC external or AC adapter • 1+1 power supply backup • PoE++ • Forwarding performance: 42 Mpps • Switching capacity*:56Gbps /520 Gbps
 CloudEngine S5735I-S16T2S4XN-V2	• 16 x 10/100/1000Base-T Ethernet ports, 2 x GE SFP ports, 4 x 10GE SFP+ ports, 1 x DI/DO, 1 x RS485 • DC external or AC adapter • 1+1 power supply backup • Forwarding performance: 87 Mpps • Switching capacity*:116 Gbps/520 Gbps
 CloudEngine S5735I-S8T8P2S4XN-V2	• 8 x 10/100/1000Base-T Ethernet ports, 8 x 10/100/1000Base-T Ethernet ports(PoE+), 2 x GE SFP ports, 4 x 10GE SFP+ ports, 1 x DI/DO, 1 x RS485 • DC external or AC adapter • PoE+ • 1+1 power supply backup • Forwarding performance: 87 Mpps • Switching capacity*:116 Gbps/520 Gbps

*Note：The value before the slash (/) refers to the device's switching capability, while the value after the slash (/) means the system's switching capability.

**Note：‘-T ’means Trusted Platform Module(HTM), support hardware root of trust and measurement startup.

Power Supply

Technical specifications of the power supplies applicable to the CloudEngine S573I5-S-V2 series

Power Module	Technical Specifications	Applied Switch Model

Power Module	Technical Specifications	Applied Switch Model
 PAC60S12-AN	• Dimensions (H x W x D): 150 mm x 40 mm x 130 mm • Weight: 0.9 kg • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 100V DC to 250 V DC • Maximum input voltage range: – 90 V AC to 290 V AC – 88 V DC to 300 V DC • Rated input current: – 2 A • Rated output voltage: – 12 V DC • Rated output current: 5 A • Rated output power: – 60 W • Hot swap: Supported • Operating temperature: -40℃ to 70℃ • Thermal Efficiency: 85% • Maximum Heat Dissipation (BTU): 30.71	• CloudEngine S5735I-S8T4SN-V2 • CloudEngine S5735I-S8T4XN-V2 • CloudEngine S5735I-S8T4XN-T-V2
 PAC240S56-CN	• Dimensions (H x W x D): 150 mm x 60 mm x 133 mm • Weight: 1.47 kg • Rated input voltage range: – 100 V AC to 240 V AC, 50/60 Hz – 100V DC to 250 V DC • Maximum input voltage range: – 90 V AC to 290 V AC, 45/66 Hz – 77 V DC to 300 V DC • Maximum Input current: – 100V AC～240V AC：3 A – 100V DC～138V DC：2.5 A – 138V DC～250V DC：2 A • Rated output voltage: – 56 V DC • Rated output power: – 240W total (PoE output 220W) • Hot swap: Supported • Operating temperature: -40℃ to 70℃ • Thermal Efficiency: 90% • Maximum Heat Dissipation (BTU): 81.88	• CloudEngine S5735I-S8U4XN-V2 • CloudEngine S5735I-S8U2XN-V2

The following table lists its power supply configurations.

Power supply configurations of CloudEngine S5735I-S-V2

Module	Power Module 1	Power Module 2	Available PoE Power	Maximum Number of Ports (Fully Loaded)
CloudEngine S5735I-S8U4XN-V2	PAC240S56-CN	-	210 W	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 7 • 802.3bt (60 W per port): 3 • 802.3bt (90 W per port): 2
	PAC240S56-CN	PAC240S56-CN	210 W	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 7 • 802.3bt (60 W per port): 3 • 802.3bt (90 W per port): 2
	External power module with 56 V DC power supply	-	400 W at most	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 8 • 802.3bt (60 W per port): 6 • 802.3bt (90 W per port): 4
	External power module with 56 V DC power supply	External power module with 56 V DC power supply	400 W at most	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 8 • 802.3bt (60 W per port): 6 • 802.3bt (90 W per port): 4
CloudEngine S5735I-S8U2XN-V2	PAC240S56-CN		210 W	• 802.3af (15.4 W per port): 13 • 802.3at (30 W per port): 7 • 802.3bt (60 W per port): 3 • 802.3bt (90 W per port): 2
	PAC240S56-CN	PAC240S56-CN	240 W	• 802.3af (15.4 W per port): 15 • 802.3at (30 W per port): 8 • 802.3bt (60 W per port): 4 • 802.3bt (90 W per port): 2
CloudEngine S5735I-S8T8P2S4XN-V2	PAC240S56-CN	-	200 W	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 6
	PAC240S56-CN	PAC240S56-CN	240 W	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 8
	External power module with 56 V DC power supply	-	240 W (maximum)	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 8
	External power module with 56 V DC power supply	External power module with 56 V DC power supply	240 W (maximum)	• 802.3af (15.4 W per port): 8 • 802.3at (30 W per port): 8

Product Features and Highlights

Powerful Service Processing Capability

- CloudEngine S5735I-S-V2 supports a broad set of Layer 2/Layer 3 multicast protocols, such as PIM SM, PIM DM, PIM SSM, and IGMP snooping. This capability is ideal for high-definition video backhaul and video conferencing access.

- CloudEngine S5735I-S-V2 provides multiple Layer 3 features including OSPF, IS-IS, BGP, and VRRP, meeting enterprises' access and aggregation service needs and enabling a variety of voice, video, and data applications.

Multiple Security Control Mechanisms

- CloudEngine S5735I-S-V2 supports MAC address authentication, 802.1X authentication, and implements dynamic delivery of policies (VLAN, QoS, and ACL) to users.
- CloudEngine S5735I-S-V2 provides a series of mechanisms to defend against DoS attacks and user-targeted attacks. DoS attacks are targeted at switches and include SYN flood, Land, Smurf, and ICMP flood attacks. User-targeted attacks include bogus DHCP server attacks, IP/MAC address spoofing, DHCP request flood, and changing of the DHCP CHADDR value.
- CloudEngine S5735I-S-V2 sets up and maintains a DHCP snooping binding table, and discards the packets that do not match the table entries. The DHCP snooping trusted port feature ensures that users connect only to the authorized DHCP server.
- CloudEngine S5735I-S-V2 supports strict ARP learning, which protects a network against ARP spoofing attacks to ensure that users can connect to the Internet normally.
- CloudEngine S5735I-H-V2 supports policy association, user permission policy management and policy execution, and user permission association switchover based on the authentication status.
- CloudEngine S5735I-S16T2S4XN-V2 & CloudEngine S5735I-S8T8P2S4XN-V2 supports Media Access Control Security (MACsec) with all downlink ports and uplink ports . It provides identity authentication, data encryption, integrity check, and replay protection to protect Ethernet frames and prevent attack packets.

Multiple Reliability Mechanisms

- CloudEngine S5735I-S-V2 supports a single power module or two power modules. When two power modules are used, the power modules work in 1+1 backup mode. The can be directly connected to an external DC power supply or powered by a power module.
- In addition to traditional Spanning Tree Protocol (STP), Rapid Spanning Tree Protocol (RSTP), and Multiple Spanning Tree Protocol (MSTP), the CloudEngineS5735I-S-V2 supports Huawei-developed Smart Ethernet Protection (SEP) technology, Media Redundancy Protocol (MRP) technology and the latest Ethernet Ring Protection Switching (ERPS) standard. ERPS is defined in ITU-T G.8032. It implements 10ms fast protection switching based on traditional Ethernet MAC and bridging functions.
- CloudEngine S5735I-S-V2 supports Smart Link, which implements backup of uplinks. One CloudEngine S5735I-S switch can connect to multiple aggregation switches through multiple links, significantly improving reliability of access devices.

Easy Network deployment

- CloudEngine S5735I-S-V2 supports Huawei Easy Operation, a solution that provides zero-touch deployment, replacement of faulty devices without additional configuration, USB-based deployment, batch device configuration, and batch remote upgrade. The capabilities facilitate device deployment, upgrade, service provisioning, and other management and maintenance operations, and also greatly reduce O&M costs. CloudEngine S5735I-S-V2 can be managed using SNMP v1/v2c/v3, CLI, web-based network management system, or SSH v2.0. Additionally, it supports RMON, multiple log hosts, port traffic statistics collection, and network quality analysis, which facilitate network optimization and reconstruction.
- CloudEngine S5735I-S-V2 uses the Packet Conservation Algorithm for Internet (iPCA) technology that changes the traditional method of using simulated traffic for fault location. iPCA technology can monitor network quality for any service flow anywhere and anytime, without extra costs. It can detect temporary service interruptions in a very short time and can identify faulty ports accurately. This cutting-edge fault detection technology turns "extensive management" to "fine granular management."

Note: CloudEngine S5735I-S8T4SN-V2 doesn't support USB port.

Mature IPv6 Technologies

- CloudEngine S5735I-S-V2 supports IPv4/IPv6 dual stack, IPv6 RIPng, BGP4+, OSPFv3.
- CloudEngine S5735I-S-V2 can be deployed on a pure IPv4 network, a pure IPv6 network, or a shared IPv4/IPv6 network, helping achieve IPv4-to-IPv6 transition.

Intelligent Stack (iStack)

- CloudEngine S5735I-S-V2 supports intelligent stack (iStack). This technology combines multiple switches into a logical switch. Member switches in a stack implement redundancy backup to improve device reliability and use inter-device link aggregation to improve link reliability.

- iStack provides high network scalability. You can increase ports, bandwidth, and processing capacity of a stack by simply adding member switches to the stack.
- iStack also simplifies device configuration and management. After a stack is set up, multiple physical switches are virtualized into one logical device. You can log in to any member switch in the stack to manage all the member switches in the stack. CloudEngine S5735I-S-V2 support stacking through electrical ports.

Note: DI/DO & RS485 ports are unavailable when member switches in a stack.

PoE Function

CloudEngine S5735I-S-V2 PoE models can support PoE++(up to 90W power supply), Meeting high-power power supply requirements for Wi-Fi 6 APs, IP cameras, and Video phones

- **Perpetual PoE**: When a PoE switch is abnormal Power-off or the software version is upgraded, the power supply to PDs is not interrupted. This capability ensures that PDs are not powered off during the switch reboot.
- **Fast PoE**: PoE switches can supply power to PDs within seconds after they are powered on. This is different from common switches that generally take 1 to 3 minutes to start to supply power to PDs. When a PoE switch reboots due to a power failure, the PoE switch continues to supply power to the PDs immediately after being powered on without waiting until it finishes reboot. This greatly shortens the power failure time of PDs.

Intelligent O&M

- CloudEngine S5735I-S-V2 provides telemetry technology to collect device data in real time and send the data to Huawei campus network analyzer CampusInsight. The CampusInsight analyzes network data based on the intelligent fault identification algorithm, accurately displays the real-time network status, effectively demarcates and locates faults in a timely manner, and identifies network problems that affect user experience, accurately guaranteeing user experience.
- CloudEngine S5735I-S-V2 supports WebMaster solution, WebMaster is an embedded local management system. It automatically discovers network topology and provides full-network visibility, including topology views, network elements (NEs), and device alarms. The solution also enables unified management of switches, AR routers, and APs—without requiring an external network management system.WebMaster offers a graphical interface for intuitive operations, including guided service provisioning, one-click batch upgrades, zero-configuration device replacement, and automatic authentication and onboarding of NEs. It can also detect loops caused by incorrect cable connections and eliminate them automatically.With its rich feature set, WebMaster enables the entire campus network to be visible, manageable, and controllable—all through a single device.

Intelligent Upgrade

- CloudEngine S5735I-S-V2 supports the intelligent upgrade feature. Specifically, CloudEngine S5735I-S-V2 obtains the version upgrade path and downloads the newest version for upgrade from the Huawei Online Upgrade Platform (HOUP). The entire upgrade process is highly automated and achieves one-click upgrade. In addition, preloading the version is supported, which greatly shortens the upgrade time and service interruption time.
- The intelligent upgrade feature greatly simplifies device upgrade operations and makes it possible for the customer to upgrade the version independently. This greatly reduces the customer's maintenance costs. In addition, the upgrade policies on the HOUP platform standardize the upgrade operations, which greatly reduces the risk of upgrade failures.

Cloud Management

- The Huawei cloud management platform allows users to configure, monitor, and inspect switches on the cloud, reducing on-site deployment and O&M manpower costs and decreasing network OPEX. Huawei switches support both cloud management and on-premise management modes. These two management modes can be flexibly switched as required to achieve smooth evolution while maximizing return on investment (ROI).

OPS(Open Programmability System)

- CloudEngine S5735I-S-V2 supports Open Programmability System (OPS), an open programmable system based on the Python language. IT administrators can program the O&M functions of a CloudEngine S5735I-S-V2 switch through Python scripts to quickly innovate functions and implement intelligent O&M.

Licensing

CloudEngine S5735I-S-V2 supports both the traditional feature-based licensing mode and the latest Huawei IDN One Software (N1 mode for short) licensing mode. The N1 mode is ideal for deploying Huawei CloudCampus Solution in the on-premises scenario, as it greatly enhances the customer experiences in purchasing and upgrading software services with simplicity.

Software Package Features in N1 Mode

Switch Functions	N1 Basic Software	N1 Foundation Software Package	N1 Advanced Software Package
Basic network functions: Layer 2 functions, IPv4, IPv6 and others Note: For details, see the Service Features	√	√	√
Basic network automation based on the iMaster NCE-Campus: • NE management: Device management, topology management and discovery • User access authentication	×	√	√
Advanced network automation and intelligent O&M: IPCA, CampusInsight basic functions	×	×	√

Product Specifications

Functions and Features

Item	Description
MAC address table	IEEE 802.1d compliance
	32K MAC entries
	MAC address learning and aging
	Static, dynamic, and blackhole MAC address entries
	Packet filtering based on source MAC addresses
VLAN	4K VLANs simultaneously
	1K VLANif interface simultaneously
	Sub-VLAN, Super-VLAN, Mux VLAN, Voice VLAN
	QinQ and enhanced selective QinQ
	VLAN Stacking, VLAN Mapping
	LNP, VCMP, GVRP
Ethernet loop protection	Smart Link tree topology and Smart Link multi-instance, providing millisecond-level protection switchover
	SEP
	STP (IEEE 802.1d), RSTP (IEEE 802.1w), and MSTP (IEEE 802.1s)
	ERPS (G.8032)

Item	Description
	MRP (IEC62439-2)
	BPDU protection, root protection, and loop protection
	LBDT
	Y.1731
IP routing	Static route, RIPv1/v2, RIPng, OSPF, OSPFv3, ECMP, IS-IS, IS-ISv6, BGP, BGP4+, VRRP, and VRRP6
	Up to 8192 FIBv4 entries
	Up to 3072 FIBv6 entries
IPv6 features	Up to 3072 ND entries
	Path MTU (PMTU)
	IPv6 ping, IPv6 tracert, and IPv6 Telnet
Reliability	LACP
	VRRP
	BFD
	LLDP
	Forwarding latency＜4μs
Multicast	PIM DM, PIM SM, PIM SSM, , PIMv6
	IGMP v1/v2/v3, IGMP v1/v2/v3 snooping and IGMP fast leave
	Multicast load balancing among member ports of a trunk
	Port-based multicast traffic statistics
	Multicast VLAN
QoS/ACL	Rate limiting on packets sent and received by a port
	Packet redirection
	Port-based traffic policing and two-rate three-color CAR
	Eight queues on each port
	DRR, SP and DRR+SP queue scheduling algorithms
	Re-marking of the 802.1p priority and DSCP priority
	Packet filtering at Layer 2 to Layer 4, filtering out invalid frames based on the source MAC address, destination MAC address, source IP address, destination IP address, TCP/UDP port number, protocol type, and VLAN ID
	Rate limiting in each queue and traffic shaping on ports
	Network Slicing（VLAN）
Industrial protocol	Profinet RT, Ethernet/IP, Modbus TCP, OPC UA and GOOSE mainstream industrial protocol forwarding
	Profinet configuration protocol and PI CC-B.
Security	Hierarchical user management and password protection
	DoS attack defense, ARP attack defense, and ICMP attack defense

Item	Description
	Binding of the IP address, MAC address, port number, and VLAN ID
	Port isolation, port security, and sticky MAC
	Blackhole MAC address entries
	Limit on the number of learned MAC addresses
	IEEE 802.1x authentication and limit on the number of users on a port
	AAA authentication, RADIUS authentication, HWTACACS authentication, and NAC
	SSH v2.0
	HTTPS
	CPU defense
	Blacklist and whitelist
	IEEE 802.1x authentication, MAC address authentication
	DHCPv4 client/relay/server/snooping
	DHCPv6 client/relay
	Attack source tracing and punishment for IPv6 packets such as ND, DHCPv6
	ND snooping
	MACsec AES256 encryption*
	Cyber security certification (IEC62443-4-2 SL2)
Management and maintenance	iStack**
	Cloud management based on Netconf/Yang
	Virtual cable test
	SNMP v1/v2c/v3
	RMON
	Web-based NMS
	System logs and alarms of different levels
	802.3az EEE
	GVRP
	Native-IP IFIT (iPCA2.0), sFlow, NQA, Telemetry
	1588v2 TC
	WebMaster Management
	Terminal identification
Interoperability	Supports VBST (Compatible with PVST/PVST+/RPVST)

**Note: The S5735I-S8T8P2S4XN-V2 and S5735I-S16T2S4XN-V2 models support this function.*

***Note: iStack will be supported in R22C10 version, DI/DO & RS485 ports are unavailable when member switches in a stack.*

Hardware Specifications

Hardware specifications of the CloudEngine S5735I-S8T4SN-V2/-S8T4XN-V2/-S8T4XN-T-V2/-S8U4XN-V2 models

Item		CloudEngine S5735I-S8T4SN-V2	CloudEngine S5735I-S8T4XN-V2	CloudEngine S5735I-S8U4XN-V2
Physical specifications	Dimensions (H x W x D, mm)	150.0 mm x 46.0 mm x 133.0 mm	150.0 mm x 86.0 mm x 133.0 mm	150.0 mm x 86.0 mm x 133.0 mm
	Chassis weight (including packaging)	1.23 kg	2.11 kg	2.21 kg
Fixed port	GE port	8	8	8(PoE++)
	GE SFP port	4	NA	NA
	10GE SFP+ port	NA	4	4
Management port	Console port (RJ45)	Supported	Supported	Supported
	USB port	Not supported	USB 2.0	USB 2.0
CPU	Frequency	1.1 GHz	1.1 GHz	1.1 GHz
	Cores	2	2	2
Storage	Memory (RAM)	2 GB	2 GB	2 GB
	Flash memory	1 GB in total. To view the available flash memory size, run the display	1 GB in total. To view the available flash memory size, run the display	1 GB in total. To view the available flash memory size, run the display
Power supply system	Power supply type	60W AC (AC power adapter) or DC external	60W AC (AC power adapter) or DC external	240W AC (AC PoE power adapter) or DC external
	Power supply redundancy	1:1 hot backup	1:1 hot backup	1:1 hot backup
	Rated voltage range	• DC input: 12V DC～48V DC	• DC input: 12V DC～48V DC	• DC input: 56V DC
	Maximum voltage range	• DC input: 9.6V DC～60V DC	• DC input: 9.6V DC～60V DC	• DC input: 54–57 V DC (PoE/PoE+/PoE++) or 48 V DC (PoE)
	Maximum input current	2 A	2 A	8 A
	Maximum power consumption of the device	18.59 W	20.35 W	• 21.7 W (without PDs; When with PDs, max PD power consumption is 400 W)*
	Typical power consumption	17.44 W	19.94 W	20.45 W
Heat dissipation system	Heat dissipation mode	Natural heat dissipation	Natural heat dissipation	Natural heat dissipation
	Number of fan modules	0	0	0

Item		CloudEngine S5735I-S8T4SN-V2	CloudEngine S5735I-S8T4XN-V2	CloudEngine S5735I-S8U4XN-V2
	Airflow	NA	NA	NA
	Maximum heat dissipation of the device (BTU/hour)	63.4	69.4	• 74.01 (Without PDs; When with PDs, max PD heat dissipation is 1364.1 BTU/hour)
Environment parameters	Long-term operating temperature	0–1800 m altitude, industry optical modules: -40°C to +75°C	0–1800 m altitude, industry optical modules: -40°C to +75°C Configured with PEN optical modules: 0°C～+60°C	0–1800 m altitude, industry optical modules: -40°C to +75°C Configured with PEN optical modules: 0°C～+60°C
	Short-term operating temperature[3]	NA	NA	NA
	Storage temperature	-40°C to +85°C	-40°C to +85°C	-40°C to +85°C
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	5% to 95% (non-condensing)
	Operating altitude	5000 m	5000 m	5000 m
	Noise under normal temperature (sound power)	Noise-free (no fans)	Noise-free (no fans)	Noise-free (no fans)
	Noise under high temperature (sound power)	Noise-free (no fans)	Noise-free (no fans)	Noise-free (no fans)
	Noise under normal temperature (sound pressure)	Noise-free (no fans)	Noise-free (no fans)	Noise-free (no fans)
	Ingress protection level	IP40	IP40	IP40
	Surge protection specification (RJ45 service port)	±7 kV in common mode	±7 kV in common mode	±7 kV in common mode
	Surge protection specification (power port)	• Using DC power modules: ±2 kV in differential mode, ±1 kV in common mode	• Using DC power modules: ±2 kV in differential mode, ±1 kV in common mode	• Using DC power modules: ±2 kV in differential mode, ±1 kV in common mode
Reliability	MTBF	104.86 years (918,573.6 hours)	101.33 years (887,650.8 hours)	89.43 years (783,406.8 hours)
	MTTR (hours)	0.5	0.5	0.5
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification

Hardware specifications of the CloudEngine S5735I-S16T2S4XN-V2/S5735I-S8T8P2S4XN-V2 models

Item		CloudEngine S5735I-S8U2XN-V2	CloudEngine S5735I-S16T2S4XN-V2	CloudEngine S5735I-S8T8P2S4XN-V2
Physical specifications	Dimensions (H x W x D, mm)	150.0 mm x 44.0 mm x 133.0 mm	150.0 mm x 127.0 mm x 133.0 mm	150.0 mm x 127.0 mm x 133.0 mm
	Chassis weight (including packaging)	1.3kg	3.02kg	3.07kg
Fixed port	GE port	8	16	16(8 PoE+)
	GE SFP port	NA	2	2
	10GE SFP+ port	2	4	4
Management port	Console port (RJ45)	Supported	Supported	Supported
	USB port	USB 2.0	Not supported	USB 2.0
CPU	Frequency	1.1 GHz	1.1 GHz	1.1 GHz
	Cores	2	2	2
Storage	Memory (RAM)	2 GB	2 GB	2 GB
	Flash memory	1 GB in total. To view the available flash memory size, run the display	1 GB in total. To view the available flash memory size, run the display	1 GB in total. To view the available flash memory size, run the display
Power supply system	Power supply type	240W AC (AC PoE power adapter) or DC external	60W AC (AC power adapter) or DC external	240W AC (AC power adapter) or DC external
	Power supply redundancy	1:1 hot backup	1:1 hot backup	1:1 hot backup
	Rated voltage range	DC input: 56V DC	DC input: 12V DC～48V DC	DC input: 56V DC
	Maximum voltage range	DC input: 54–57 V DC (PoE/PoE+/PoE++) or 48 V DC (PoE)	DC input: 9.6V DC～60V DC	DC input: 54–57 V DC (PoE/PoE+) or 48 V DC (PoE)
	Maximum input current	6A	3 A	5 A
	Maximum power consumption of the device	• 18.3W (without PDs; When with PDs, max PD power consumption is 240 W)	31.3W @10.8V 27.5W @48V	• 29.3W (without PDs; When with PDs, max PD power consumption is 240 W)*
	Typical power consumption	11.4W	22.3W	23.2W
Heat dissipation system	Heat dissipation mode	Natural heat dissipation	Natural heat dissipation	Natural heat dissipation
	Number of fan modules	0	0	0

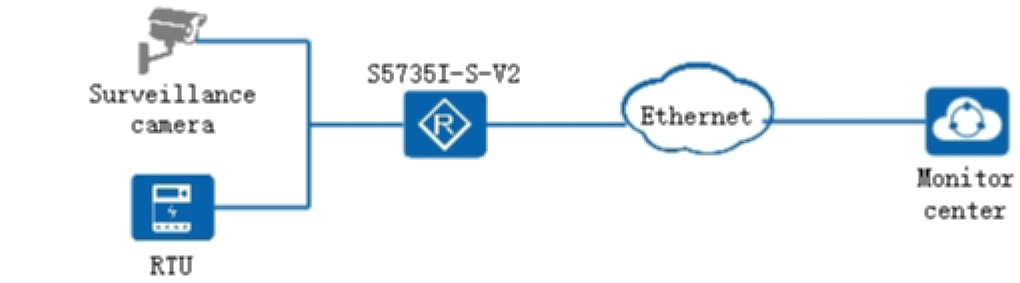
Item		CloudEngine S5735I-S8U2XN-V2	CloudEngine S5735I-S16T2S4XN-V2	CloudEngine S5735I-S8T8P2S4XN-V2
	Airflow	NA	NA	NA
	Maximum heat dissipation of the device (BTU/hour)	• 62.5 (Without PDs; When with PDs, max PD heat dissipation is 818.9 BTU/hour)	106.80 @ 10.8 V 93.83 @ 48 V	• 79.16 (Without PDs; When with PDs, max PD heat dissipation is 818.9 BTU/hour)
Environment parameters	Long-term operating temperature	0–1800 m (0–5906 ft.) altitude, industrial optical modules: -40°C to +75°C Configured with PEN optical modules: 0°C～+60°C	0–1800 m (0–5906 ft.) altitude, industrial optical modules: -40°C to +75°C Configured with PEN optical modules: 0°C～+60°C	0–1800 m (0–5906 ft.) altitude, industrial optical modules: -40°C to +75°C Configured with PEN optical modules: 0°C～+60°C
	Storage temperature	-40°C to +85°C	-40°C to +85°C	-40°C to +85°C
	Relative humidity	5% to 95% (non-condensing)	5% to 95% (non-condensing)	5% to 95% (non-condensing)
	Operating altitude	5000 m	5000 m	5000 m
	Noise under normal temperature (sound power)	Noise-free (no fans)	Noise-free (no fans)	Noise-free (no fans)
	Noise under high temperature (sound power)	Noise-free (no fans)	Noise-free (no fans)	Noise-free (no fans)
	Noise under normal temperature (sound pressure)	Noise-free (no fans)	Noise-free (no fans)	Noise-free (no fans)
	Ingress protection level	IP40	IP40	IP40
	Surge protection specification (RJ45 service port)	±6 kV in common mode	±6 kV in common mode	±6 kV in common mode
	Surge protection specification (power port)	• Using DC power modules: ±4 kV in differential mode, ±2 kV in common mode	• Using DC power modules: ±2 kV in differential mode, ±1 kV in common mode	• Using DC power modules: ±2 kV in differential mode, ±1 kV in common mode
Reliability	MTBF	73 years (639,480 hours)	62.87 years (550,741.2 hours)	56.17 years (492,049.2 hou)
	MTTR (hours)	0.5	0.5	0.5
	Availability	> 0.99999	> 0.99999	> 0.99999
Certification		• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification	• EMC certification • Safety certification • Manufacturing certification

**Note:The maximum PoE output power varies with the temperature. For details, check the product documentation.*

Networking and Applications

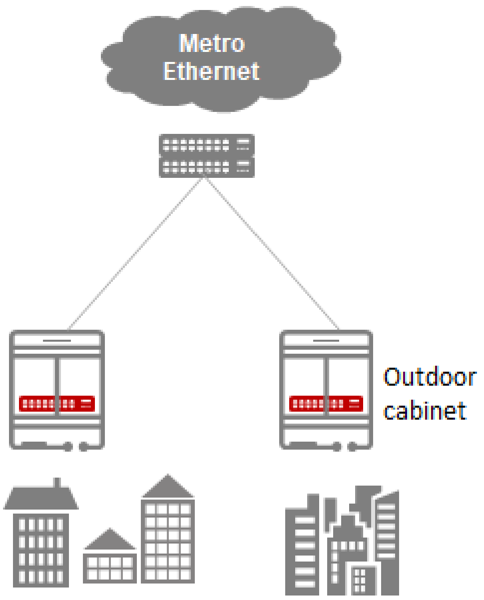
Video surveillance application, outdoor cabinet

CloudEngine S5735I-S-V2 series switches supports extended operating temperature range, with professional surge protection capabilities, suitable for outdoor environment. CloudEngine S5735I-S-V2 series switch can be used for safe city scenario to provide remote access for the camera.



ETTx scenario

CloudEngine S5735I-S-V2 series switches supports extended operating temperature and provides GE access and 10GE uplinks for ETTx access scenarios.



Safety and Regulatory Compliance

Certification Category	Description
Certification Standards	• CE （EN 55032 、EN 55035 、EN 300386、EN62368-1 ） • NRTL （UL62368-1） • IC （ICES-003 ） • RCM（AS/NZS CIPSR32 ） • IEC61850-3/IEEE1613 • CQC（GB4943,GB9254） • VCCI（VCCI-CISPR 32） • RoHS&REACH&WEEE SDOC

Certification Category	Description
	• IEC62443-4-2 SL2
Vibration and shock, environmental testing	• IEC61850-3 • IEEE1613 • EN 50155 • EN 50121-3-2 • EN 50125-3 • NEMA TS2 • ISTA 2A-2011 • IEC 60068-2-64 • IEC 60068-2-27 • IEC 60068-2-31 • ETSI EN 300 019-2-2
EMC	• EN 55032，CLASS A • EN 55035，CLASS A • EN 50155 • IEEE1613 • EN61850-3 • EN 50121-1 • EN 50121-4 • EN 61000-3-2 • EN 61000-3-3 • AS/NZS CISPR 32 CLASS A • CISPR 32 • CISPR 35 • VCCI-CISPR 32 , CLASS A • FCC CFR Title 47, Part 15, Subpart B , Class A • ICES-003 Issue 7 CLASS A • ETSI EN 300386 • IEC 61000-4-2 （ESD • IEC 61000-4-3 （RS） • IEC 61000-4-4 （EFT） • IEC 61000-4-5 （Surge） • IEC 61000-4-6 • IEC 61000-4-8 • IEC 61000-4-10 • IEC 61000-4-16 Conducted CM Disturbances (30V, Cont/ 300V, 1 sec) • IEC 61000-4-17 Ripple Immunity DC Power (10%) • IEC 61000-4-18 Damped Oscillatory Wave (2.5kV, 1MHz) • IEC 61000-4-29 DC Voltage Dips and Interruptions
Safety	• EN 62368-1 • UL 62368-1 • IEC 62368-1 • IEC 61850-3/IEEE 1613

Certification Category	Description
Environment	• ETSI EN 300 019-2-1 • ETSI EN 300 019-2-3 • IEC 60068-2-1 • IEC 60068-2-2 • IEC 60068-2-14 • IEC 60068-2-78 • IEC 60068-2-30 • IEC 60068-2-39

NOTE

- EMC: electromagnetic compatibility
- CISPR: International Special Committee on Radio Interference
- EN: European Standard
- ETSI: European Telecommunications Standards Institute
- CFR: Code of Federal Regulations
- FCC: Federal Communication Commission
- IEC: International Electrotechnical Commission
- AS/NZS: Australian/New Zealand Standard
- VCCI: Voluntary Control Council for Interference
- UL: Underwriters Laboratories
- CSA: Canadian Standards Association
- IEEE: Institute of Electrical and Electronics Engineers
- RoHS: restriction of the use of certain hazardous substances
- REACH: Registration Evaluation Authorization and Restriction of Chemicals
- WEEE: Waste Electrical and Electronic Equipment

MIB and Standards Compliance

Supported MIBs

Category	MIB
Public MIB	• BRIDGE-MIB • DISMAN-NSLOOKUP-MIB • DISMAN-PING-MIB • DISMAN-TRACEROUTE-MIB • ENTITY-MIB • EtherLike-MIB • IF-MIB • IP-FORWARD-MIB • IPv6-MIB • LAG-MIB • LLDP-EXT-DOT1-MIB • LLDP-EXT-DOT3-MIB • LLDP-MIB • NOTIFICATION-LOG-MIB

Category	MIB
	• NQA-MIB • OSPF-TRAP-MIB • P-BRIDGE-MIB • Q-BRIDGE-MIB • RFC1213-MIB • RIPv2-MIB • RMON-MIB • SAVI-MIB • SNMP-FRAMEWORK-MIB • SNMP-MPD-MIB • SNMP-NOTIFICATION-MIB • SNMP-TARGET-MIB • SNMP-USER-BASED-SM-MIB • SNMPv2-MIB • TCP-MIB • UDP-MIB
Huawei-proprietary MIB	• HUAWEI-AAA-MIB • HUAWEI-ACL-MIB • HUAWEI-ALARM-MIB • HUAWEI-ALARM-RELIABILITY-MIB • HUAWEI-BASE-TRAP-MIB • HUAWEI-BRAS-RADIUS-MIB • HUAWEI-BRAS-SRVCFG-EAP-MIB • HUAWEI-BRAS-SRVCFG-STATICUSER-MIB • HUAWEI-CBQOS-MIB • HUAWEI-CDP-COMPLIANCE-MIB • HUAWEI-CONFIG-MAN-MIB • HUAWEI-CPU-MIB • HUAWEI-DAD-TRAP-MIB • HUAWEI-DC-MIB • HUAWEI-DATASYNC-MIB • HUAWEI-DEVICE-MIB • HUAWEI-DHCPR-MIB • HUAWEI-DHCPS-MIB • HUAWEI-DHCP-SNOOPING-MIB • HUAWEI-DIE-MIB • HUAWEI-DNS-MIB • HUAWEI-DLDP-MIB • HUAWEI-ELMI-MIB • HUAWEI-ERPS-MIB • HUAWEI-ERRORDOWN-MIB • HUAWEI-ENERGYMNGT-MIB • HUAWEI-EASY-OPERATION-MIB • HUAWEI-ENTITY-EXTENT-MIB

Category	MIB
	• HUAWEI-ENTITY-TRAP-MIB • HUAWEI-ETHARP-MIB • HUAWEI-ETHOAM-MIB • HUAWEI-FLASH-MAN-MIB • HUAWEI-FWD-RES-TRAP-MIB • HUAWEI-GARP-APP-MIB • HUAWEI-GTSM-MIB • HUAWEI-HGMP-MIB • HUAWEI-HWTACACS-MIB • HUAWEI-IF-EXT-MIB • HUAWEI-INFOCENTER-MIB • HUAWEI-IPPOOL-MIB • HUAWEI-IPV6-MIB • HUAWEI-ISOLATE-MIB • HUAWEI-L2IF-MIB • HUAWEI-L2MAM-MIB • HUAWEI-L2VLAN-MIB • HUAWEI_LDT-MIB • HUAWEI-LLDP-MIB • HUAWEI-MAC-AUTHEN-MIB • HUAWEI-MEMORY-MIB • HUAWEI-MFF-MIB • HUAWEI-MFLP-MIB • HUAWEI-MSTP-MIB • HUAWEI-MULTICAST-MIB • HUAWEI-NAP-MIB • HUAWEI-NTPV3-MIB • HUAWEI-PERFORMANCE-MIB • HUAWEI-PORT-MIB • HUAWEI-PORTAL-MIB • HUAWEI-QINQ-MIB • HUAWEI-RIPv2-EXT-MIB • HUAWEI-RM-EXT-MIB • HUAWEI-RRPP-MIB • HUAWEI-SECURITY-MIB • HUAWEI-SEP-MIB • HUAWEI-SNMP-EXT-MIB • HUAWEI-SSH-MIB • HUAWEI-STACK-MIB • HUAWEI-SWITCH-L2MAM-EXT-MIB • HUAWEI-SWITCH-SRV-TRAP-MIB • HUAWEI-SYS-MAN-MIB • HUAWEI-TCP-MIB • HUAWEI-TFTPC-MIB • HUAWEI-TRNG-MIB

Category	MIB
	• HUAWEI-XQOS-MIB

Standard Compliance

Standard Organization	Standard or Protocol
IETF	• RFC 768 User Datagram Protocol (UDP) • RFC 792 Internet Control Message Protocol (ICMP) • RFC 793 Transmission Control Protocol (TCP) • RFC 826 Ethernet Address Resolution Protocol (ARP) • RFC 854 Telnet Protocol Specification • RFC 951 Bootstrap Protocol (BOOTP) • RFC 959 File Transfer Protocol (FTP) • RFC 1058 Routing Information Protocol (RIP) • RFC 1112 Host extensions for IP multicasting • RFC 1157 A Simple Network Management Protocol (SNMP) • RFC 1256 ICMP Router Discovery • RFC 1305 Network Time Protocol Version 3 (NTP) • RFC 1349 Internet Protocol (IP) • RFC 1493 Definitions of Managed Objects for Bridges • RFC 1542 Clarifications and Extensions for the Bootstrap Protocol • RFC 1643 Ethernet Interface MIB • RFC 1757 Remote Network Monitoring (RMON) • RFC 1901 Introduction to Community-based SNMPv2 • RFC 1902-1907 SNMP v2 • RFC 1981 Path MTU Discovery for IP version 6 • RFC 2131 Dynamic Host Configuration Protocol (DHCP) • RFC 2328 OSPF Version 2 • RFC 2453 RIP Version 2 • RFC 2460 Internet Protocol, Version 6 Specification (IPv6) • RFC 2461 Neighbor Discovery for IP Version 6 (IPv6) • RFC 2462 IPv6 Stateless Address Auto configuration • RFC 2463 Internet Control Message Protocol for IPv6 (ICMPv6) • RFC 2474 Differentiated Services Field (DS Field) • RFC 2740 OSPF for IPv6 (OSPFv3) • RFC 2863 The Interfaces Group MIB • RFC 2597 Assured Forwarding PHB Group • RFC 2598 An Expedited Forwarding PHB • RFC 2571 SNMP Management Frameworks • RFC 2865 Remote Authentication Dial In User Service (RADIUS) • RFC 3046 DHCP Option82 • RFC 3376 Internet Group Management Protocol, Version 3 (IGMPv3) • RFC 3513 IP Version 6 Addressing Architecture • RFC 3579 RADIUS Support For EAP • RFC 4271 A Border Gateway Protocol 4 (BGP-4)

Standard Organization	Standard or Protocol
	• RFC 4760 Multiprotocol Extensions for BGP-4 • draft-grant-tacacs-02 TACACS+
IEEE	• IEEE 802.1D Media Access Control (MAC) Bridges • IEEE 802.1p Traffic Class Expediting and Dynamic Multicast Filtering • IEEE 802.1Q Virtual Bridged Local Area Networks • IEEE 802.1ad Provider Bridges • IEEE 802.2 Logical Link Control • IEEE Std 802.3 CSMA/CD • IEEE Std 802.3u 100BASE-T specification • IEEE Std 802.3ab 1000BASE-T specification • IEEE Std 802.3ad Aggregation of Multiple Link Segments • IEEE Std 802.3ae 10GE WEN/LAN Standard • IEEE Std 802.3x Full Duplex and flow control • IEEE Std 802.3z Gigabit Ethernet Standard • IEEE802.1ax/IEEE802.3ad Link Aggregation • IEEE 802.1ab Link Layer Discovery Protocol • IEEE 802.1D Spanning Tree Protocol • IEEE 802.1w Rapid Spanning Tree Protocol • IEEE 802.1s Multiple Spanning Tree Protocol • IEEE 802.1x Port based network access control protocol • IEEE 802.3af DTE Power via MIDI • IEEE 802.3at DTE Power via the MDI Enhancements • IEEE 802.3 Ethernet • IEEE 802.3u Fast Ethernet • IEEE 802.3z Gigabit Ethernet • IEEE 802.3ab Gigabit Ethernet over Copper • IEEE 802.3ad Link Aggregation
ITU	• ITU SG13 Y.17ethoam • ITU SG13 QoS control Ethernet-Based IP Access • ITU-T Y.1731 ETH OAM performance monitor
ISO	• ISO 10589 IS-IS Routing Protocol
MEF	• MEF 2 Requirements and Framework for Ethernet Service Protection • MEF 9 Abstract Test Suite for Ethernet Services at the UNI • MEF 10.2 Ethernet Services Attributes Phase 2 • MEF 11 UNI Requirements and Framework • MEF 13 UNI Type 1 Implementation Agreement • MEF 15 Requirements for Management of Metro Ethernet Phase 1 Network Elements • MEF 17 Service OAM Framework and Requirements • MEF 20 UNI Type 2 Implementation Agreement • MEF 23 Class of Service Phase 1 Implementation Agreement • XMODEM/YMODEM Protocol Reference

Ordering Information

The following table lists ordering information of the CloudEngine S5735I-S-V2 series switches.

Model	Product Description
CloudEngine S5735I-S8T4SN-V2	CloudEngine S5735I-S8T4SN-V2(8x10/100/1000BASE-T ports, 4xGE SFP ports, AC power)
CloudEngine S5735I-S8T4XN-V2	CloudEngine S5735I-S8T4XN-T-V2(8x10/100/1000BASE-T ports, 4x10GE SFP+ ports, AC power)
CloudEngine S5735I-S8T4XN-T-V2	CloudEngine S5735I-S8T4XN-T-V2(8x10/100/1000BASE-T ports, 4x10GE SFP+ ports, HTM, AC power)
CloudEngine S5735I-S8U4XN-V2	CloudEngine S5735I-S8U4XN-V2(8x10/100/1000BASE-T ports, 4x10GE SFP+ ports,PoE++, AC power)
CloudEngine S5735I-S8U2XN-V2	S5735I-S8U2XN-V2 (8*10/100/1000BASE-T ports, PoE++, 2*10GE SFP+ ports, DIN Rail Mounting, Dual redundant 54 to 57V DC supply, Fanless, Without power module)
CloudEngine S5735I-S16T2S4XN-V2	CloudEngine S5735I-S16T2S4XN-V2 (16*10/100/1000BASE-T ports,2*GE SFP ports,4*10GE SFP+ ports, DIN Rail Mounting, Dual redundant 9.6 to 60V DC power, Fanless)
CloudEngine S5735I-S8T8P2S4XN-V2	CloudEngine S5735I-S8T8P2S4XN-V2 (16*10/100/1000BASE-T ports,8*POE+,2*GE SFP ports, 4*10GE SFP+ ports, DIN Rail Mounting, Dual redundant 54 to 57V DC power, Fanless)
PAC60S12-AN	Industrial 60 W AC power module,DIN RAIL
PAC240S56-CN	Industrial 240W PoE power module, DIN RAIL
N1-S57S-M-Lic	S57XX-S Series Basic SW,Per Device
N1-S57S-M-SnS1Y	S57XX-S Series Basic SW,SnS,Per Device,1Year
N1-S57S-F-Lic	N1-CloudCampus,Foundation,S57XX-S Series,Per Device
N1-S57S-F-SnS1Y	N1-CloudCampus,Foundation,S57XX-S Series,SnS,Per Device,1Year
N1-S57S-A-Lic	N1-CloudCampus,Advanced,S57XX-S Series,Per Device
N1-S57S-A-SnS1Y	N1-CloudCampus,Advanced,S57XX-S Series,SnS,Per Device,1Year
N1-S57S-FToA-Lic	N1-Upgrade-Foundation to Advanced,S57XX-S,Per Device
N1-S57S-FToA-SnS1Y	N1-Upgrade-Foundation to Advanced,S57XX-S,SnS,Per Device,1Year

More Information

For more information about Huawei Campus Switches, visit http://e.huawei.com or contact us in the following ways:

- Global service hotline: http://e.huawei.com/en/service-hotline
- Logging in to the Huawei Enterprise technical support website: http://support.huawei.com/enterprise/
- Sending an email to the customer service mailbox: support_e@huawei.com

Huawei Technologies Co., Ltd.

Address:Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China

Website:e.huawei.com

10.8 10GE SFP+ Optical Modules

10.8.1 OMXD30000 (02318169)

26,33 &54 - Optical Transceiver,SFP+,10G,Multi-mode Module(850nm,0.3km,LC)

Table 10-41 OMXD30000 specifications

Item	Value
Basic Information	
Module name	OMXD30000
Part Number	02318169
Model	OMXD30000
Form factor	SFP+
Application standard	10GBASE-SR
Connector type	LC
Optical fiber type	MMF
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
Digital diagnostic monitoring (DDM)	Supported
Transmission rate [bit/s]	10 Gbit/s
Target transmission distance [km]	Multimode fiber (with modal bandwidth of 160 MHz*km and diameter of 62.5 µm): 0.026 km Multimode fiber (OM1): 0.033 km Multimode fiber (with modal bandwidth of 400 MHz*km and diameter of 50 µm): 0.066 km Multimode fiber (OM2): 0.082 km Multimode fiber (OM3): 0.3 km Multimode fiber (OM4): 0.4 km
Transmitter Optical Characteristics	
Center wavelength [nm]	850 nm
Maximum Tx optical power [dBm]	-1.0 dBm
Minimum Tx optical power [dBm]	-7.3 dBm
Minimum extinction ratio [dB]	3.0 dB
Receiver Optical Characteristics	

Item	Value
Rx sensitivity [dBm]	-11.1 dBm
Overload power [dBm]	-1.0 dBm

10.8.2 OSX010000 (02318170)

Table 10-42 OSX010000 specifications

Item	Value
Basic Information	
Module name	OSX010000
Part Number	02318170
Model	OSX010000
Form factor	SFP+
Application standard	10GBASE-LR
Connector type	LC
Optical fiber type	SMF
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
Digital diagnostic monitoring (DDM)	Supported
Transmission rate [bit/s]	10 Gbit/s
Target transmission distance [km]	Single-mode fiber: 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1310 nm
Maximum Tx optical power [dBm]	0.5 dBm
Minimum Tx optical power [dBm]	-8.2 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity [dBm]	-12.6 dBm
Overload power [dBm]	0.5 dBm

Item	Value
Rx sensitivity [dBm]	-10.5 dBm
Overload power [dBm]	2.3 dBm

NOTE

This module can only be used on a switch running V200R022C10 or a later version.

10.13.13 QSFP-40G-LX4-MM

43 - 40GBase-LX4-MM Optical Transceiver,QSFP+,40GE,Multimode(1310nm,0.15km,LC)

Table 10-97 QSFP-40G-LX4-MM specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LX4-MM
Part Number	02313NUG
Model	QSFP-40G-LX4-MM
Form factor	QSFP+
Application standard	40GBASE-LX4
Connector type	LC
Optical fiber type	MMF
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
Digital diagnostic monitoring (DDM)	Supported
Transmission rate [bit/s]	40 Gbit/s
Target transmission distance [km]	Multimode fiber (OM3, diameter: 50 µm): 150 m Multimode fiber (OM4, diameter: 50 µm): 150 m
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power [dBm]	2.3 dBm
Minimum Tx optical power [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB
Receiver Optical Characteristics	
Rx sensitivity [dBm]	-10.5 dBm

Item	Value
Overload power [dBm]	3.5 dBm
NOTE Limitations: - In actual applications, the number of connectors in an optical fiber link cannot exceed 4. - This module is sensitive to fiber link contamination. During deployment, ensure that the fiber end face meets the fiber application standard. For details, refer to the requirements for single-mode connectors in the end face requirements for fiber ceramic ferrules under "Cables" > "Fiber Jumpers."	

NOTE

This module can only be used on a switch running V200R022C10 or a later version.

10.13.14 QSFP-40G-LR4 (02313URY)

Table 10-98 QSFP-40G-LR4 specifications

Item	Value
Basic Information	
Module name	QSFP-40G-LR4
Part Number	02313URY
Model	QSFP-40G-LR4
Form factor	QSFP+
Application standard	40GBASE-LR4
Connector type	LC
Optical fiber type	SMF
Working case temperature [°C(°F)]	0°C to 70°C (32°F to 158°F)
Digital diagnostic monitoring (DDM)	Supported
Transmission rate [bit/s]	40 Gbit/s
Target transmission distance [km]	Single-mode fiber: 10 km
Transmitter Optical Characteristics	
Center wavelength [nm]	1271 nm,1291 nm,1311 nm,1331 nm
Maximum Tx optical power [dBm]	2.3 dBm
Minimum Tx optical power [dBm]	-7.0 dBm
Minimum extinction ratio [dB]	3.5 dB

PowerEdge R350

Spec Sheet

DELL Technologies

Deliver powerful compute with a single processor server

52 - R350 DELL SERVER R350 INTEL XEON 16G 2TB

The Dell EMC PowerEdge R350 addresses evolving compute demands with an easy to manage rack server designed for businesses looking for affordable enterprise features.

Your Innovation Engine for businesses of all sizes

The Dell EMC PowerEdge R350, powered by Intel® Xeon® E-2300 processors, delivers increased performance and is designed for productivity and data intensive applications. It supports 3200 MT/s DDR4 speeds and 32 GB DIMMs, up to 128 GB for memory intensive workloads. In addition, to address substantial throughput improvements, the PowerEdge R350 supports PCIe Gen 4 and offers enhanced efficiency to support increasing power and thermal requirements. This makes the PowerEdge R350 a powerful and versatile server for small to mid-sized businesses to enable a variety of workloads ranging from business-critical to cloud infrastructure. It is also widely used for point-of-sale transactions and enterprise level requirements for data analysis and virtualization.

Increase efficiency and accelerate operations with autonomous collaboration

The Dell EMC OpenManage systems management portfolio tames the complexity of managing and securing IT infrastructure. Using Dell Technologies' intuitive end-to-end tools, IT can deliver a secure, integrated experience by reducing process and information silos in order to focus on growing the business. The Dell EMC OpenManage portfolio is the key to your innovation engine, unlocking the tools and automation that help you scale, manage, and protect your technology environment.

- Built-in telemetry streaming, thermal management, and RESTful API with Redfish offer streamlined visibility and control for better server management
- Intelligent automation lets you enable cooperation between human actions and system capabilities for added productivity
- Integrated change management capabilities for update planning and seamless, zero-touch configuration and implementation
- Full-stack management integration with Microsoft, VMware, ServiceNow, Ansible and many other tools

Protect your data assets and infrastructure with proactive resilience

The Dell EMC PowerEdge R350 server is designed with a cyber-resilient architecture, integrating security deeply into every phase in the lifecycle, from design to retirement.

- Operate your workloads on a secure platform anchored by cryptographically trusted booting and silicon root of trust
- Maintain server firmware safety with digitally signed firmware packages
- Prevent unauthorized configuration or firmware change with system lockdown
- Securely and quickly wipe all data from storage media including hard drives, SSDs and system memory with System Erase

PowerEdge R350

The Dell EMC PowerEdge R350 offers streamlined productivity, high-speed memory and capacity, powerful compute to address common business applications. Ideal for inside or outside the data center:

- Small to mid-sized businesses
- Remote office/branch office
- Collaboration and sharing
- Database support and management

Feature	Technical Specifications
Processor	One Intel Xeon E-2300 series processor with up to 8 cores or one Intel Pentium processor with up to 2 cores
Memory	• Four DDR4 DIMM slots, supports UDIMM 128 GB max, speeds up to 3200 MT/s • Supports unregistered ECC DDR4 DIMMs only NOTE: For Pentium processor, maximum memory speed supported is 2666 MT/s.
Storage controllers	• Internal controllers (RAID): PERC H355, PERC H355f, PERC H345, PERC H345f, H755, H755f, HBA355i, HBA355f, S150 • Internal Boot: Internal Dual SD Module or USB or Boot Optimized Storage Subsystem (BOSS-S2): HWRAID 2 x M.2 SSDs • External HBAs (non-RAID): HBA355e
Drive Bays	Front bays: • Up to 4 x 3.5-inch SAS/SATA (HDD/SSD) max 64 TB • Up to 8 x 2.5-inch SAS/SATA (HDD/SSD) max 128 TB
Power Supplies	• 600W Platinum 100-240 VAC or 240 VDC redundant, hot swap
Cooling options	Air cooling
Fans	Up to four fans with full redundancy
Dimensions	• Height – 42.8 mm (1.69 inches) • Width – 482 mm (18.98 inches) • Depth – 585.3 mm (23.04 inches) without bezel 598.94 mm (23.58 inches) with bezel
Form Factor	1U rack server
Embedded Management	• iDRAC9 • iDRAC Direct • iDRAC RESTful API with Redfish • iDRAC Service Module
Bezel	Optional LCD bezel or security bezel
OpenManage Software	• OpenManage Enterprise • OpenManage Power Manager plugin • OpenManage SupportAssist plugin • OpenManage Update Manager plugin
Mobility	OpenManage Mobile
Integrations and Connections	OpenManage Integrations • BMC Truesight • Microsoft System Center • Red Hat Ansible Modules • VMware vCenter and vRealize Operations Manager OpenManage Connections • IBM Tivoli Netcool/OMNIbus • IBM Tivoli Network Manager IP Edition • Micro Focus Operations Manager • Nagios Core • Nagios XI
Security	• Cryptographically signed firmware • Secure Boot • Secure Erase • Silicon Root of Trust • System Lockdown (requires iDRAC9 Enterprise or Datacenter) • TPM 1.2/2.0 FIPS, CC-TCG certified, TCM 2.0 optional
Embedded NIC	2 x 1 GbE LOM
Network Options	NA
Ports	Front Ports • 1 x iDRAC Direct (Micro-AB USB) port • 1 x USB 2.0 Internal Ports • 1 x USB 3.0 (optional) Rear Ports • 1 x USB 2.0 • 1 x iDRAC ethernet port • 1 x USB 3.0 • 1 x Serial port • 1 x VGA
PCIe	3 PCIe slots: • 1x8 Gen4 (x16 connector) low profile, half length • 1x8 Gen4 (x8 connector) low profile, half length • 1x4 Gen4 (x8 connector) for dedicated PERC only
Operating System and Hypervisors	• Canonical Ubuntu Server LTS • Citrix Hypervisor • Microsoft Windows Server with Hyper-V • Red Hat Enterprise Linux • SUSE Linux Enterprise Server For specifications and interoperability details, see Dell.com/OSsupport.
OEM-ready version available	From bezel to BIOS to packaging, your servers can look and feel as if they were designed and built by you. For more information, visit Dell.com/OEM.

Recommended support and services

Dell ProSupport Plus for critical systems or Dell ProSupport for premium hardware and software support for your PowerEdge solution. Consulting and deployment offerings are also available. Contact your Dell representative today for more information. Availability and terms of Dell Services vary by region. For more information, visit Dell.com/ServiceDescriptions.

APEX Flex on demand

Acquire the technology you need to support your changing business with payments that scale to match actual usage. For more information, visit: www.delltechnologies.com/en-us/payment-solutions/flexible-consumption/flex-on-demand.

Discover more about PowerEdge servers

Learn more about our PowerEdge servers	Learn more about our systems management solutions	Search our Resource Library	Follow PowerEdge servers on Twitter	Contact a Dell Technologies Expert for Sales or Support

December 2021